Evaluating Baseline Detectors of Hijacked LLM Agents Under Rare Attacks and Distribution Shift

Mahmoud Abbasi and Juan M. Núñez V.

***Abstract*—Detectors can flag hijacked large language model (LLM) agents from their tool-call traces. In eleven audited detector evaluations, attacks make up 16% to 86% of the test samples, none varies attack prevalence, and training and test data mostly share a generator. We ask what such evaluations predict about deployment. On AgentDrift and AgentDojo we evaluate logistic regression on tool-identity, behavioral and combined features, a gated recurrent unit (GRU) sequence model, a content reference based on term frequency-inverse document frequency (TF-IDF) and a released LLM action monitor (AlignmentCheck) with two backbones, under a graded series of shifts with thresholds fixed on training data. Three results follow. At 2% prevalence the strongest in-distribution model reaches an area under the receiver operating characteristic curve (AUROC) of 0.879 but a precision of 0.076 at its training threshold, and that precision moves from 0.039 to 0.117 with the mix of negatives alone. As the share of test tools absent from training rises, detector rankings cross near half coverage and reverse below it, because unseen tools reach untrained inputs, and training-time dropout largely removes the reversal. Training thresholds also over-alert, but a reimplemented frozen-encoder detector does so without untrained inputs. No detector trained on one benchmark transfers to the other (AUROC 0.47 to 0.57), including the content reference, which reaches 0.996 within AgentDojo mostly by matching fixed attack strings, whereas the released monitor, trained on neither, reaches 0.930 with an open backbone and 0.934 with a commercial one. With the open backbone its native decision would still mostly raise false alerts at 2% prevalence (precision 0.096 on AgentDojo). We do not propose a detector. We report what these evaluation choices hide.**

***Index Terms*—Anomaly detection, class imbalance, distribution shift, evaluation methodology, intrusion detection, large language model agents, precision-recall analysis, prompt injection.**

I. Introduction

Agents built on large language models (LLMs) carry out user tasks by calling tools such as mail clients, payment services, file systems and web browsers, and the output of each call returns to the model as context. Instructions planted in that output can redirect the agent toward an attacker's goal, a threat known as indirect prompt injection [1]. Benchmarks such as InjecAgent [2] and AgentDojo [3] show that such injections still succeed against current models at rates that cannot be ignored. A hijacked agent has to act through its tools, so its sequence of tool calls is a natural place to look for evidence of compromise. Recent work proposes rule-based, automaton-based, graph, sequence and verifier detectors that score such traces [4]–[13].

How these detectors are evaluated matters as much as how they are built. In the eleven evaluations we audited (Section II), test sets contain 16% to 86% attacks, only four works report a threshold-free metric, none varies attack prevalence, and training and test data usually come from the same generator. In operation, successful hijacks may be a small and unknown share of the sessions that a security team audits, and the tools, tasks and domains an agent meets change over time and between organizations.

When attacks are rare, even a small false-positive rate yields alerts that are mostly false [14]. The receiver operating characteristic (ROC) does not depend on prevalence while precision does [15], [16], although the choice between ROC and precision-recall summaries is itself debated [17]. Learned security detectors also degrade under distribution shift and profit from biased splits [18]–[20]. Tool-call detectors face a further shift that is easy to measure, because one can count the share of test tool calls whose tool appeared in training.

MCPShield [9] shows that random splits can inflate the area under the ROC curve (AUROC) by up to 25.8 points relative to task-disjoint splits and traces the gap to metadata features, a one-hot tool name with an argument hash and the response length, that learn task-specific tool signatures. Agent-Sentry [6] transfers its pipeline to native AgentDojo and AgentDyn [21] traces and then fits and tests on those traces by grouped cross-validation, which amounts to a within-target control. TrajAD [11] compares leave-one-domain-out training with a fully supervised upper bound, and Fomin [22] makes the leave-one-dataset-out case for classifiers of malicious prompts. No audited evaluation measures detection at controlled attack prevalence or reports tool-vocabulary coverage for every split. None contains a domain shift in which no test tool occurs in training, or a transfer from a synthetic generator to recorded agent runs. Each of these elements has antecedents. Our contribution is to run them jointly on one set of detectors, with thresholds fixed on training data, so that the effect of each shift can be read against the others.

We evaluate logistic regression on tool identity (LR-A), on behavioral features (LR-B) and on both (LR-C), a gated recurrent unit (GRU) sequence model and a content reference based on term frequency-inverse document frequency (TF-IDF). As a released reference we add AlignmentCheck, the action monitor of LlamaFirewall [13], with an open and a commercial LLM backbone. Five research questions guide the study. RQ1 asks how the prevalence-weighted area under the precision-recall curve (AUPRC) and precision at a training-derived threshold compare with AUROC as attack prevalence falls from 10% to 1%, and how much they depend on the mix of negatives. RQ2 concerns whether reuse of generation worlds or tasks inflates the standard estimate. RQ3 examines how ranking and operating points change when a domain and

M. Abbasi (corresponding author, e-mail: mahmoudabbasi@usal.es) and J. M. Núñez V. are with the BISITE Research Group, University of Salamanca, 37007 Salamanca, Spain.

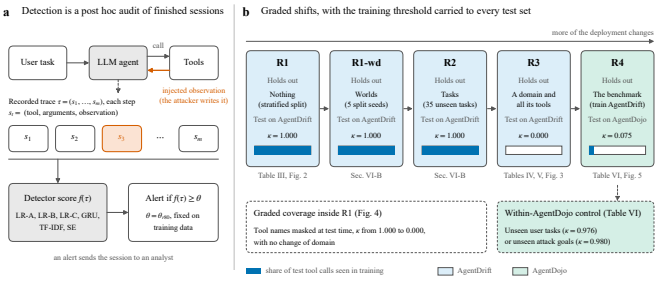


Fig. 1. Detection setting and evaluation design. (a) The attacker controls the content of a tool observation, and the detector scores the recorded trace of each finished session against a threshold fixed on training data. (b) The five settings, from the standard split R1 to cross-benchmark transfer R4, with the tool-vocabulary coverage κ of each test set (bar, as in Table II) and the table or figure that reports it. Dashed boxes are the two controls, graded coverage inside R1 and the within-AgentDojo control. Thresholds are never re-tuned on test data.

all its tools are held out, and why. RQ4 tests whether a detector trained on one benchmark keeps its signal on another once a within-target control separates a failure to transfer from features that cannot detect the target attacks. RQ5 asks which of these conclusions hold for a detector that reads content.

The paper makes four contributions. First, it defines an evaluation protocol (Fig. 1). The protocol reports prevalence-weighted AUPRC, which is the calibrated average precision of Siblini et al. [23], and treats the negative-class mix as a second mixing parameter. It also reports tool-vocabulary coverage for every setting, applies a graded series of world, task, domain and benchmark shifts, and adds a within-target control with checks on the composition of the target benchmark. Second, it follows detector behavior across the whole range of tool-vocabulary overlap rather than at full and zero coverage alone. The in-distribution ranking of our detectors holds while most test tools are known, becomes indistinguishable once about half of them are new and reverses below that, because unseen tools reach untrained inputs. Dropout during training largely removes the reversal. The training thresholds of the affected detectors degrade steadily over the same range, and a reimplemented frozen-encoder detector shows that this failure also arises without untrained inputs. Threshold failure turns out to be detector-specific, and the behavioral signal that survives zero coverage is mostly an observation-length cue written by the attacker. Third, it shows that both benchmarks are nearly separable by surface content, AgentDrift through its generator templates and AgentDojo through fixed attack strings. No trained detector transfers from AgentDrift to AgentDojo, not even a content detector that is near perfect in distribution, and the within-target control succeeds for some representations only. A released action monitor that is trained on neither benchmark does transfer. Fourth, it turns these findings into recommendations. We do not propose a detector. Our claims concern our own baselines, one content reference and one reimplementation, not the published detectors of Section II.

II. Related Work

A. Detectors Over Agent Tool-Call Traces

We identified the audited works through targeted searches of arXiv and Google Scholar up to September 2026, looking for detectors of compromised LLM agents that operate on tool-call traces. A work was included when it proposed or evaluated a trajectory-level or step-level detector of compromised or anomalous agent behavior on agent traces. Eleven works met this criterion. The audit is not a systematic review, since we followed no registered search protocol, and relevant detectors may have been missed. We left out MELON [24] because it re-executes the live agent rather than scoring recorded traces. TraceSafe [25] and ATBench [26] were excluded as guardrail benchmarks, and AgentDoG [27] as a safety guard model evaluated on such benchmarks. StepGuard [28], a step-level guard model posted inside our search window, was identified only during revision, so we discuss it but did not add it to the audit. TrajAD and Trajectory Guard [8] target procedural errors and plan anomalies rather than injections. Their protocols still bear directly on our questions, since TrajAD holds out a domain and Trajectory Guard uses external test sets without benign traces.

The audited detectors range from rule, automaton and LLM-overseer monitors, such as TraceAegis [4], Praetor [5], AgentSentry and SentinelAgent [12], to learned sequence, graph and verifier models, such as DriftNet [7], Trajectory Guard, Skynet [10], MCPShield and TrajAD. LlamaFirewall [13] pairs the PromptGuard 2 input classifier with AlignmentCheck, a few-shot LLM auditor that reasons over the whole execution trace without fine-tuning. AgentDrift [29] is a benchmark rather than a detector, and its surface-feature logistic regression only checks that the benchmark cannot be solved from surface cues.

Table S18 of the supplement records the protocol of each audited work, with every cell checked against the arXiv full text of the version named in its note. In the test sets that mix both classes, attacks or anomalies make up between 16.5% (Skynet on Agent-SafetyBench [30]) and 86.3% (MCPShield on RAS-Eval [31]) of the samples, by our arithmetic from the reported class counts. Four works report a threshold-free measure. The others report metrics at one operating point or no quantitative metric, and none varies the class ratio. Eight works test drift in a limited form, such as a task-disjoint split, unseen anomaly types or a prompt-style shift, and three not at all. Only MCPShield reports repeated seeds, and Praetor reports Clopper-Pearson intervals for single runs.

B. Agent-Security Benchmarks and Prompt-Injection Defenses

InjecAgent and AgentDojo measure how often injected tool outputs redirect an agent, and ASB [32], AgentDyn and RAS-Eval extend the measurement to more scenarios and to dynamic or real tool execution. These benchmarks report attack success and utility for live agents. They do not provide labeled corpora for training trace detectors, which AgentDrift does. Agent-SafetyBench, ATBench and TraceSafe instead ask whether a judge recognizes unsafe behavior in a recorded trajectory. Many defenses, MELON among them, act on the live agent or its inputs. By contrast, the audited detectors

and ours score recorded traces, which is closer to intrusion detection over logs. For prompt classifiers, Fomin finds that pooled cross-validation reports AUC 8.0 to 16.5 points higher than leave-one-dataset-out evaluation.

C. In-Distribution Versus Out-of-Distribution Performance and Prior Shift

Out-of-distribution accuracy is often almost linearly correlated with in-distribution accuracy [33]. Teney et al. [34] show, however, that the correlation can be inverted on real-world data, so that selecting by in-distribution accuracy selects against robustness. Shortcut learning likewise yields decision rules that fail under harder test conditions [35]. Our configurations are fixed without hyperparameter search and our thresholds are set on training data, so no test-domain information enters model selection. We measure the size of each shift with a classifier two-sample test [36]. Because precision depends on the class prior, AUPRC values from test sets with different class ratios cannot be compared directly. Siblini et al. [23] calibrate precision and average precision to a reference prior in closed form, and Boyd et al. [37] compare estimators and confidence intervals for the area under the precision-recall curve. McDermott et al. question the general preference for AUPRC under imbalance, so we report AUROC next to it.

D. Distribution Shift in Security Machine Learning

Sommer and Paxson [18] place anomaly-based intrusion detection outside the closed-world assumption of most machine learning. Arp et al. [20] catalogue pitfalls such as sampling bias, spurious correlations and inappropriate baselines. TESSERACT [19] shows how biased splits inflate results in malware classification, and conformal evaluation rejects decisions on drifting samples [38]. These works treat drift mainly as a function of time. Tool-call detectors also face a shift that is easier to quantify, because tools absent from training can appear at test time, and the coverage κ of Section III-D records how often they do.

E. Positioning of This Study

We neither propose a detector nor re-evaluate the audited detectors as released. DriftNet's code and weights have not been released, so we reimplemented its architecture from the paper as one further detector (Section VI-D). None of our numbers is DriftNet's own. Agent-Sentry's within-target control uses two benchmarks that share agents, tools and environments. TrajAD's held-out domain, on which F1 is 83.09% against 83.84% for the supervised upper bound, comes from the same generation pipeline as its training data. The task-disjoint splits of Agent-Sentry, DriftNet and MCPShield hold out tasks rather than tools. Our feature ablation extends MCPShield's finding on tool-name metadata, and its probe of the generating model, to a combined model for which no test tool was seen in training. Praetor remains the only audited work that evaluates adaptive attackers.

III. Benchmarks and Problem Formulation

A. AgentDrift

AgentDrift [29] contains 12,536 synthetic trajectories from five domains (email, banking, web, coding and medical), generated according to its README and datasheet by Llama-3.3-70B-Instruct (quantized) at temperature 0.8 with prompts and scripts that are not released (Table I). It ships a stratified split with 8,775 training, 1,880 validation and 1,881 test trajectories and a task-disjoint split with 9,081 training and 1,722 test trajectories. Every trajectory carries a world record (user, e-mail address, company, contacts and date). The 10,656 training and test trajectories contain 1,133 worlds. Labeling each trajectory with its world's majority label is correct for 0.883 [0.860, 0.909] of them, against 0.558 for the majority class. The release documents several template regularities. By our count, an ext-audit.com address appears in 0.281 of the 7,036 poisoned observations, and 0.985 of hard negatives carry at least one of the three markers the README names. Banking has 256 full hijacks against 720 in each other domain, because the validator cut off its generation job. The banking fold therefore differs from the others in hijack mix (34.7% full hijacks against 58.5% to 60.6%) and prevalence (0.401 against 0.522), which confounds comparisons across folds that involve banking.

B. AgentDojo

We use the recorded runs without defenses of `gpt-4o-2024-05-13`, `claude-3-5-sonnet-20241022`, `gemini-1.5-pro-002`, `gpt-4-turbo-2024-04-09` and `gemini-2.0-flash-001` from the AgentDojo repository [3] over the workspace, travel, banking and Slack suites. A run without an attack is benign, and a run with an attack is compromised when the benchmark's security check reports that the attacker's goal was reached and unsuccessful otherwise. Runs with fewer than two tool calls are removed (Table I), and the removal is asymmetric across classes and models. It takes, for example, 77 of 132 `gemini-2.0-flash-001` benign runs. Because trace length alone separates compromised from benign traces with an AUROC of 0.675, the filter shapes the negatives. Population P0 has a native prevalence (the attack share of the unresampled pool) of 0.854, 117 user-task identifiers and 30 injection goals. Of its traces, 69.8% contain no tool seen in AgentDrift. The 4,987 retained unsuccessful attacks are scored separately. Three properties act as confounds. First, 65 benign traces ran an injection task as the user task, and removing them gives population C1. Second, `gpt-4o-2024-05-13` contributes 1,866 of the 2,294 compromised traces (81.3%). Third, 101 compromised traces, all from that model, come from denial-of-service (DoS) variants, which AgentDojo labels compromised exactly when the user task was not completed.

An earlier version of this study paired each tool call with its observation through the tool-call identifier. Gemini runs leave that identifier empty and some GPT runs reuse it, so affected steps received another step's observation. The earlier extractor also cut observations at 2,000 characters without saying so.

TABLE I
CHARACTERISTICS OF THE EVALUATED BENCHMARKS

Property	AgentDrift [29]	AgentDojo [3] (population P0)
Version and license	Release 1.0 (record schema 2.0), commit `014a514f`, CC BY 4.0	Commit `089ed468`, MIT
Domains or suites	5 (email, banking, web, coding, medical)	4 (workspace, travel, banking, Slack)
Distinct tools	103, disjoint across domains	67, of which 7 occur in AgentDrift (κ = 0.075)
Negative class	4,000 benign and 1,500 hard negatives, plus 1,500 failed attacks used as training negatives and scored separately at test	392 benign
Positive class	5,536 attacked (3,136 full, 1,500 partial, 900 delayed hijacks)	2,294 compromised (security check succeeded), 101 of them DoS
Excluded runs	None	Runs with fewer than two tool calls (236 benign, 70 compromised and 2,260 unsuccessful-attack runs)
Trajectory length (tool calls)	3 to 11 (mean 5.67)	2 to 29 (mean 5.24)

The corrected extractor pairs calls and observations by position, checked by a unit test, and stores observations uncut. The fix changes 413 of the 2,686 P0 traces by observation content and 408 by per-step observation length. Section S2 and Table S1 give the changes by model, the extent of the cut, and the earlier and corrected values. Every AgentDojo number here comes from the corrected extraction.

C. Threat and Deployment Model

We consider an indirect injector who controls the content of tool outputs that the agent reads, has no access to the agent's model or to the detector, and does not adapt to the detector. The goal is to make the agent act on the attacker's behalf, for example by sending data or money, or, in the DoS variants, to stop it from completing its task. The defender scores the recorded trace of each finished session once, and an alert sends the session to an analyst. Our setting is therefore a post hoc audit. Recall counts hijacked sessions identified afterwards rather than prevented actions, and alerts are counted per 1,000 audited sessions. The prevalence grid of 1% to 10% (Section IV-B) stands for the share of audited sessions that contain a successful hijack, and its lower end is set by our pool sizes.

The content-blind detectors read at most tool names, argument counts and observation lengths. The content reference also reads argument values and observation text, including the injected text. The attacker directly writes the maximum and mean observation length (Section V-A), since the injected observation is one of the observations. The other behavioral features depend on the actions the hijack induces, which the attacker influences through the injected goal but does not set. Argument counts and tool identities are limited by the tools available to the agent.

D. Trace-Level Detection Task

A trajectory is a sequence of tool-call steps

$$\tau = (s_1, \ldots, s_m), \qquad s_t = (a_t, u_t, o_t), \tag{1}$$

where a_t is the tool name, u_t the arguments and o_t the observation returned by the tool. The label $y(\tau) \in \{0, 1\}$ is 1 for an attacked AgentDrift trajectory or a compromised AgentDojo trace. A detector assigns each trajectory of the form (1) a score $f(\tau) \in \mathbb{R}$ and flags it when the score reaches a threshold θ,

$$\hat{y}_\theta(\tau) = \mathbf{1}\left[f(\tau) \geq \theta\right]. \tag{2}$$

In training, the failed attacks and hard negatives of AgentDrift count as negatives. For AgentDrift, evaluation uses pool E, which holds benign, hard-negative and attacked test trajectories, and failed attacks are scored separately. For AgentDojo it uses P0. Every AUROC and AUPRC in the paper refers to these pools unless another population is named. With $\mathcal{V}$ the tool vocabulary of the benign training trajectories, the tool-vocabulary coverage of a test set $\mathcal{D}$ is the share of its tool calls whose tool appears in $\mathcal{V}$,

$$\kappa(\mathcal{D}; \mathcal{V}) = \frac{\sum_{\tau \in \mathcal{D}} \sum_{t=1}^{m(\tau)} \mathbf{1}\left[a_t \in \mathcal{V}\right]}{\sum_{\tau \in \mathcal{D}} m(\tau)}. \tag{3}$$

E. Distribution-Shift Settings

Table II defines the settings and gives κ from (3), and Fig. 1(b) shows how they relate. R1 is the standard stratified split. R1-wd re-splits the pooled training and test trajectories over five split seeds so that no world is shared. R2 is the task-disjoint split, and R3 holds out one domain, which forces $\kappa = 0$. R4 trains on AgentDrift and tests on AgentDojo, changing the ecosystem, the trace generator, attack construction and label semantics at once. The within-AgentDojo control trains and tests on P0 by five-fold cross-validation, repeated with fold-assignment seeds 0 to 4. Folds are grouped by suite and user task (UT) or, for compromised traces, by suite and injection goal (IT). UT-F adds the unsuccessful attacks as training negatives. P0F adds them to P0 as evaluation negatives, and AgentDojo leave-one-suite-out (LOSO) is a recorded-trace analogue of R3. Consecutive settings differ in more than one factor, so we compare them descriptively.

IV. Evaluation Framework

A. Pre-Specified Analysis Plans and What Was Known

All analyses follow four pre-specified internal analysis plans, none of them externally registered, and Table S3 records what was known when each plan was written and every deviation from it. File modification times date each plan but cannot show that it was not edited later, and only part of each plan was prospective. The R4 plan was last written after the transfer and within-AgentDojo results were known, so its one-sided near-chance clause (Section IV-F) is post hoc. The ablation plan was prospective only for LR-A and for the R1 and R3 results of LR-B, and the GRU plan preceded every GRU result but followed all linear results. A revision plan was written after an independent pre-submission review had reported exploratory versions of most analyses, so the revision analyses are confirmatory re-runs of analyses whose approximate outcome was known, not prospective tests. The released monitor of Section V-F is the exception. Its plan was registered by hash before any monitor call, and its outcomes were prospective.

TABLE II
EVALUATION SETTINGS WITH TOOL-VOCABULARY COVERAGE AND MEASURED SHIFT

Setting	Training data	Test pool (n_-/n_+)	Tool coverage κ	Shift statistic (C2ST AUROC)	Shift introduced
R1	Stratified training split (8,775)	Stratified test split, 825/831	1.000	0.501 [0.485, 0.515]	None (reference)
R1-wd	Pooled corpus with no shared world, split seeds 0 to 4 (8,711 to 8,764)	786 to 811 / 863 to 924	1.000	0.553 to 0.564	World identity
R2	Task-disjoint training split (9,081)	35 unseen tasks, 729/775	1.000	0.551 [0.537, 0.565]	Task
R3	Four domains (8,446, or 8,840 with banking held out)	Held-out domain, 935/1,020 (banking 935/626)	0.000	0.622 (email) to 0.866 (web)	Domain and all its tools
R4	All pooled AgentDrift (10,656)	AgentDojo P0, 392/2,294 (P0F adds 4,987 negatives)	0.075	0.992 [0.991, 0.993]	Benchmark, ecosystem, generator, labels
Within, UT	Four fifths of P0, 5 repetitions	Held-out folds of P0	0.976	0.721 (0.616 to 0.844)	Unseen user tasks
Within, IT	As UT, grouped by injection goal	Held-out folds of P0	0.980	0.722 (0.631 to 0.799)	Unseen attack goals
Within, UT-F	As UT, unsuccessful attacks as training negatives	Held-out folds of P0	0.976	0.681 (0.589 to 0.785)	Training population
LOSO	Three suites of P0 (1,677 to 2,401)	Held-out suite, 56 to 128 / 182 to 904	0.000 (banking, Slack), 0.084 (travel), 0.362 (workspace)	0.887 to 0.971	Suite and most of its tools

The shift statistic is the cross-validated AUROC of a logistic regression on representation B that separates training from test rows (0.5 means indistinguishable). It ignores the tool-vocabulary shift. Ranges are over seeds, folds or suites, and within-AgentDojo rows give the mean and range over 25 cross-validation pairs. Table S2 gives every value.

B. Controlled Prevalence and Prevalence-Weighted AUPRC

Prevalence shift leaves the class-conditional score distributions unchanged, so precision at prevalence π follows from the true- and false-positive rates at a threshold,

$$\mathrm{Prec}(\theta;\pi) = \frac{\pi\,\mathrm{TPR}(\theta)}{\pi\,\mathrm{TPR}(\theta) + (1-\pi)\,\mathrm{FPR}(\theta)}. \tag{4}$$

Our primary AUPRC estimand is the prevalence-weighted average precision over the thresholds θ_k of the full test pool,

$$\mathrm{wAUPRC@}\pi = \sum_k \big(\mathrm{TPR}(\theta_k) - \mathrm{TPR}(\theta_{k-1})\big)\,\mathrm{Prec}(\theta_k;\pi). \tag{5}$$

It equals average precision with each positive weighted by $\pi/(1-\pi)\cdot n_-/n_+$, where n_- and n_+ are the numbers of negatives and positives. It is also the calibrated average precision of Siblini et al. [23]. We report π of 1%, 2%, 5% and 10%. Because (5) uses every positive at every prevalence, its intervals come from a bootstrap of the full pool. An earlier version of this study drew only as many positives per replicate as the target prevalence allows (8 to 92 in R1). That estimator targets the expected average precision of so small a test set and is biased upward at low prevalence, so we keep it only to show how variable a small test set is.

C. Threshold Selection

Operating points apply thresholds fixed without test data through (2). The main one,

$$\theta_{r80} = Q_{0.20}\big(\{f(\tau) \mid \tau \in \mathcal{D}_{\mathrm{train}},\ y(\tau)=1\}\big), \tag{6}$$

is the 20th percentile of in-sample training-positive scores, so it flags 80% of training positives. For the GRU we also report θ_{val80}, the same quantile on validation positives, and for every detector $\theta = 0.5$. In the within-AgentDojo cross-validation, thresholds are recomputed on each training fold.

D. Uncertainty, Clustering and Multiplicity

Intervals are 95% percentile intervals from 2,000 bootstrap replicates of the final test predictions [39], stratified by class for AgentDrift trajectories. Following the cluster bootstrap [40], AgentDojo intervals resample the 117 suite and user-task clusters (attack-goal clusters for the positives in IT), and R2 intervals resample its 35 test tasks. Where a detector has several fits, such as cross-validation repetitions or GRU training seeds, each replicate draws one fit. The GRU interval therefore describes a randomly drawn single-seed deployment, while its point estimate is the mean over seeds. No interval includes the variability of retraining on new data. Paired differences resample the same rows for both detectors, and differences between settings use independent bootstraps.

We call a difference distinguishable from zero when its 95% interval excludes zero, and an AUROC distinguishable from chance when its interval excludes 0.5. We also apply the Holm correction [41] within pre-defined families. These are the five R3 folds, the comparisons at one prevalence, and the representations of one R4 configuration. A result whose decision depends on the resampling unit, on the correction or on Monte Carlo error is called borderline. Each table draws its numbers from one analysis run, so intervals for the same quantity can differ in the third decimal between tables.

E. Domain-Matched Comparator, Threshold Failure and Reversal

An R3 AUROC ranks trajectories within one domain, whereas the pooled R1 AUROC also ranks across domains. We therefore compare each R3 fold with R1dm, which is the R1 model scored on the R1 test pool restricted to the held-out domain. Threshold failure is a change in recall or FPR at θ_{r80} of (6), relative to R1dm, that is distinguishable from zero. A reversal in a fold has two requirements after Holm correction within the five folds. One detector must hold an in-distribution advantage over another that is distinguishable from zero in R1dm, and the R3 paired difference must be

distinguishable from zero with the opposite sign. AUROC is the primary metric and wAUPRC@10% the secondary one.

F. Cross-Benchmark Decision Rule

The plan for R4 calls within-AgentDojo detection reasonable when AUROC is at least 0.80 with a lower 95% bound of at least 0.70, and wAUPRC@10% is at least 0.30 with a lower bound above 0.10. Transfer is near chance when its AUROC interval includes 0.5 or its AUROC is below 0.60. The second clause is one-sided and post hoc (Section IV-A), and under it an interval lying entirely below 0.5 also counts as near chance. We rename the plan's three cases to avoid a clash with representations A, B and C. Transfer failure (Case A) requires reasonable within-AgentDojo detection under UT on P0 and C1, near-chance transfer and a within-minus-transfer difference distinguishable from zero. Weak representation (Case B) applies when within detection under UT is not reasonable for P0 or C1 and transfer is also poor, and population-dependent (Case C) when within detection is reasonable only in the controlled populations. Mixed or other outcomes are reported as observed, including UT reasonable but IT not reasonable. We apply the rule literally, per representation, over the P0, C1, model-balanced and within-model populations. The operational target is compromise, and treating every injection attempt as positive is only an exploratory target.

V. Detectors and Representations

A. Representations

Representation A counts how often each tool of $\mathcal{V}$ is called, so a trajectory with $\kappa = 0$ maps to the zero vector. Representation B has ten features that do not name tools. They are the trajectory length m, the number of distinct tools, tool diversity (distinct tools divided by m), the number of immediate repeats, the count and fraction of calls whose tool name contains one of fourteen write-like keywords (such as send, delete, transfer or share), the maximum and mean number of arguments per call, and the maximum and mean observation length in characters. One non-blind annotator marked which AgentDrift tools change state. Against that annotation the keyword flag has a precision of 0.957 and a recall of 0.440, and annotation-based features change LR-B's AUROC by at most 0.024 (Table S4). Representation C concatenates A and B.

B. Logistic Regression and Unsupervised References

Logistic regression on representations A, B and C gives LR-A, LR-B and LR-C. It uses L2 regularization with inverse strength 1, the lbfgs solver, class weights inversely proportional to class frequency and features standardized with training statistics. Two unsupervised references are fitted on benign training trajectories. The first is a first-order Markov model of tool transitions with add-0.5 smoothing, which scores the mean negative log-probability of transitions and is constant when every tool is unseen. The second is an Isolation Forest [42] with 200 trees on representation C.

C. Content-Aware Reference

The TF-IDF detector is a class-weighted logistic regression on TF-IDF features [43] of the concatenated text of every step, which includes the tool name, argument keys and values and the observation text. It uses lowercase word unigrams and bigrams with a minimum document frequency of 2, at most 50,000 features and sublinear term frequency. Sensitivity variants remove all digits, mask the strings of the trajectory's world record, or keep only the words of tool names.

D. Ablation and Dropout Variants

LR-B-noobs and LR-C-noobs drop the two observation-length features. They were chosen after exploratory results were known and are post hoc. LR-B-minus drops the two keyword features. LR-C-drop is trained on the training set together with a copy whose tool block is zeroed, which is equivalent to dropping the tool block with probability 0.5.

E. GRU Sequence Detector

The GRU [44] is a learned analogue of LR-C with more capacity, used as a robustness check. Each tool call becomes a token (0 for padding, 1 for the unknown token `<UNK>` and one index per tool of the benign training trajectories). Tokens are embedded in 32 dimensions and passed to a one-layer GRU with 64 hidden units. Its final hidden state is concatenated with the ten standardized behavioral features and fed to a linear output layer, giving about 22k parameters. Sequences keep at most their last 16 calls. Training minimizes class-weighted binary cross-entropy with Adam [45] (learning rate 10^{-3}, coupled weight decay 10^{-4}), with batch size 64, gradient clipping at norm 1.0 and at most 30 epochs. Early stopping uses the AUPRC of a stratified 10% validation hold-out (patience 5). We use training seeds 0, 1 and 2 and no pretrained language model. In AgentDrift `<UNK>` never occurs in training, so its embedding receives no gradient from the loss and stays an untrained random vector. Within AgentDojo it occurs in up to 334 steps of 327 compromised training traces per fold. Removing those traces changes the within-AgentDojo AUROC by less than 0.01. GRU-drop0.2, with the pre-specified $p = 0.2$, and GRU-drop0.1 replace each training token by `<UNK>` independently with probability p.

F. Released Action Monitor

AlignmentCheck is the action monitor of LlamaFirewall [13]. It shows an LLM the user's request and the agent's actions so far and asks whether the latest action pursues a goal that the user did not set. We use its released prompt and output schema verbatim and scan each step against the trace before it, as the released scanner does, so a trace of m steps costs m calls. Each step is rendered as the released demonstration renders an action, with the tool name, the arguments and an empty thought. No tool observation is shown, so the monitor never reads the injected text. The AgentDojo user request is taken from the recorded run files. An output that cannot be parsed takes the released default, which flags the step, and the native decision flags a trace when any step is flagged.

We run the prompt with two backbones. MON-L is Llama 3.3 70B Instruct [46] in a 4-bit quantization [47], served with vLLM [48] at temperature 0 with output constrained to the schema. Its graded step score is the probability of `true` at the conclusion token, and its trace score is the maximum over steps. MON-A is Claude Sonnet 5.5 (`claude-sonnet-5-5`), queried through a batch interface at its lowest thinking setting and with default sampling, because the interface rejects a temperature for this model. Its coarse trace score is the number of flagged steps. The monitor is not trained, so a setting changes only its test rows and the training positives from which θ_{r80} is read, and the within-minus-transfer contrast of Section IV-F does not apply to it. MON-L runs in every setting, MON-A on the R1 test set and on all AgentDojo traces. Two sensitivity arms of MON-L add the agent's own text as the thought (MON-L-th) or replace masked tool names by `unknown_tool` (MON-L-mask). None of this is LlamaFirewall as deployed. There is no PromptGuard layer, the scan runs after the fact instead of blocking actions, and neither backbone is Meta's default. AgentDrift was generated by a Llama 3.3 70B model, so MON-L may be familiar with its traces, and the AgentDojo results carry the claims. The plan was registered by hash before any monitor call, and Section S14 lists its five hashed amendments, each made before any monitor result was analysed.

VI. Experimental Results

A. Attack Prevalence and the Negative Mix (RQ1)

Detection quality that looks good by AUROC degrades quickly as attacks become rarer (Table III and Fig. 2). LR-C has an AUROC of 0.879, yet at 2% prevalence its wAUPRC is 0.232 and its precision at θ_{r80} is 0.076, with 12.2 false alerts for every true one. The GRU ranks better but reaches a wAUPRC of only 0.382 at 2%, and its test recall at θ_{r80} falls short of the 0.80 target. At θ_{val80} its recall is 0.775 [0.745, 0.804] at a precision of 0.179 [0.147, 0.225] (Table S5). At a fixed test recall of 0.8, its FPR is 0.082 against 0.210 for LR-C, although this comparison is descriptive because it uses test labels.

Using all positives settles comparisons that subsampling left open. The subsampled LR-C value at 2% is 0.259 [0.105, 0.466], an interval 2.46 times wider than the weighted one. After Holm correction within each prevalence, 3, 5, 5 and 5 of five paired comparisons are distinguishable from zero at 1%, 2%, 5% and 10%, against 1, 1, 4 and 5 with subsampling. LR-C exceeds the Markov model at every prevalence (+0.132 [+0.093, +0.235] at 1%). Two comparisons at 2% are borderline. These are LR-C − LR-B (+0.081 [+0.003, +0.168], Holm p = 0.044, not distinguishable under world-cluster resampling) and GRU − LR-C (+0.150 [+0.023, +0.296], Holm p = 0.040), and neither is distinguishable at 1%.

The negative mix matters as much as prevalence (Table S6). At θ_{r80}, LR-C flags 0.118 of benign trajectories, 0.387 of hard negatives and 0.218 of failed attacks. Its precision at 2% is therefore 0.117 [0.097, 0.145] against benign negatives only, 0.076 with the pool-E mix of 27% hard negatives, and 0.039 [0.034, 0.047] against hard negatives only. LR-A, LR-B, LR-C, the Markov model and the GRU all flag hard negatives more often than benign trajectories. LR-C recalls 0.830 of full, 0.727 of partial and 0.642 of delayed hijacks. Its AUROC advantage over LR-B is distinguishable on full hijacks (+0.092 [+0.075, +0.109]) but not on partial or delayed ones.

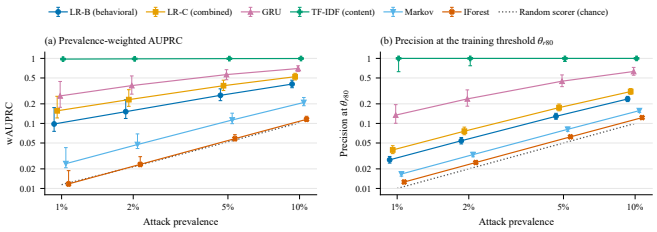


Fig. 2. Detection quality in R1 as attack prevalence falls. (a) Prevalence-weighted AUPRC. (b) Precision at the training threshold θ_{r80}, with the bar of TF-IDF extending to its implied lower bound. Both axes are logarithmic, bars are 95% bootstrap intervals, and the dotted line is chance (the random-scorer mean in (a) and the prevalence in (b)).

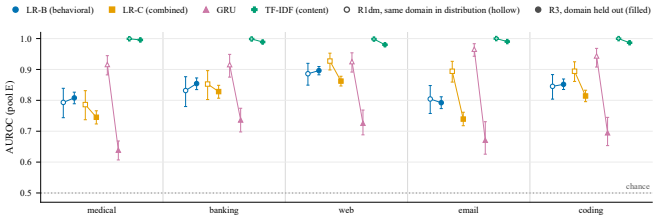


Fig. 3. AUROC of LR-B, LR-C, the GRU and TF-IDF on each held-out domain. Hollow markers give the R1 model on the same domain (R1dm) and filled markers the model trained without it (R3). Bars are 95% intervals, and the dotted line is chance.

Below the floor set by our pool size, (4) gives LR-C a precision at θ_{r80} of 0.0040 at 0.1% prevalence and 0.00040 at 0.01%. A threshold that flags at most 1% of training negatives catches 0.276 [0.244, 0.307] of R1 attacks at a test FPR of 0.017, whereas the GRU and TF-IDF exceed that budget on test data (FPR 0.070 and 0.033).

B. World and Task Reuse (RQ2)

R1 reuses generation worlds. For 99.3% of its test trajectories the world also occurs in training, and the training-world attack rate predicts the test label with an AUROC of 0.963. Nested, regularized probes, however, do not identify worlds beyond what the domain gives (representation A 0.096 against a domain-only floor of 0.097, Table S7). The world-disjoint control shows little inflation (Table S8). Over five split seeds, no decrease in pool-E AUROC is distinguishable from zero in any of 25 detector and seed pairs, and LR-A gains in every seed (+0.039 to +0.045). LR-B is lower than in R1 in all five seeds (−0.027 to −0.008). Its largest loss is distinguishable only when the resampling accounts for trajectories shared between the test sets, and not after Holm correction over seeds, so we treat it as borderline. The split also raises the email share of pool E from 0.208 to between 0.553 and 0.621.

On unseen tasks with known tools, no change is distinguishable in either direction. With task-cluster resampling, R2 − R1 AUROC is +0.023 [−0.005, +0.049] for LR-C and −0.020 [−0.054, +0.008] for the GRU, and LR-C's wAUPRC@10%

TABLE III
DETECTION QUALITY IN R1 AT CONTROLLED ATTACK PREVALENCE, WITH PREVALENCE-WEIGHTED AUPRC AND PRECISION AT THE TRAINING THRESHOLD θ_{r80}

		wAUPRC at attack prevalence				Precision at θ_{r80}				Alerts
Detector	AUROC	1%	2%	5%	10%	2%	10%	Recall	FPR	per 1,000 (2%)
LR-A (tool identity)	0.777 [0.755, 0.799]	0.089 [0.069, 0.164]	0.140 [0.112, 0.214]	0.255 [0.213, 0.327]	0.382 [0.334, 0.445]	0.043 [0.040, 0.048]	0.198 [0.183, 0.216]	0.744	0.335	343
LR-B (behavioral)	0.830 [0.810, 0.849]	0.098 [0.076, 0.175]	0.152 [0.119, 0.231]	0.271 [0.224, 0.340]	0.403 [0.354, 0.462]	0.054 [0.048, 0.061]	0.238 [0.217, 0.260]	0.795	0.284	294
LR-C (combined)	0.879 [0.862, 0.895]	0.156 [0.121, 0.261]	0.232 [0.183, 0.330]	0.380 [0.321, 0.462]	0.523 [0.466, 0.590]	0.076 [0.067, 0.088]	0.309 [0.281, 0.344]	0.771	0.192	203
GRU	0.936 [0.922, 0.947]	0.265 [0.177, 0.442]	0.382 [0.282, 0.537]	0.564 [0.470, 0.671]	0.699 [0.628, 0.770]	0.238 [0.184, 0.328]	0.629 [0.551, 0.727]	0.682	0.045	58
TF-IDF (content)	0.999 [0.999, 1.000]	0.975 [0.960, 0.991]	0.984 [0.973, 0.994]	0.992 [0.986, 0.997]	0.996 [0.992, 0.998]	1.000 ($\geq$ 0.771)	1.000 ($\geq$ 0.948)	0.763	0.000 ($\leq$ 0.004)	15
Markov	0.724 [0.700, 0.748]	0.024 [0.021, 0.042]	0.047 [0.041, 0.069]	0.113 [0.099, 0.144]	0.210 [0.187, 0.250]	0.033 [0.031, 0.036]	0.157 [0.147, 0.169]	0.810	0.482	489
IForest	0.588 [0.560, 0.615]	0.012 [0.011, 0.019]	0.023 [0.022, 0.031]	0.058 [0.054, 0.067]	0.115 [0.107, 0.129]	0.025 [0.024, 0.027]	0.123 [0.116, 0.130]	0.785	0.623	626
Random scorer	0.500	0.011 [0.009, 0.016]	0.022 [0.019, 0.027]	0.052 [0.047, 0.059]	0.102 [0.093, 0.114]	0.020	0.100	n/a	n/a	n/a
MON-L at θ_{r80}	0.856 [0.837, 0.874]	0.048 [0.040, 0.067]	0.089 [0.075, 0.117]	0.198 [0.171, 0.243]	0.338 [0.298, 0.395]	0.083 [0.072, 0.096]	0.330 [0.298, 0.366]	0.769	0.173	185
MON-L, native decision	as above					0.073 [0.064, 0.082]	0.299 [0.273, 0.328]	0.847	0.221	233
MON-A, native decision	0.817 [0.796, 0.837] ‡	0.038 [0.032, 0.060] ‡	0.072 [0.062, 0.100] ‡	0.164 [0.143, 0.202] ‡	0.287 [0.256, 0.333] ‡	0.054 [0.049, 0.060]	0.237 [0.219, 0.257]	0.874	0.313	324

Pool E of R1 has 831 attacked and 825 negative trajectories (600 benign and 225 hard negatives). Precision and alerts per 1,000 trajectories follow (4) from recall and FPR at θ_{r80} under the pool-E negative mix. Table S6 gives their intervals by negative class. TF-IDF flags no pool-E negative, so its FPR is the exact 95% upper bound and its precision the lower bound implied by it. The random-scorer row gives the mean and the 2.5th to 97.5th percentiles over 2,000 random scorers. The monitor rows are the AlignmentCheck prompt with the Llama 3.3 70B (MON-L) and Claude Sonnet 5.5 (MON-A) backbones (Section V-F), at θ_{r80} and at the released native decision, on the same bootstrap replicates. ‡ MON-A ranks traces by the number of flagged steps, so its AUROC and wAUPRC are coarse.

TABLE IV
AUROC UNDER DOMAIN SHIFT (R3) AGAINST THE DOMAIN-MATCHED R1 VALUE (R1DM), WITH PAIRED DIFFERENCES AND THE PRE-SPECIFIED REVERSAL RULE

	LR-B (behavioral)		LR-C (combined)		GRU		TF-IDF	LR-C − LR-B		GRU − LR-C	
Held-out domain	R1dm	R3	R1dm	R3	R1dm	R3	(content)	R1dm	R3	R1dm	R3
R1 pooled	0.830		0.879		0.936		0.999	$+0.049^{*}$ [+0.034, +0.064]		$+0.057^{*}$ [+0.042, +0.072]	
medical	0.793	0.808	0.786	0.745	0.916	$0.639^{\downarrow}$	0.996	-0.007 [−0.035, +0.023]	-0.063^{*} [−0.074, −0.053]	$+0.130^{*}$ [+0.089, +0.173]	$-0.107^{*\dagger}$ [−0.130, −0.087]
banking	0.832	0.854	0.853	0.828	0.915	$0.736^{\downarrow}$	0.989	$+0.021$ [−0.014, +0.053]	-0.026^{*} [−0.038, −0.014]	$+0.063^{*}$ [+0.017, +0.111]	$-0.092^{*\dagger}$ [−0.133, −0.061]
web	0.886	0.896	0.927	$0.862^{\downarrow}$	0.925	$0.726^{\downarrow}$	0.980	$+0.041^{*}$ [+0.014, +0.069]	$-0.034^{*\dagger}$ [−0.046, −0.023]	-0.002 [−0.030, +0.023]	-0.137^{*} [−0.168, −0.100]
email	0.804	0.792	0.894	$0.739^{\downarrow}$	0.966	$0.671^{\downarrow}$	0.990	$+0.090^{*}$ [+0.046, +0.136]	$-0.053^{*\dagger}$ [−0.063, −0.043]	$+0.071^{*}$ [+0.042, +0.103]	$-0.069^{*\dagger}$ [−0.106, −0.009]
coding	0.845	0.852	0.894	$0.814^{\downarrow}$	0.944	$0.695^{\downarrow}$	0.987	$+0.049^{*}$ [+0.019, +0.081]	$-0.038^{*\dagger}$ [−0.048, −0.026]	$+0.049^{*}$ [+0.009, +0.086]	$-0.119^{*\dagger}$ [−0.155, −0.078]

R1dm is the R1 model scored on the R1 test pool of the held-out domain (111 to 180 positives, 165 negatives), and R3 is the model trained without that domain (626 to 1,020 positives, 935 negatives). Single AUROCs are points, with intervals in Fig. 3. * marks a difference distinguishable from zero after Holm correction over five domains, ↓ an R3 decrease against R1dm distinguishable after Holm correction, and † a reversal by the rule of Section IV-E. The R1dm and R3 pools share 345 trajectories per domain (276 in banking), so the independent intervals of R3 − R1dm are conservative. Comparisons across folds that involve banking are confounded by its hijack mix (Section III-A).

gain of +0.092 [+0.002, +0.174] is borderline. Unlike MCP-Shield's tool-name metadata, which falls from 0.898 to 0.640 under task-disjoint splits, LR-A scores 0.788 in R2 against 0.777 in R1. The reason is that all tasks of an AgentDrift domain share one tool set, so a task-disjoint split keeps full coverage and cannot remove task-specific tool signatures. R2 also reuses worlds as often as R1, and its 35 test tasks give wide intervals.

C. Domain Shift, Ranking and Thresholds (RQ3)

Holding out a domain lowers the AUROC of LR-C and the GRU but leaves LR-B unchanged (Table IV and Fig. 3). Against its own domain's R1 score, the AUROC decrease of LR-C is distinguishable after Holm correction in web, email and coding (−0.065 to −0.155) but not in medical or banking. The pooled comparator would have pointed to medical, banking, email and coding instead and would have suggested that web keeps its AUROC, which it does not. On wAUPRC@10% the decrease is distinguishable in all five folds. LR-B changes in no fold, whereas the GRU falls in every fold.

The in-distribution orderings do not survive. After Holm correction, LR-C ranks below LR-B in all five held-out domains, and the GRU ranks below LR-C in all five for each of its training seeds. In R1dm, the in-distribution advantage of LR-C over LR-B is distinguishable only in web, email and coding, and that of the GRU over LR-C in every domain except

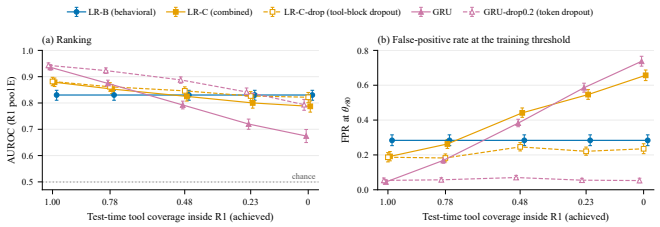


Fig. 4. Removing tool identity at test time inside R1. (a) AUROC and (b) FPR at θ_{r80} against the achieved tool coverage. Dashed lines with hollow markers are the dropout variants, and bars are 95% intervals conditional on the drawn masks and seeds.

web. The rule of Section IV-E therefore finds a C-versus-B reversal in three domains and a GRU-versus-C reversal in four. The GRU plan did not anticipate the GRU reversal, so we treat it as exploratory. World-cluster resampling gives the same folds, and a joint bootstrap over trajectories shared by the R1dm and R3 pools changes no decision. Standardizing the hijack-pattern composition lowers banking's R3 AUROC by 0.037 to 0.051, moves no other fold by more than 0.008 and leaves LR-C below LR-B in every fold. The shift statistic does not track these losses, since email has the smallest behavioral shift in Table II but the largest LR-C loss.

Thresholds fail in a detector-specific way (Table S9 and Fig. S1). LR-C over-alerts in every fold. Its FPR at θ_{r80} rises from between 0.121 and 0.321 in R1dm to between 0.503 and 0.722 in R3, and its recall from between 0.613 and 0.906 to between 0.828 and 0.990. At $\theta = 0.5$ its R3 FPR is 0.538 to 0.733. LR-A, the Isolation Forest, GRU-drop0.2 and TF-IDF under-alert in every fold. The scores of LR-A and the Markov model are constant at $\kappa = 0$ (AUROC exactly 0.500), and every trajectory exceeds the Markov threshold. The GRU with pooled seeds over-alerts in three folds after Holm correction and in four before it, with an R3 FPR of 0.438 to 0.764 against at most 0.071 in R1dm. Each seed's own threshold fails in every fold, and the seeds disagree (banking recall 0.903, 0.006 and 0.957). LR-B's threshold carries over in four of five domains. It fails only in medical, where recall falls from 0.694 to 0.560 (change −0.135 [−0.208, −0.059]), and its FPR rise in email is not distinguishable from zero (+0.061 [−0.018, +0.138]).

A threshold set on 200 unlabeled held-out trajectories to reproduce the R1 alert rate partly restores LR-C's operating point (Table S10). Under the pre-specified reading, which uses a sample at native prevalence, its fold-mean FPR falls to 0.217 to 0.312 with recall of 0.609 to 0.770. A post hoc reading with traffic at 2% prevalence gives an FPR of 0.179 to 0.221. The remedy fails for the GRU, whose recall under the post hoc reading collapses to 0.123 to 0.273.

D. Mechanism of the Domain-Shift Failures (RQ3)

Three measurements bear on these results (Table V), and a fourth, with a detector that has no untrained input, tests how far they reach. The first is a pair of offsets. At $\kappa = 0$ the tool block of LR-C is an all-zero raw vector, which the training scaler maps to a constant c. Writing the logit as $g(\tau) = w_A^\top z_A(\tau) + w_B^\top z_B(\tau) + b$ with standardized blocks z_A and z_B, every held-out trajectory receives

$$g(\tau) = w_B^\top z_B(\tau) + \big(w_A^\top c + b\big). \tag{7}$$

The offset $w_A^\top c$ in (7) is +1.653 to +1.899 logits, a tool contribution that only 10.7% to 15.0% of training trajectories reach. Because it is constant within a fold, it cannot change the ranking. It does, however, push every held-out trajectory toward the attack side, so LR-C's recall and FPR rise together. The GRU shows the same effect more strongly. Adam's coupled weight decay alone shrinks the norm of its untrained <UNK> embedding from 5.250 to 6.477 at initialization to 0.095 to 2.332, well below the median trained tool-embedding norms of 4.146 to 4.824. Through the tool pathway, held-out negatives then receive a mean logit 6.523 higher than training negatives (positive in 49 of 50 fold and seed cells), against 0.476 in R1. Dropout trains the missing inputs and shrinks both offsets. These are measurements of one component of each logit. They show that the untrained inputs add an offset in these two detectors. They do not show that an offset of this size needs an untrained input. The fourth measurement shows that it does not.

The second measurement removes tool identity at test time inside R1, with no change of domain, and follows both orderings across the range of coverage rather than at its two ends (Fig. 4 and Table S11). Masking a random share of the 103 training tool names gives achieved coverages of 1.000, 0.775, 0.483, 0.225 and 0.000. LR-C falls from 0.879 through 0.853 and 0.825 to 0.787 [0.765, 0.808], and its FPR at θ_{r80} rises from 0.192 through 0.263 and 0.442 to 0.657, a change of +0.466 [+0.433, +0.497]. The GRU falls further, from 0.936 to 0.674, and its FPR rises from 0.045 to 0.737. LR-B holds at 0.830 throughout, since no tool feature enters it. Both reversals are therefore graded rather than a cliff at zero. LR-C − LR-B is +0.049 [+0.034, +0.064] at full coverage, −0.005 [−0.016, +0.006] at 0.483 and −0.043 [−0.055, −0.031] at zero, and GRU − LR-C follows the same path from +0.057 [+0.043, +0.071] to −0.113 [−0.129, −0.096]. For both pairs the crossing lies between coverages of 0.775 and 0.483, so a deployment in which about half the tools are new is already enough to remove the in-distribution advantage, well before the disjoint-vocabulary case of R3. The cross-benchmark setting of Section VI-E sits near the lower end of this range at $\kappa = 0.075$. Dropout keeps FPR nearly flat but does not remove the ranking loss.

The third measurement shows that dropout during training largely removes the R3 reversals at no in-distribution cost. GRU-drop0.2 meets the reversal rule against LR-C in no fold and lies above LR-C in banking. At $p = 0.1$ the reversal survives in medical only. LR-C-drop − LR-B still meets the rule in web, email and coding, but the two scores are almost rank-identical (Spearman correlation of at least 0.997), so we read it as a near tie. Dropout changes the direction of threshold failure rather than removing it, since GRU-drop0.2 under-alerts. GRU − LR-C stays below zero in every fold, whether the interval is the mixture interval of Section IV-D, a two-level bootstrap over seeds and trajectories or a seed-averaged interval. In email, however, the per-seed differences

TABLE V
OUT-OF-VOCABULARY MECHANISM AND DROPOUT REMEDIES, WITH TOOL-PATHWAY OFFSETS, DROPOUT AUROC AND PAIRED DIFFERENCES IN R1 AND EACH R3 FOLD

	Tool-block offset $w_A^\top c$		GRU tool-pathway offset		Tool-block dropout			Token dropout ($p = 0.2$)		
Setting	LR-C	LR-C-drop	GRU	GRU-drop0.2	LR-C-drop AUROC	LR-C-drop − LR-B	LR-C-drop − LR-C	GRU-drop0.2 AUROC	GRU-drop0.2 − LR-C	GRU-drop0.2 − GRU
R1	n/a	n/a	+0.476	+0.243	0.882	$+0.051^{*}$ [+0.036, +0.067]	$+0.003^{*}$ [+0.001, +0.005]	0.943	$+0.064^{*}$ [+0.050, +0.079]	+0.007 [−0.002, +0.016]
R3 medical	+1.794	−0.057	+5.862	+2.420	0.797	-0.011^{*} [−0.013, −0.009]	$+0.051^{*}$ [+0.042, +0.061]	0.747	+0.001 [−0.017, +0.019]	$+0.108^{*}$ [+0.081, +0.139]
R3 banking	+1.899	−0.057	+5.679	+2.260	0.853	−0.001 [−0.003, +0.001]	$+0.025^{*}$ [+0.015, +0.035]	0.848	$+0.019^{*}$ [+0.004, +0.033]	$+0.112^{*}$ [+0.076, +0.147]
R3 web	+1.702	−0.033	+7.376	+2.280	0.894	$-0.002^{*\dagger}$ [−0.004, −0.001]	$+0.032^{*}$ [+0.022, +0.043]	0.870	+0.008 [−0.010, +0.021]	$+0.144^{*}$ [+0.102, +0.181]
R3 email	+1.790	−0.021	+6.826	+1.774	0.777	$-0.015^{*\dagger}$ [−0.017, −0.013]	$+0.038^{*}$ [+0.029, +0.047]	0.766	+0.026 [−0.007, +0.055]	$+0.095^{*}$ [+0.020, +0.150]
R3 coding	+1.653	−0.032	+6.872	+2.159	0.847	$-0.005^{*\dagger}$ [−0.007, −0.003]	$+0.033^{*}$ [+0.022, +0.042]	0.811	−0.003 [−0.036, +0.019]	$+0.116^{*}$ [+0.054, +0.163]

The GRU offset is the mean tool-pathway contribution on held-out negatives minus that on training negatives, averaged over ten seeds for the GRU and three seeds for GRU-drop0.2. The contrast GRU-drop0.2 − GRU pairs seeds 0 to 2. AUROC values and offsets are points. * marks a paired difference distinguishable from zero after Holm correction over five folds (without correction in the R1 row, which gives the in-distribution cost of each remedy), and † a reversal by the rule of Section IV-E that persists after the remedy.

TABLE VI
CROSS-BENCHMARK TRANSFER TO AGENTDOJO ON THE CORRECTED EXTRACTION, WITHIN-AGENTDOJO CONTROLS AND THE PRE-SPECIFIED OUTCOME PER REPRESENTATION

	Transfer	Within AgentDojo, UT		Within, IT	Transfer,	Within UT	
Detector	AUROC	AUROC	− transfer	AUROC	unsuccessful as negatives	wAUPRC 10%	Pre-specified outcome
LR-A (tool identity)	0.566 [0.515, 0.615]	0.788 [0.740, 0.828]	$+0.223^{*}$ [+0.159, +0.285]	0.690 [0.545, 0.823]	0.585 [0.536, 0.635]	0.303 [0.228, 0.488]	Mixed (weak representation and population-dependent, reasonable on C1 only)
LR-B (behavioral)	0.482 [0.436, 0.521]	0.741 [0.688, 0.792]	$+0.263^{*}$ [+0.198, +0.323]	0.719 [0.642, 0.794]	0.626 [0.575, 0.669]	0.228 [0.183, 0.314]	Weak representation (no population reaches the bar)
LR-C (combined)	0.473 [0.427, 0.512]	0.797 [0.747, 0.842]	$+0.327^{*}$ [+0.268, +0.382]	0.688 [0.547, 0.823]	0.626 [0.574, 0.670]	0.298 [0.228, 0.453]	Mixed (weak representation and population-dependent, reasonable on C1 only)
GRU	0.513 [0.464, 0.551]	0.840 [0.784, 0.884]	$+0.329^{*}$ [+0.264, +0.388]	0.645 [0.499, 0.783]	0.645 [0.593, 0.689]	0.411 [0.317, 0.607]	Transfer failure under UT, but within not reasonable under IT
TF-IDF (content)	0.520 [0.473, 0.565]	0.996 [0.992, 0.999]	$+0.477^{*}$ [+0.430, +0.523]	0.987 [0.974, 0.996]	0.626 [0.578, 0.666]	0.980 [0.959, 0.996]	Transfer failure under UT, also reasonable under IT
SE (reimplementation)	0.481 [0.399, 0.560]	0.956 [0.916, 0.984]	$+0.478^{*}$ [+0.389, +0.563]	not run	0.572 [0.490, 0.657]	0.727 [0.501, 0.934]	Transfer failure under UT, IT not run
MON-L (Llama 3.3 70B w4a16)	0.930 [0.909, 0.951]	untrained, as transfer	n/a	n/a	0.916 [0.897, 0.932]	0.751 [0.679, 0.854]	M1, transfers
MON-A (Claude Sonnet 5.5)	0.934 [0.915, 0.952]	untrained, as transfer	n/a	n/a	0.918 [0.899, 0.936]	0.811 [0.755, 0.872]	M1, transfers

Transfer models are trained on AgentDrift, and within models by cross-validation grouped by user task (UT) or attack goal (IT). Intervals resample suite and user-task clusters (attack-goal clusters for IT), and the paired gap uses the same clusters for both scores and is centered on the bootstrap mean. * marks a gap distinguishable from zero after Holm correction over the six representations. The outcome applies the rule of Section IV-F over the P0, C1, model-balanced and within-model populations. SE is our reimplementation of the DriftNet architecture. Its within-AgentDojo scores average ten repetitions, against five for the other rows, and it was not fitted under IT. The monitor rows are untrained, so the within-AgentDojo columns do not apply to them, and their outcome is that of Section VI-G.

range from −0.096 to −0.025 (Table S12, which also gives the earlier version's post hoc linear head on B trained with the GRU procedure).

The fourth measurement repeats the ladder with a detector that has no untrained input. SE is our reimplementation of the DriftNet architecture [7], whose code and weights have not been released, so none of its numbers is DriftNet's own. It embeds each step with a frozen sentence encoder and passes the sequence to a small transformer trained on AgentDrift (Table S19). A held-out tool name is then an ordinary point in a space the encoder already covers. The analysis plan was fixed before any embedding or fit existed. It predicted no reversal and a small offset. The first prediction held and the second did not. On the full detection logit, held-out negatives sit 5.161, 5.795 and 7.420 logits above training negatives in medical, coding and web, averaged over ten seeds. In banking and email the offset is 0.524 and 0.692, close to SE's in-distribution value of 0.643. The large offsets match the GRU's tool-pathway offsets of 5.679 to 7.376 in Table V. The full-logit estimator differs from the one in Table V. Applied post hoc to the GRU, it gives 5.6 to 7.4 in every fold and 0.661 in R1, within 0.51 of the GRU's tool-pathway values, so the two detectors sit on one scale. The training thresholds fail with the offsets. At θ_{r80} SE's FPR rises from 0.001 in R1 to 0.046, 0.049 and 0.106 in medical, coding and web, while in banking and email recall falls to 0.345 and 0.620. The ranking

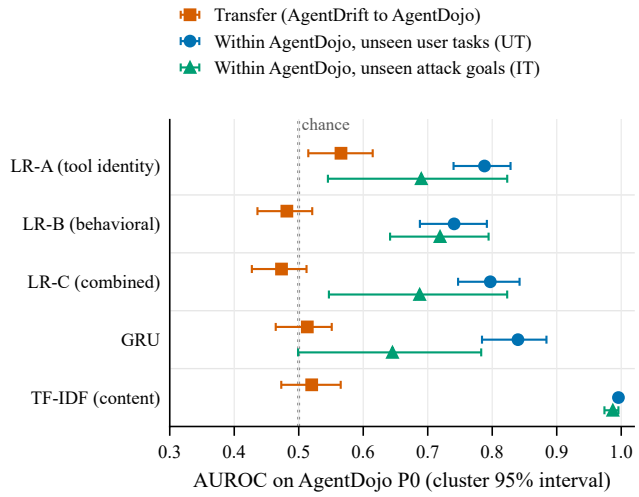


Fig. 5. AUROC on AgentDojo P0 for transfer from AgentDrift and for the within-AgentDojo controls on unseen user tasks (UT) and unseen attack goals (IT). Bars are 95% cluster-bootstrap intervals, and the dotted line is chance.

behaves as the first three measurements predict. SE stays above LR-B in every fold and meets the reversal rule in none. The margin is distinguishable from zero after Holm correction in four folds, but not in web (+0.008 [−0.056, +0.043]). Masking tool names inside R1 barely moves SE, from 0.998 at full coverage to 0.994 at zero, where LR-C and the GRU lose 0.092 and 0.262. That sweep is a weak shift for SE, though. Masking moves the R1 test steps to a mean cosine distance of 0.126 from their nearest training step, against 0.26 to 0.41 in R3. SE also loses 0.040 to 0.093 AUROC from domain-matched R1 to R3, a loss comparable to LR-C's. Our reading is narrow. Untrained inputs fit the ranking reversals and the graded loss with coverage, and SE agrees with that account without proving it, because SE differs from LR-C in more than its inputs. They are one source of the threshold offsets. A domain shift in a trained input space produces offsets of the same size, so recalibration on target data is needed whatever the representation. The plan's bar for a small offset was +0.5, which lies below SE's own R1 value. The decision therefore rests on the three large folds, not on banking and email.

What remains of the behavioral signal at zero coverage is largely an observation-length cue (Table S13). The mean observation length carries the largest standardized weight in LR-B (−1.861 [−1.965, −1.770]) and in LR-C's behavioral block. On its own it ranks attacked trajectories below benign ones in every AgentDrift population. Even benign-labeled steps of attacked trajectories are short (median 59 characters, and 73 in hard negatives, against 98 in benign trajectories), which points to the generator. Without the two observation features, LR-B loses 0.112 to 0.221 AUROC in every R3 fold, while the two features alone come within 0.053 of full LR-B. LR-C-noobs still ranks below LR-B-noobs in all five folds on wAUPRC@10% and in four on AUROC. The attacker writes this cue. Appending characters to the injected observation of each attacked R1 trajectory lowers LR-C's recall at θ_{r80} from 0.771 to 0.634 with 100 characters and to 0.033 with 1,200, and the GRU's from 0.682 to 0.000. SE reads the observation through its embedding rather than as a length, and the same padding lowers its recall from 0.796 to 0.531 and its AUROC from 0.998 to 0.883 (Table S19). We tested only the effect on scores, not whether padded injections would still succeed. Within AgentDojo the cue points the other way (mean observation AUROC 0.558), and padding slightly raises LR-C's recall.

On recorded traces the C-versus-B reversal does not replicate consistently (Table S14). In AgentDojo LOSO no suite meets the reversal rule, because the in-distribution C − B advantage is not distinguishable from zero in any suite (pooled +0.056 [−0.003, +0.111]). In the two zero-coverage suites the held-out sign differs (banking −0.059 [−0.172, +0.034], Slack +0.044 [+0.016, +0.076]). Rule 4 of the ablation plan asks whether the combined model degrades more than the behavioral one from R1 to R4. It is met (+0.057 [+0.042, +0.073]), but this is a weak test because both transfer near chance.

E. Cross-Benchmark Transfer (RQ4)

No detector trained on AgentDrift transfers to AgentDojo (Table VI and Fig. 5). On P0 the transfer intervals of LR-B, LR-C and the GRU include 0.5, and LR-A (0.5655) counts as near chance only through the post hoc one-sided clause. Transfer wAUPRC@10% lies near the chance mean of 0.103 for the first three and slightly above it for LR-A (0.138 [0.121, 0.178]). At its training threshold, transferred LR-C flags 0.224 of benign and 0.225 of compromised traces. Composition does not explain the failure (Table S15). With positives reweighted to the benign model mix at native counts, transfer moves by at most a few hundredths (LR-C 0.486 [0.435, 0.530]). LR-A − LR-C stays distinguishable from zero (+0.094 [+0.020, +0.168] on P0), although LR-A's balanced transfer is only 0.555. LR-C transfers at 0.476 on the important_instructions template, the only one run for every model, and at 0.309 to 0.666 across the 15 templates of `gpt-4o-2024-05-13`. Without the observation features, transfer rises to 0.630 for LR-B-noobs and 0.657 for LR-C-noobs. This is consistent with a reversed and out-of-range cue (54.1% of P0 traces have a longer maximum observation than any AgentDrift trace), but these variants are post hoc. SE, our reimplementation of the DriftNet architecture, also fails to transfer (0.481 [0.399, 0.560]) while reaching 0.956 [0.916, 0.984] within AgentDojo, and zeroing its four world features changes neither number detectably (Table S19).

The within-target control succeeds for some representations only. Under UT the within-minus-transfer gap is distinguishable after Holm correction for every representation, but among the content-blind detectors only the GRU clears the reasonable bar on P0. Under IT these detectors reach 0.645 to 0.719, and the gap remains distinguishable only for LR-B, with LR-C borderline (+0.220 [+0.038, +0.394], Holm p = 0.051). The last column of Table VI gives the outcomes. LR-A and LR-C are mixed because the conditions for weak representation and for population-dependent both hold. LR-C misses the P0 bar by 0.003 AUROC and 0.002 wAUPRC@10% (0.797 and

0.2985, shown as 0.298). It meets the bar on C1 (0.832) and reaches the 0.80 point criterion without the 101 DoS positives (0.813 [0.763, 0.859]), so its outcome is borderline. Behavioral detectors rarely catch the DoS positives (within-UT LR-C recall of 0.176 against 0.745 on the others), and removing them leaves transfer unchanged.

The population matters in a second way. With the unsuccessful attacks as additional evaluation negatives (P0F), transfer rises to 0.626 for LR-C and 0.645 for the GRU and is no longer near chance, because the AgentDrift-trained models score unsuccessful attacks below benign runs. SE rises less, to 0.572 [0.490, 0.657], an interval that still includes 0.5. Trained within AgentDojo with unsuccessful attacks as negatives (UT-F), LR-C reaches 0.819 and the GRU 0.869 on P0F. This does not count as transfer failure in the rule's sense, because transfer is no longer near chance. With every injection attempt as positive, an exploratory target, the content-blind representations score only 0.584 to 0.613 within AgentDojo. The LR-C transfer interval widens from [0.441, 0.508] under trajectory resampling to [0.329, 0.599] with attack-goal clusters (Table S16). A regularized probe does not identify the generating model from B or C (B 0.488 [0.402, 0.578]). The stronger signal reported in the earlier version came from the extraction error.

Among the outcomes of the GRU plan, Outcome 1 (strong R1 performance, a decrease in at least three R3 folds and near-chance transfer on P0 and under model balancing) is indeterminate at the boundary. Its balanced interval is [0.501, 0.569] under the pre-specified stratified bootstrap and [0.487, 0.575] under the cluster bootstrap. Outcome 2 (a substantial improvement under shift) is not met, and Outcome 3 (near-chance transfer with strong R1 performance) is met. Outcome 4 (large seed variance) is triggered on the operating point, but under criteria defined after the fact (R3 banking FPR of 0.619, 0.000 and 0.694 across seeds).

F. Content-Aware Reference (RQ5)

The TF-IDF reference changes several conclusions (Table S17). In distribution it is near perfect (R1 AUROC 0.999, no false positive at θ_{r80}) and exceeds LR-C in all seven in-distribution settings, including the five world-disjoint seeds. Under R3 its AUROC stays between 0.980 and 0.996, above LR-C in every fold after Holm correction, with no reversal and no rise in FPR. The difficulty at low prevalence, the FPR explosion and the ranking reversal therefore belong to the content-blind detectors rather than to the benchmark. Threshold non-transfer holds for content too, in the recall direction (R3 recall 0.272 to 0.696 against 0.685 to 0.817 in R1dm). TF-IDF recalls 0.733 to 0.988 of full hijacks but only 0.038 to 0.345 of partial and 0.006 to 0.225 of delayed ones. The words of tool names alone score 0.520 to 0.663 in R3, below LR-B in every fold.

AgentDrift is nearly separable by surface content, because every trajectory category carries its own generator template. The token "2026", which has the strongest negative weight, occurs in 0.742 of benign and 0.848 of failed-attack training trajectories against none of the hard negatives and 0.046 of attacked ones. The hard-negative and failed-attack templates are just as specific. Yet none of the ten single-token rules selected on training data exceeds a test AUROC of 0.780. Masking world strings, anonymizing names or removing digits changes the R1 AUROC by at most 0.001, so the separation is spread over many template tokens.

Content does not rescue transfer. TF-IDF transfers with an AUROC of 0.520 [0.471, 0.564] and flags no trace at its training threshold. Its advantage over LR-C there (+0.047 [−0.002, +0.096], p = 0.057 under cluster resampling) is borderline. Within AgentDojo it is again near perfect, reaching 0.987 even on unseen attack goals and 0.989 to 0.998 in LOSO. It largely detects the presence of an injection, however. Five fixed attack strings occur in 0.996 of compromised and 0.947 of unsuccessful-attack traces and in no benign trace, and used as a fixed rule they give an AUROC of 0.998. With unsuccessful attacks as training negatives TF-IDF falls to 0.941 [0.911, 0.966], although it still ranks compromised traces above unsuccessful attacks (AUROC 0.808).

G. A Released Action Monitor (RQ1 and RQ4)

A released monitor that is trained on neither benchmark does transfer. On P0 MON-L reaches an AUROC of 0.930 [0.909, 0.951], and 0.940 [0.928, 0.952] with positives reweighted to the benign model mix, and it exceeds every AgentDrift-trained detector on the same rows (differences +0.410 [+0.359, +0.468] to +0.457 [+0.410, +0.509]). The failure of Section VI-E is therefore a property of detectors trained on one benchmark, not of the task. On the AgentDrift R1 test set, which is new to the prompt, it reaches 0.856 [0.838, 0.874], against 0.879 for LR-C trained on that benchmark (difference −0.023 [−0.043, −0.002]). Under the held-out domains of R3 it reaches 0.829 to 0.925 and ranks above LR-C and the GRU in every fold, although below TF-IDF, so an untrained monitor is also spared the domain-shift failures of Section VI-C. Meta evaluated the prompt on AgentDojo, so the AgentDojo result may be optimistic for the prompt (Section VIII).

Its native decision is not usable at low prevalence (RQ1). It flags 0.900 of compromised and 0.173 of benign P0 traces and 0.847 and 0.221 of the corresponding R1 test traces, so its implied precision at 2% prevalence is 0.096 [0.071, 0.141] on P0 and 0.073 [0.065, 0.082] on R1, below 0.5 in both. The released decision rule, like a training threshold, would mostly raise false alerts at a 2% attack rate.

The released decision keeps a similar FPR on the two benchmarks (R1 negatives minus P0 benign traces +0.047 [−0.029, +0.117]), but a threshold read from training data does not port, even for an untrained detector. On P0 the threshold read from AgentDojo training folds instead of AgentDrift training positives changes recall by −0.075 [−0.103, −0.049] and FPR by −0.064 [−0.105, −0.033], so the threshold claim of Section VI-C extends to a detector without untrained inputs.

Judging actions does separate attempted from successful hijacks better than reading content. MON-L flags 0.212 [0.145, 0.287] of the 4,987 unsuccessful attacks, against 0.308 [0.238, 0.375] for TF-IDF trained within AgentDojo, a difference of

−0.096 [−0.186, −0.006], an interval that only just excludes zero. MON-A flags 0.063 [0.047, 0.083] of them (difference −0.245 [−0.316, −0.176]). Relative to its recall on P0 the monitor flags unsuccessful attacks at a ratio of 0.235, against 0.395 for TF-IDF. This comparison was added after registration because the registered comparator, TF-IDF at its transfer threshold, flags no AgentDojo trace.

The commercial backbone changes one reading. MON-A reaches an AUROC of 0.934 [0.915, 0.952] on P0 and 0.817 [0.796, 0.837] on R1 from its coarse step count. Its native decision flags 0.883 of compromised and 0.038 of benign P0 traces, so its implied precision at 2% is 0.320 [0.203, 0.591] on P0, an interval that reaches above 0.5, but it flags 0.313 of R1 negatives and reaches only 0.054 [0.049, 0.060] there. The pre-specified reading that the native decision is unusable at 2% therefore holds for MON-L only, and whether it is usable depends on the backbone and the benchmark. MON-A's 174 refusals on P0 take the default flag, and excluding them lowers its P0 FPR to 0.028. The two backbones agree on trace flags with Cohen's κ of 0.69. MON-L remains the reference because it is reproducible. The advantage of MON-L over MON-A on R1 minus that on P0 is +0.043 [+0.019, +0.067], which suggests that familiarity with its own generator helps MON-L on AgentDrift, although the backbones also differ in capability.

Masking every tool name in R1 moves the AUROC of MON-L by −0.091 [−0.109, −0.074], more than 0.05, so the monitor relies on what tool names mean, although with every name masked it still reaches 0.765 [0.743, 0.788]. Adding the agent's own text as the thought gives 0.944 [0.924, 0.961] on P0, and every MON-L call returned a valid output. Table S20 and Table S21 give every setting and arm.

VII. Discussion

A. Summary of Principal Findings

This study asked how evaluation choices change the picture of tool-call detectors for hijacked LLM agents. On attack prevalence (RQ1), a high AUROC did not imply a usable operating point once attacks were rare, and precision depended as much on the mix of negatives as on prevalence. Reuse of generation worlds or tasks (RQ2) inflated the standard estimate very little. Under a domain shift with no shared tools (RQ3), the in-distribution ranking of the content-blind detectors reversed, and thresholds failed in a way that differed from detector to detector. Across benchmarks (RQ4), no trained detector transferred from AgentDrift to AgentDojo, and the within-target control supported a failure of transfer for some representations only. The content reference (RQ5) was near perfect within each benchmark but did not transfer either. A released action monitor, trained on neither benchmark, did transfer, so the failure belongs to detectors trained on one benchmark. Its native decision would mostly raise false alerts at 2% prevalence. These findings concern our own baselines and one content reference, and several of them rest on borderline decisions (Section IV-D).

B. Interpretation and Relation to Prior Work

Prevalence and the negative class. For the content-blind detectors, a high AUROC coexisted with alert streams that low prevalence would fill with false alarms, so AUROC describes them far too favorably. This is the base-rate effect that Axelsson [14] described for intrusion detection, and it follows from the prevalence dependence of precision [15], [16]. The headline precision also moved threefold with the share of hard negatives, an effect that prevalence alone does not capture. Prevalence-weighted average precision [23] makes AUPRC comparable across prevalences, but a precision figure means little unless the negative mix is stated.

Tool coverage and unseen inputs. The rankings of LR-C over LR-B and of the GRU over LR-C failed on held-out AgentDrift domains mainly because of inputs never seen in training. The domain still costs something on its own, since SE, which has no untrained input, lost 0.040 to 0.093 AUROC without reversing. Its thresholds also failed in three held-out domains, with offsets as large as the GRU's, so the offsets have a wider cause than unseen tools. MCPShield [9] showed that tool-name metadata features inflate estimates under random splits. Our results add that the same features can harm a combined model when no test tool was seen in training, because unseen tools reach untrained inputs (Section VI-D). This is a concrete and correctable form of the inversion of in-distribution and out-of-distribution rankings discussed by Teney et al. [34], since dropout during training largely removed it. The recorded-trace replication showed no consistent reversal, so the size and sign of the effect probably depend on how disjoint the vocabularies are and on what the behavioral weights learned.

Attacker-written cues and template content. The behavioral signal that survived zero coverage rested largely on observation length. An indirect injector writes this quantity, and its association with the label reversed between benchmarks, so it offers little assurance under the threat model of Section III-C. Content features instead separate generator templates on AgentDrift and mostly detect the presence of an injection within AgentDojo. Both cases fit the pattern of shortcut learning [35], in which a model relies on a cue that separates the classes in one dataset but does not define the behavior of interest.

We identified these cues by changing them rather than by inferring them from weights, and the four interventions sit in different sections, so we gather them here. Deleting the two observation-length features cost LR-B between 0.112 and 0.221 AUROC in every held-out domain, while those two features alone came within 0.053 of the full model (Section VI-D). Appending characters to the injected observation, which an attacker controls, lowered LR-C recall at θ_{r80} from 0.771 to 0.033 at 1,200 characters and the GRU to 0.000, and the same padding slightly raised recall inside AgentDojo, where the cue points the other way (Section VI-D). SE was hurt less, with recall falling from 0.796 to 0.531 at 1,200 characters. Masking the world strings of a trajectory, anonymizing names or removing digits moved the content reference by at most 0.001 AUROC, and no single-token rule selected on training data exceeded a test AUROC of 0.780, so its separation is spread across many template tokens rather than carried by a few (Section VI-F). Removing the observation features instead raised cross-benchmark transfer, to 0.630 for LR-B and 0.657

for LR-C, although those variants are post hoc (Section VI-E). A detector can therefore pass a disjoint-vocabulary split while resting on a quantity the attacker writes, which is an argument for reporting such interventions next to the headline numbers.

Reading a failed transfer. The within-target control remains necessary to interpret near-chance transfer. In our study, however, it supports a failure of transfer only for the GRU and the content reference and points to weak features for LR-B. For LR-C the outcome turns on small margins, on the population and on how DoS runs are labeled, so a transfer claim needs the target population stated. Agent-Sentry [6] applies a similar control to benchmarks that share agents, tools and environments, whereas our two benchmarks differ at once in label semantics, attack construction, task sources, trace origin and tool vocabulary. Our evidence therefore cannot say which difference causes the failure. For prompt classifiers, Fomin [22] found that pooled cross-validation overstates leave-one-dataset-out AUC by 8.0 to 16.5 points. Our within-minus-transfer gaps of +0.223 to +0.477 AUROC are larger, but they combine several shifts that Fomin's setting keeps apart.

C. Implications for Evaluation Practice

We draw seven recommendations for evaluating post hoc detectors of agent compromise.

1) Report prevalence-weighted average precision at stated prevalences next to AUROC, with intervals from the full test pool.
2) Report FPR by negative class and recall by attack pattern, and state the negative mix behind any precision figure.
3) Report tool-vocabulary coverage for every split.
4) Train tool-indexed detectors with dropout on tool inputs when test tools may be unseen, and report threshold behavior per detector, since thresholds can fail in either direction.
5) Include a content baseline, even a simple one, to show how far a benchmark is separable by surface content.
6) Pair every cross-benchmark claim with a within-target control, report its outcome per representation and per population, and state how failed attempts are labeled.
7) Without target labels, recalibrate thresholds on unlabeled target traffic. This remedy is partial, since it restores an alert budget but not ranking quality, and it assumes that the prevalence of the calibration sample is known.

D. Implications for Benchmark Design

The content reference shows that both benchmarks can be solved largely from surface text. On AgentDrift each trajectory category carries its own generator template, and within AgentDojo five fixed attack strings separate compromised from benign traces almost perfectly (Section VI-F). A detector evaluated on either benchmark may therefore score well by recognizing how the data were generated rather than how a hijacked agent behaves, and this caveat also applies to published detectors evaluated on these benchmarks. Benchmarks for trace detectors would be more informative if hard negatives shared the templates of attacked traces, if attack text were varied rather than drawn from a few fixed strings, if generation prompts and scripts were released, and if tool overlap across splits were documented.

E. Implications for Deployment

For a security team that audits agent sessions after the fact, the results suggest caution in three respects. First, a detector validated on one benchmark should not be assumed to work on the team's own agents and tools, and its threshold should be checked on local traffic before its alerts are trusted. Second, label-free recalibration can bring the alert rate back to a budget, but it does not restore ranking quality and failed for the GRU (Section VI-C). Third, content features gave the strongest scores within each benchmark, yet on these benchmarks they mostly recognized generator templates and fixed injection strings. An adaptive attacker could change such cues, a case that we did not test and that only Praetor [5] among the audited works evaluates.

VIII. Limitations and Threats to Validity

Internal validity. Beyond the confounds of Sections III-B and III-E, the GRU was rerun at ten seeds for Table V and the SE comparison, and seeds 0 to 2 reproduce the three-seed cache. Elsewhere the GRU uses three seeds, whose means differ from the ten-seed means by at most 0.015 AUROC. GRU-drop0.2 stays at three seeds. The full-logit GRU offsets of Section VI-D, the ten-seed tool-pathway offsets of Table V and the cluster intervals of the SE row of Table VI were computed after the SE values were seen. Several choices (the noobs variants, the post hoc label-free reading and the criteria for GRU Outcome 4) were made after related results were seen, and several decisions are borderline. The breakdown by negative class is missing for R1-wd, IT, UT-F and LOSO, and FPR budgets were evaluated in R1 only.

Construct validity. We score whole trajectories after the fact and evaluate neither step-level localization nor scoring before each call. The two benchmarks encode different notions of compromise, DoS runs are labeled by task failure, and the state-change features rest on a keyword list (Section V-A).

External validity. Our only frozen sentence encoder is a reimplementation of one published architecture, trained without the THOUGHT field. At R2 its F1 is 0.973 against a published 0.983, in a setting that is not like for like. We include one released monitor, the AlignmentCheck prompt, with two backbones of our choice, a 4-bit Llama 3.3 70B and Claude Sonnet 5.5, without its PromptGuard layer and run after the fact on recorded actions, so its results bound the released prompt and not LlamaFirewall as deployed (Section V-F). Its graded score is read from token probabilities that the released scanner does not use, its open backbone shares a model family with the AgentDrift generator, and Meta may have refined the prompt with AgentDojo in view. We include no released content guard such as PromptGuard 2, which a separate study evaluates, and no published trace detector as released. The GRU is not run in LOSO, and it has no dropout arm in R1-wd, R2 or within AgentDojo. AgentDrift is synthetic, comes from one generator, is nearly separable by surface content

and has fully disjoint tool sets across domains, so R3 is an extreme vocabulary shift. We study one benchmark pair in one direction with a small AgentDojo benign pool (392 traces) and no adaptive attacker, so the findings may not carry over to other detectors, agent ecosystems or deployment traffic.

IX. Reproducibility and Data Availability

A. Code and Data Availability

The code, the derived data and the result files are publicly available at https://github.com/Mahmoud-Abbasi-svg/agent-trace-detector-eval. We use AgentDrift release 1.0 at commit `014a514f` and the AgentDojo recorded runs at commit `089ed468`, with the environment variables `AGENTDRIFT` and `AGENTDOJO` pointing to local clones. All results were computed on CPUs. Apart from SE, they used CPython 3.10.4, numpy 2.2.6, scipy 1.15.3, scikit-learn 1.7.2, torch 2.11.0+cpu and matplotlib 3.10.8, and the lock file `revision/requirements-lock.txt` pins these packages with their runtime dependencies. SE ran on one node of a Slurm cluster with CPython 3.13.5, torch 2.11.0, transformers 5.14.1, sentence-transformers 6.0.1, scikit-learn 1.7.2 and numpy 2.2.6. Its environment and run record are in `revision/sencoder_run`, and a GPU was used only to check that GPU and CPU embeddings agree. MON-L ran on one NVIDIA H100 80GB HBM3 with CPython 3.13.5, vLLM 0.29.0, torch 2.13.0+cu130 and transformers 5.18.0, and MON-A through the provider's batch interface. Every monitor call, its raw output and its score are released in `revision/monitor_run`. Section S11 of the supplement gives the remaining model settings, every seed and the execution order, and scripts generate all tables and figures from the result files.

B. Ethics and Data Statement

Both benchmarks are public, AgentDrift under CC BY 4.0 and AgentDojo under the MIT license. AgentDrift is synthetic, and its documentation notes that some generated company names coincide with real organizations, although no content describes any real organization. We add no personal data and will redistribute the extracted AgentDojo traces with the extraction code under the MIT license and with attribution. Our observation-length finding concerns simple baselines and a quantity that indirect injectors already control. We report it so that evaluations test for it, and we release no attack artifact beyond the padding procedure described here.

C. Declaration of AI Use

The authors used a generative AI assistant during the preparation of this paper. The authors reviewed all content and take full responsibility for it.

X. Conclusion

We evaluated our own tool-call detectors for hijacked LLM agents and a content reference at controlled attack prevalence and along world, task, domain and benchmark shifts. At low prevalence AUROC overstated operational usefulness, and precision depended as much on the negative mix as on prevalence. Holding out worlds or tasks, by contrast, cost little. Under a disjoint-vocabulary domain shift, the ranking of the content-blind detectors changed because unseen tools reached untrained inputs, and dropout during training largely remedied this. A reimplemented frozen-encoder detector kept its ranking but not its thresholds, so threshold failure under domain shift does not need an untrained input. Threshold failure proved detector-specific, and the surviving behavioral signal was mostly an attacker-written length cue. No trained detector transferred from AgentDrift to AgentDojo, and the within-target control supported a failure of transfer for some representations only. A released action monitor trained on neither benchmark did transfer. Both benchmarks were also nearly separable by surface content, so strong detection scores on them should be read with care. Finding which benchmark difference causes that failure will require benchmark pairs that vary one factor at a time.

References

[1] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz, "Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection," in *Proc. 16th ACM Workshop Artif. Intell. Secur. (AISec)*, Copenhagen, Denmark, 2023, pp. 79–90, doi: 10.1145/3605764.3623985.

[2] Q. Zhan, Z. Liang, Z. Ying, and D. Kang, "InjecAgent: Benchmarking indirect prompt injections in tool-integrated large language model agents," in *Findings Assoc. Comput. Linguistics: ACL 2024*, Bangkok, Thailand, 2024, pp. 10471–10506, doi: 10.18653/v1/2024.findings-acl.624.

[3] E. Debenedetti, J. Zhang, M. Balunović, L. Beurer-Kellner, M. Fischer, and F. Tramèr, "AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), Datasets Benchmarks Track*, vol. 37, Vancouver, BC, Canada, 2024, pp. 82895–82920, doi: 10.52202/079017-2636.

[4] J. Liu et al., "TraceAegis: Securing LLM-based agents via hierarchical and behavioral anomaly detection," arXiv:2510.11203v1, Oct. 2025.

[5] H. Dang, "Enforcing benign trajectories: A behavioral firewall for structured-workflow AI agents," arXiv:2604.26274v1, Apr. 2026.

[6] R. Sequeira, S. Damianakis, U. Iqbal, and K. Psounis, "Agent-Sentry: Bounding LLM agents via execution provenance," arXiv:2603.22868v2, May 2026.

[7] A. Pinjari and M. P. Saint-Germain, "DriftNet: A dual-head trajectory transformer for detecting and localizing prompt injection in LLM agents," arXiv:2609.10892v1, Sep. 2026.

[8] L. Advani, "Trajectory Guard: A lightweight, sequence-aware model for real-time anomaly detection in agentic AI," arXiv:2601.00516v1, Jan. 2026.

[9] S. Zavrak, "Content-aware attack detection in LLM agent tool-call traffic: An empirical study of features, architectures, and evaluation protocols," arXiv:2605.11053v3, May 2026.

[10] C. Zhang et al., "Skynet: Workflow-level anomaly detection for agentic AI via semantic and structural modeling," arXiv:2609.06835v2, Sep. 2026.

[11] Y. Liu et al., "TrajAD: Trajectory anomaly detection for trustworthy LLM agents," arXiv:2602.06443v1, Feb. 2026.

[12] X. He, D. Wu, Y. Zhai, and K. Sun, "SentinelAgent: Graph-based anomaly detection in multi-agent systems," arXiv:2505.24201v1, May 2025.

[13] S. Chennabasappa et al., "LlamaFirewall: An open source guardrail system for building secure AI agents," arXiv:2505.03574v1, May 2025.

[14] S. Axelsson, "The base-rate fallacy and the difficulty of intrusion detection," *ACM Trans. Inf. Syst. Secur.*, vol. 3, no. 3, pp. 186–205, Aug. 2000, doi: 10.1145/357830.357849.

[15] J. Davis and M. Goadrich, "The relationship between precision-recall and ROC curves," in *Proc. 23rd Int. Conf. Mach. Learn. (ICML)*, Pittsburgh, PA, USA, 2006, pp. 233–240, doi: 10.1145/1143844.1143874.

[16] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLoS ONE*, vol. 10, no. 3, Mar. 2015, Art. no. e0118432, doi: 10.1371/journal.pone.0118432.

[17] M. B. A. McDermott, H. Zhang, L. H. Hansen, G. Angelotti, and J. Gallifant, "A closer look at AUROC and AUPRC under class imbalance," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 37, Vancouver, BC, Canada, 2024, pp. 44102–44163, doi: 10.52202/079017-1400.

[18] R. Sommer and V. Paxson, "Outside the closed world: On using machine learning for network intrusion detection," in *Proc. IEEE Symp. Secur. Privacy (SP)*, Oakland, CA, USA, 2010, pp. 305–316, doi: 10.1109/SP.2010.25.

[19] F. Pendlebury, F. Pierazzi, R. Jordaney, J. Kinder, and L. Cavallaro, "TESSERACT: Eliminating experimental bias in malware classification across space and time," in *Proc. 28th USENIX Secur. Symp. (USENIX Security)*, Santa Clara, CA, USA, Aug. 2019, pp. 729–746.

[20] D. Arp et al., "Dos and don'ts of machine learning in computer security," in *Proc. 31st USENIX Secur. Symp. (USENIX Security)*, Boston, MA, USA, Aug. 2022, pp. 3971–3988.

[21] H. Li, R. Wen, S. Shi, N. Zhang, Y. Vorobeychik, and C. Xiao, "AgentDyn: Are your agent security defenses deployable in real-world dynamic environments?" arXiv:2602.03117, 2026.

[22] M. Fomin, "When benchmarks lie: Evaluating malicious prompt classifiers under true distribution shift," arXiv:2602.14161, 2026.

[23] W. Siblini, J. Fréry, L. He-Guelton, F. Oblé, and Y.-Q. Wang, "Master your metrics with calibration," in *Advances in Intelligent Data Analysis XVIII (IDA 2020)*, ser. Lecture Notes in Computer Science, vol. 12080, Konstanz, Germany, 2020, pp. 457–469, doi: 10.1007/978-3-030-44584-3_36.

[24] K. Zhu, X. Yang, J. Wang, W. Guo, and W. Y. Wang, "MELON: Provable defense against indirect prompt injection attacks in AI agents," in *Proc. 42nd Int. Conf. Mach. Learn. (ICML)*, ser. PMLR, vol. 267, 2025, pp. 80310–80329.

[25] Y.-S. Chen, S.-Y. Huang, C.-L. Yang, and Y.-N. Chen, "TraceSafe: A systematic assessment of LLM guardrails on multi-step tool-calling trajectories," in *Proc. 3rd Conf. Lang. Model. (COLM)*, San Francisco, CA, USA, 2026, arXiv:2604.07223.

[26] Y. Li et al., "ATBench: A diverse and realistic agent trajectory benchmark for safety evaluation and diagnosis," in *Proc. 3rd Conf. Lang. Model. (COLM)*, San Francisco, CA, USA, 2026, arXiv:2604.02022.

[27] D. Liu et al., "AgentDoG: A diagnostic guardrail framework for AI agent safety and security," arXiv:2601.18491, 2026.

[28] Z. Zheng et al., "StepGuard: Learning step-level guardrails with scalable supervision and safety-utility balancing," arXiv:2608.24777, 2026.

[29] A. Pinjari and M. P. Saint-Germain, "AgentDrift: A step-labeled benchmark of injection-hijacked LLM agent trajectories," arXiv:2609.06972v1, Sep. 2026.

[30] Z. Zhang et al., "Agent-SafetyBench: Evaluating the safety of LLM agents," arXiv:2412.14470, 2024.

[31] Y. Fu, X. Yuan, and D. Wang, "RAS-Eval: A comprehensive benchmark for security evaluation of LLM agents in real-world environments," arXiv:2506.15253, 2025.

[32] H. Zhang et al., "Agent Security Bench (ASB): Formalizing and benchmarking attacks and defenses in LLM-based agents," in *Proc. 13th Int. Conf. Learn. Represent. (ICLR)*, Singapore, 2025.

[33] J. Miller et al., "Accuracy on the line: On the strong correlation between out-of-distribution and in-distribution generalization," in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, ser. PMLR, vol. 139, 2021, pp. 7721–7735.

[34] D. Teney, Y. Lin, S. J. Oh, and E. Abbasnejad, "ID and OOD performance are sometimes inversely correlated on real-world datasets," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 36, 2023, pp. 71703–71722, doi: 10.52202/075280-3139.

[35] R. Geirhos et al., "Shortcut learning in deep neural networks," *Nature Mach. Intell.*, vol. 2, no. 11, pp. 665–673, Nov. 2020, doi: 10.1038/s42256-020-00257-z.

[36] D. Lopez-Paz and M. Oquab, "Revisiting classifier two-sample tests," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2017. [Online]. Available: https://arxiv.org/abs/1610.06545

[37] K. Boyd, K. H. Eng, and C. D. Page, "Area under the precision-recall curve: Point estimates and confidence intervals," in *Proc. Eur. Conf. Mach. Learn. Knowl. Discovery Databases (ECML PKDD)*, ser. Lecture Notes in Computer Science, vol. 8190, 2013, pp. 451–466, doi: 10.1007/978-3-642-40994-3_29.

[38] F. Barbero, F. Pendlebury, F. Pierazzi, and L. Cavallaro, "Transcending TRANSCEND: Revisiting malware classification in the presence of concept drift," in *Proc. IEEE Symp. Secur. Privacy (SP)*, San Francisco, CA, USA, 2022, pp. 805–823, doi: 10.1109/SP46214.2022.9833659.

[39] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman & Hall, 1993.

[40] C. A. Field and A. H. Welsh, "Bootstrapping clustered data," *J. R. Stat. Soc. B, Stat. Methodol.*, vol. 69, no. 3, pp. 369–390, Jun. 2007, doi: 10.1111/j.1467-9868.2007.00593.x.

[41] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Statist.*, vol. 6, no. 2, pp. 65–70, 1979.

[42] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. 8th IEEE Int. Conf. Data Mining (ICDM)*, Pisa, Italy, 2008, pp. 413–422, doi: 10.1109/ICDM.2008.17.

[43] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," *Inf. Process. Manage.*, vol. 24, no. 5, pp. 513–523, 1988, doi: 10.1016/0306-4573(88)90021-0.

[44] K. Cho et al., "Learning phrase representations using RNN encoder-decoder for statistical machine translation," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, Doha, Qatar, Oct. 2014, pp. 1724–1734, doi: 10.3115/v1/D14-1179.

[45] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Represent. (ICLR)*, San Diego, CA, USA, 2015. [Online]. Available: https://arxiv.org/abs/1412.6980

[46] A. Grattafiori et al., "The Llama 3 herd of models," arXiv:2407.21783, Jul. 2024.

[47] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh, "GPTQ: Accurate post-training quantization for generative pre-trained transformers," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023.

[48] W. Kwon et al., "Efficient memory management for large language model serving with PagedAttention," in *Proc. 29th Symp. Operating Syst. Principles (SOSP)*, 2023, pp. 611–626.

Supplementary Material for Evaluating Baseline Detectors of Hijacked LLM Agents Under Rare Attacks and Distribution Shift

This document supplements the main paper. It gives the detail behind every supplementary table and figure that the main text cites, the complete list of deviations from the pre-specified analysis plans, and the instructions needed to reproduce every number. Section, table, figure and equation numbers without the prefix S refer to the main paper. Every number in the tables below is read by script from the result files of the revision. The prose repeats values from those tables or from the stream summaries named in Section S11.

S1. Conventions

The supplement uses the notation of the main paper. The within-AgentDojo control has three designs. UT groups the cross-validation folds by suite and user task, IT groups the compromised traces by suite and injection goal, and UT-F is UT with the unsuccessful attacks added as training negatives. P0F is the evaluation population that adds the 4,987 retained unsuccessful attacks to P0 as negatives. The result files store the three designs under the keys WUT, WIT and WUT_F, and the tables print the names used in the paper. The three outcomes of the cross-benchmark rule are called transfer failure, weak representation and population-dependent, which correspond to Cases A, B and C of the R4 plan (Section IV-F).

Values are point estimates followed by 95% percentile intervals from 2,000 bootstrap replicates [39]. Unless a note says otherwise, AgentDrift intervals resample trajectories stratified by class. AgentDojo intervals resample the 117 suite and user-task clusters [40], or the attack-goal clusters for the positives of IT, and R2 intervals resample its 35 test tasks. A difference is distinguishable from zero when its interval excludes zero, and an AUROC is distinguishable from chance when its interval excludes 0.5 (Section IV-D). The marks are the same in every table. An asterisk (*) marks a difference distinguishable from zero after Holm correction [41] within the family named in the note, and † marks a reversal by the rule of Section IV-E. The arrow $\downarrow$ marks an R3 decrease against R1dm that is distinguishable after Holm correction, and § marks a within-AgentDojo value that meets the reasonable bar of Section IV-F. Threshold failure is defined in Section IV-E and the cross-benchmark rule in Section IV-F.

Each table is drawn from one analysis run, so intervals for the same quantity can differ in the third decimal between tables. The resampling units are fixed test predictions, and no interval includes the variability of retraining on new data [49].

S2. AgentDojo Extraction and Measured Shift

The extractor of the earlier version paired each tool call with the tool message that carried the same tool-call identifier. Gemini runs store an empty identifier, so every step of a Gemini trace received the last tool message of the run. Some GPT runs reuse identifiers, so an earlier step received the observation of a later one. The two observation-length features of representation B were therefore wrong for a model-specific subset of traces. The corrected extractor pairs calls and observations by position, as the extractor's own documentation intended. The unit test `tests/test_agentdojo_alignment.py` checks the alignment. It confirms a known Gemini run, finds that positional and identifier-based pairing agree on 8,151 runs with usable identifiers, and confirms that the extracted file uses positional pairing. The test passed on 26 and 27 September 2026.

Compared by observation content, 413 of the 2,686 P0 traces change (188 from `gemini-2.0-flash-001`, 185 from `gemini-1.5-pro-002`, 37 from `gpt-4o-2024-05-13` and 3 from `gpt-4-turbo-2024-04-09`). Compared by per-step observation length, 408 change (186, 183, 36 and 3). A further 249 traces change only because the earlier extractor cut observations at 2,000 characters without saying so. The cut affected 448 of 14,079 P0 steps in 262 traces. In the corrected extraction, 500 P0 steps in 300 traces are longer than 2,000 characters. Observations are now stored uncut in both benchmarks. With the cut applied again, within-AgentDojo LR-C reaches 0.846 against 0.797 without it. No paired interval was computed for this point difference.

Table S1 compares the earlier and corrected values. LR-A does not read observations and is unchanged. The corrected extraction moves transfer toward 0.5 for LR-B, LR-C and the GRU and lowers their within-AgentDojo AUROC. Most of the within-AgentDojo drop comes from the removal of the cut. The version 0.3 scripts never produced the earlier per-model values of LR-A and the GRU or the earlier IT value of LR-A, so these were recomputed on the archived version 0.3 extraction with the version 0.3 code. The lower block lists the runs removed by the filter on fewer than two tool calls. It removes 236 of 628 benign runs but only 70 of 2,364 compromised runs, all of the latter from `gpt-4o-2024-05-13`. For `gemini-2.0-flash-001` it removes 77 of 132 benign runs.

Table S2 measures how far each setting moves the behavioral features. The statistic is a classifier two-sample test [36], the cross-validated AUROC of a class-weighted logistic regression on representation B that separates training rows from test rows. R1 is indistinguishable from its training data (0.501), and world-disjoint and task-disjoint splits move it slightly (0.551 to 0.564). The held-out domains move it by 0.622 (email) to 0.866 (web), and AgentDojo is almost perfectly separable from AgentDrift (0.992). The statistic does not track the degradation of the detectors. Email has the smallest behavioral shift of the five folds but the largest LR-C loss against R1dm (-0.155).

S3. Pre-Specified Plans and Deviations

The study followed four pre-specified internal analysis plans, none of which was externally registered. We call them the R4 plan (`R4_CONTROL.md`), the ablation plan (`PRE_ABLATION_DESIGN.md`), the

TABLE S1
AGENTDOJO EXTRACTION FIX WITH EARLIER AND CORRECTED VALUES, AND THE RUNS EXCLUDED BY THE SHORT-RUN FILTER

v0.3 → corrected	LR-A (tool identity)	LR-B (behavioral)	LR-C (combined)	GRU
Transfer AUROC	0.566 → 0.566 [0.543, 0.586]	0.447 → 0.482 [0.448, 0.516]	0.440 → 0.473 [0.441, 0.508]	0.488 → 0.513 [0.482, 0.544]
Within UT AUROC	0.788 → 0.788 [0.759, 0.816]	0.798 → 0.741 [0.708, 0.771]	0.855 → 0.797 [0.766, 0.829]	0.860 → 0.840 [0.794, 0.871]
Within IT AUROC	0.690 → 0.690 [0.616, 0.769]	0.771 → 0.719 [0.676, 0.758]	0.766 → 0.688 [0.634, 0.736]	not run → 0.645 [0.581, 0.691]
Within UT − transfer	+0.222 → +0.223 [+0.189, +0.258]	+0.351 → +0.259 [+0.214, +0.304]	+0.415 → +0.323 [+0.277, +0.370]	+0.371 → +0.326 [+0.276, +0.368]
Within IT − transfer	+0.124 → +0.124 [+0.048, +0.210]	+0.325 → +0.237 [+0.184, +0.288]	+0.326 → +0.214 [+0.151, +0.273]	not run → +0.132 [+0.062, +0.187]
Transfer FPR at θ_{r80}	0.043 → 0.043 [0.025, 0.064]	0.332 → 0.225 [0.184, 0.265]	0.347 → 0.225 [0.184, 0.265]	0.230 → 0.168 [0.115, 0.232]
Transfer, `gemini-1.5-pro-002`	0.572 → 0.572 [0.513, 0.637]	0.582 → 0.483 [0.395, 0.573]	0.593 → 0.477 [0.389, 0.566]	0.646 → 0.535 [0.449, 0.616]
Transfer, `gemini-2.0-flash-001`	0.525 → 0.525 [0.455, 0.596]	0.612 → 0.565 [0.461, 0.662]	0.640 → 0.562 [0.458, 0.662]	0.751 → 0.627 [0.530, 0.718]
Transfer, `gpt-4-turbo-2024-04-09`	0.540 → 0.540 [0.484, 0.591]	0.487 → 0.487 [0.406, 0.567]	0.485 → 0.485 [0.405, 0.564]	0.533 → 0.531 [0.456, 0.605]
Transfer, `gpt-4o-2024-05-13`	0.573 → 0.573 [0.530, 0.610]	0.456 → 0.454 [0.389, 0.515]	0.447 → 0.444 [0.380, 0.505]	0.488 → 0.482 [0.425, 0.541]
Short-run filter, excluded / all	**Benign**	**Compromised**	**Unsuccessful**	**Kept (b / c / u)**
`gpt-4o-2024-05-13`	32/124 (0.258)	70/1936 (0.036)	1216/4839 (0.251)	92 / 1866 / 3623
`claude-3-5-sonnet-20241022`	40/124 (0.323)	0/7 (0.000)	184/622 (0.296)	84 / 7 / 438
`gemini-1.5-pro-002`	45/124 (0.363)	0/107 (0.000)	185/522 (0.354)	79 / 107 / 337
`gpt-4-turbo-2024-04-09`	42/124 (0.339)	0/180 (0.000)	141/449 (0.314)	82 / 180 / 308
`gemini-2.0-flash-001`	77/132 (0.583)	0/134 (0.000)	534/815 (0.655)	55 / 134 / 281
All models	236/628 (0.376)	70/2364 (0.030)	2260/7247 (0.312)	392 / 2294 / 4987

Each cell of the upper block gives the version 0.3 value, then the corrected value with its interval under the trajectory-level bootstrap of the earlier version. These intervals therefore differ slightly from the cluster intervals of Table VI. The GRU was not run under IT in version 0.3. The lower block counts runs with fewer than two tool calls against all runs of each class, and the last column gives the retained benign, compromised and unsuccessful-attack runs.

TABLE S2
MEASURED SHIFT PER SETTING WITH A CLASSIFIER TWO-SAMPLE TEST ON REPRESENTATION B, AND TOOL-OCCURRENCE COVERAGE

Setting	n train / test	Tool coverage κ	C2ST AUROC all rows	C2ST AUROC pool E	LR-B AUROC	LR-C AUROC	LR-C R3 − R1dm
R1	8775 / 1881	1.000	0.501 [0.485, 0.515]	0.512 [0.497, 0.527]	0.830 [0.811, 0.850]	0.879 [0.863, 0.895]	n/a
R1-wd s0	8762 / 1894	1.000	0.553 [0.538, 0.566]	0.555 [0.540, 0.570]	0.821 [0.800, 0.839]	0.891 [0.875, 0.905]	n/a
R1-wd s1	8739 / 1917	1.000	0.564 [0.550, 0.578]	0.571 [0.557, 0.586]	0.803 [0.784, 0.824]	0.889 [0.874, 0.904]	n/a
R1-wd s2	8711 / 1945	1.000	0.555 [0.541, 0.570]	0.559 [0.544, 0.575]	0.822 [0.802, 0.840]	0.901 [0.886, 0.915]	n/a
R1-wd s3	8752 / 1904	1.000	0.555 [0.540, 0.569]	0.551 [0.536, 0.566]	0.818 [0.797, 0.837]	0.888 [0.872, 0.902]	n/a
R1-wd s4	8764 / 1892	1.000	0.558 [0.545, 0.573]	0.573 [0.558, 0.588]	0.819 [0.798, 0.838]	0.892 [0.875, 0.907]	n/a
R2	9081 / 1722	1.000	0.551 [0.537, 0.565]	0.554 [0.538, 0.569]	0.849 [0.829, 0.868]	0.902 [0.887, 0.917]	n/a
R3 medical	8446 / 2210	0.000	0.842 [0.834, 0.850]	0.845 [0.837, 0.853]	0.808 [0.789, 0.826]	0.745 [0.724, 0.766]	−0.041 [−0.091, +0.012]
R3 banking	8840 / 1816	0.000	0.701 [0.689, 0.713]	0.736 [0.723, 0.749]	0.854 [0.835, 0.873]	0.828 [0.807, 0.849]	−0.024 [−0.072, +0.029]
R3 web	8446 / 2210	0.000	0.866 [0.858, 0.873]	0.875 [0.868, 0.882]	0.896 [0.883, 0.909]	0.862 [0.847, 0.878]	−0.065 [−0.095, −0.033]
R3 email	8446 / 2210	0.000	0.622 [0.609, 0.635]	0.620 [0.606, 0.633]	0.792 [0.773, 0.811]	0.739 [0.718, 0.761]	−0.155 [−0.194, −0.113]
R3 coding	8446 / 2210	0.000	0.751 [0.740, 0.761]	0.754 [0.743, 0.766]	0.852 [0.835, 0.869]	0.814 [0.796, 0.833]	−0.080 [−0.117, −0.042]
R4	10656 / 7673	0.075	0.992 [0.991, 0.993]	0.992 [0.990, 0.994]	0.482	0.473	n/a
LOSO banking	1959 / 1450	0.000	0.887 [0.876, 0.898]	0.887 [0.875, 0.900]	n/a	n/a	n/a
LOSO slack	1677 / 1601	0.000	0.969 [0.964, 0.974]	0.977 [0.972, 0.981]	n/a	n/a	n/a
LOSO travel	2401 / 2083	0.084	0.891 [0.881, 0.900]	0.887 [0.865, 0.908]	n/a	n/a	n/a
LOSO workspace	2021 / 2539	0.362	0.971 [0.967, 0.974]	0.951 [0.943, 0.959]	n/a	n/a	n/a
Within UT (25 CV pairs)	2149 / 1535	0.976	0.721 (0.616 to 0.844)	n/a	n/a	n/a	n/a
Within UT-F (25 CV pairs)	6138 / 1535	0.976	0.681 (0.589 to 0.785)	n/a	n/a	n/a	n/a
Within IT (25 CV pairs)	2149 / 1535	0.980	0.722 (0.631 to 0.799)	n/a	n/a	n/a	n/a

The two-sample test standardizes representation B, fits a logistic regression with balanced class weights under five-fold stratified cross-validation (seed 0) and reports the AUROC of the out-of-fold scores with a stratified bootstrap interval. A value of 0.5 means that training and test rows are indistinguishable in the ten behavioral features. The pool-E variant drops the failed and unsuccessful attacks on both sides. The statistic ignores the tool-vocabulary shift, which κ records as the share of test tool occurrences whose tool appears in the benign training vocabulary. The R4 test rows include the unsuccessful attacks, and its LR-B and LR-C cells are transfer points. Within-AgentDojo rows give the mean and range over the 25 cross-validation pairs, with mean training and test sizes.

GRU plan (GRU_PREREGISTRATION.md, a file name kept from the earlier version) and the revision plan (REVISION_ANALYSIS_PLAN.md). All four are released with the code at https://github.com/Mahmoud-Abbasi-svg/agent-trace-detector-eval. The released monitor follows a fifth plan, MONITOR_PREREGISTRATION.md, which was registered by hash before any monitor call and is described with its amendments in Section S14. The revision plan was recorded in revision/PLAN_MANIFEST.txt on 26 September 2026 at 18.30.33, when no file existed in the revision cache. The three earlier plans have no independent timestamp. Their file modification times show when each file was last written, not when each part of it was written. They therefore cannot show that a plan was not edited after results existed.

Table S3 gives the full record in four parts. Part A states which results were known when each plan was written. Part B lists every deviation from the three earlier plans, including the items raised by the review and one found during the check (gradient clipping, V22). Part C lists how each item of the revision plan was executed, and Part D lists the corrections that the consolidation of the revision streams made after independent verification. The version 0.3 results file of the GRU and the GRU plan now carry appended correction notes. Their unmodified versions, with their original modification times, are kept in archive_v0.3.

The B− arm of the ablation plan (V8) drops the two keyword features, and version 0.3 did not run it. This arm gives LR-B-minus an AUROC of 0.827 in R1 and 0.803, 0.839, 0.899, 0.807 and 0.849 in the medical, banking, web, email and coding folds of R3 (Table S4). These values are close to those of LR-B, so the omission does not change any conclusion.

S4. Behavioral Features

Two of the ten behavioral features count calls whose tool name contains one of fourteen write-like keywords. Table S4 checks this flag against a state-change annotation of every AgentDrift and AgentDojo tool. The annotation was made by a single annotator with no second rater. That annotator was an AI assistant working for the authors and knew the keyword list. The flag is precise but catches only 22 of the 50 state-changing AgentDrift tools, with a call-level recall of 0.455. Replacing the keyword features with annotation-based ones changes the AUROC of LR-B by at most 0.024, and dropping them altogether (LR-B-minus, the B− arm of the ablation plan) changes it by at most 0.015 in R1 and R3. The keyword features are therefore not what drives the results.

S5. Operating Points and the Negative Mix

Table S5 complements the training threshold θ_{r80} of the main paper with other operating points in R1. FPR at a fixed test TPR of 0.8 places the threshold with test labels, so it is a descriptive comparison of ranking at matched recall. It shows that the GRU's advantage over LR-C (0.082 against 0.210) does not depend on where the threshold sits. Budget thresholds flag at most 1% or 5% of the benign and hard-negative training trajectories. Under them no logistic regression has a test FPR that distinguishably exceeds its budget, whereas the GRU and TF-IDF exceed both budgets. With the validation threshold θ_{val80} the GRU reaches a recall of 0.775 [0.745, 0.804] at an FPR of 0.073 [0.055, 0.092] and a precision at 2% of 0.179 [0.147, 0.225]. The threshold $\theta = 0.5$ behaves like θ_{r80} for the logistic regressions. LR-C over-alerts under R3 (FPR 0.538 to 0.733), and every detector recalls at most 0.301 of compromised traces in R4.

Table S6 breaks the operating point at θ_{r80} down by negative class and hijack pattern, as the AgentDrift documentation asks [29]. In R1, LR-A, LR-B, LR-C, the Markov model and the GRU all flag hard negatives more often than benign trajectories. Precision at 2% prevalence therefore depends on the negative mix as much as on prevalence. For LR-C it is 0.117 against benign negatives only, 0.076 with the 27% hard negatives of pool E and 0.039 against hard negatives only. Recall differs by pattern as well. TF-IDF, which flags no negative, recalls 0.974 of full hijacks but only 0.410 of delayed ones. In R3, LR-B's full-hijack recall ranges from 0.468 (medical) to 0.840 (web). In AgentDojo, transferred LR-C flags benign runs as often as compromised ones (0.224 against 0.225). Within AgentDojo, TF-IDF flags no benign run but 0.308 of the unsuccessful attacks.

S6. World and Task Reuse

Table S7 reports regularized, nested probes that ask whether the representations identify the domain, the generation world or the generating model. Representation A identifies the domain perfectly, because tool sets are disjoint across domains. For worlds, representation A reaches 0.096 against a domain-only floor of 0.097, and representation B stays below that floor. A linear probe therefore finds no world information beyond the domain. The world probe population is 81% email, and a supplementary probe on the other four domains gives the same picture. The model-identity probe on benign AgentDojo traces is not distinguishable from chance for B (0.488) or C (0.427). The signal on B reported in the earlier version (0.615) came from the extraction error of Section S2.

Table S8 reports the world-disjoint control on pool E for five split seeds and the task-disjoint split R2 with task-cluster intervals. No decrease in AUROC is distinguishable from zero in any of the 25 detector and seed pairs, and LR-A gains in every seed. LR-B is lower than in R1 in all five seeds. Its largest loss (seed 1) is distinguishable only when the resampling accounts for the trajectories that the two test pools share. It is not distinguishable after Holm correction over seeds, so it is borderline. The world-disjoint splits also change the domain mix, since email makes up 0.553 to 0.621 of their pool E against 0.208 in R1. On R2, no change against R1 is distinguishable for LR-B, LR-C or the GRU, and the wAUPRC@10% gain of LR-C is borderline.

S7. Threshold Failure under Domain Shift

Part A of Table S9 gives the threshold-failure status of every detector and every GRU seed in each held-out domain, relative to the R1 model scored on the same domain. It also gives the

TABLE S3
DEVIATIONS FROM THE PRE-SPECIFIED PLANS, PART A, WHAT WAS KNOWN WHEN EACH PLAN WAS WRITTEN

Plan	Last written	Results already known	Results not yet known	Consequence
R4 plan (plan in Part 1, results appended as Part 2)	25 Sep. 2026, 14.32.54	Transfer AUROC of LR-C (0.440) and LR-B (0.447), written at 13.42.56, then all within-AgentDojo results (UT 0.855 and 0.798, IT 0.766 and 0.771), written at 14.26.35, and LR-C in R1 to R3, written at 14.29.48	None can be established. The within-AgentDojo results were prospective only if Part 1 predates 14.26.35, which file times cannot show	The one-sided near-chance clause (AUROC < 0.60) and the Case rules were fixed with transfer known
Ablation plan	25 Sep. 2026, 14.42.08	LR-C in R1, R2 and R3 as well as LR-C and LR-B in R4 (transfer 0.440 and 0.447, within UT 0.855 and 0.798, IT), all listed in the plan as reproduction anchors	LR-A in every setting. LR-B in R1 and R3, and hence the R3 comparison of LR-C with LR-B (results written at 14.47.34)	Only LR-A and the R1 and R3 results of LR-B were prospective. Rule 4 names R4 as its primary evidence although both R4 values were known
GRU plan	25 Sep. 2026, 15.02.47	All linear results, including LR-C below LR-B in all five R3 folds (14.47.34), the world-disjoint results (14.53.36) and the R4 control (14.26.35)	Every GRU result (code at 15.05.05, results at 15.34.18)	The GRU results were prospective. The GRU diagnostic (15.44.03 to 16.02.26) came after them and is post hoc
Revision plan	26 Sep. 2026, 18.30.32, entered in the plan manifest at 18.30.33	All version 0.3 results, and the exploratory numbers of the independent review for most planned analyses (dropout arms, TF-IDF, reweighted AUPRC, domain-matched comparators, observation-length ablations and padding, unsuccessful attacks as negatives, the corrected extraction, cluster bootstraps)	No revision output existed, and the corrected extraction and the re-run of the version 0.3 scripts came after the plan	The revision analyses are confirmatory re-runs of analyses whose approximate outcome was known, not prospective tests

Times are file modification times (hours.minutes.seconds) in Central European Summer Time, read on 27 September 2026. For result files overwritten by the re-run of 26 September, the times of the archived version 0.3 copies are used. File times show only when a file was last written.

TABLE S3
DEVIATIONS FROM THE PRE-SPECIFIED PLANS, PART B, THE R4 AND ABLATION PLANS

ID	Plan and location	Planned	Executed	Effect and where reported
V1	R4 plan, line 3	Part 1 written before any within-AgentDojo result	The file was last written at 14.32.54, after the R4 control code (14.23.43) and its results (14.26.35), and Part 2 was appended to the same file	The timing of Part 1 cannot be verified, and no independent record exists (Sec. IV-A)
V2	R4 plan, line 61	Near chance if the AUROC interval includes 0.5, or AUROC < 0.60	The one-sided clause was written with transfer 0.440 [0.407, 0.471] known	Post hoc. On corrected data it alone makes LR-A near chance in every population and TF-IDF near chance on C1 (Sec. IV-F, Table VI)
V3	R4 plan, line 65	Mixed or other outcomes reported as observed, including UT reasonable but IT not reasonable	The version 0.3 manuscript omitted the clause and stated that the conditions for Case A were met	The earlier verdict misstated the rule. LR-C is now mixed, and the GRU shows transfer failure under UT with the mixed clause applying under IT (Sec. VI-E)
V4	R4 plan, lines 62 to 64	Case rules applied per population to both representations run	Case A was declared for LR-C only, and the manuscript extended the verdict to the features in general	Overgeneralized. Outcomes are now given per representation (Table VI)
V5	R4 plan, line 10	Unsuccessful attacks excluded at extraction	As planned, although AgentDrift counts failed attacks as training negatives	A design choice, not a deviation. P0F and UT-F are added (Sec. VI-E, Table S15)
V6	R4 plan, line 13	Within-model AUROC and AUPRC with their macro-average	The macro-average was computed for AUROC only	Minor, no effect
V7	R4 plan, line 16	Model-identity probe, five-fold logistic regression on benign traces	As planned (unregularized, C = 1, not nested)	No deviation. Its signal on B (0.615) came from the extraction error, and a nested regularized probe replaces it (Table S7)
V8	Ablation plan, lines 87, 90 and 92	Arms A, B, B− and C, with the paired contrast of C against B−	B− and its contrast were not run and were called not triggered, although only the A+UNK arm was optional	No effect on conclusions. LR-B-minus has R1 AUROC 0.827 and R3 AUROC 0.803, 0.839, 0.899, 0.807 and 0.849 (Table S4)
V9	Ablation plan, line 73 (Rule 2)	B degrades if the independent-bootstrap change from R1 to the shift is negative and distinguishable	No such change was computed for B, and the values were compared descriptively	The rule was not applied in version 0.3. LR-B R3 − R1dm is now distinguishable in no fold (−0.012 to +0.022, Table IV)
V10	Ablation plan, line 79 (Rule 4)	Degradation difference, with R4 as the primary evidence for Outcome 4	Not computed and not reported	The R1-referenced degradation difference is +0.057 [+0.042, +0.073], so the R4 clause is met, but the test is weak because both transfer near chance (Sec. VI-D)
V11	Ablation plan, line 91	Transfer on P0, C1 and model-balanced, with within-AgentDojo UT and IT references	Within UT only per representation, IT only for LR-B and LR-C, and model-balanced only at 10% prevalence	LR-A under IT (0.690) and native-count model-balanced values for all detectors are now reported (Tables VI and S15)
V12	Ablation plan, lines 19 to 22	1,029 worlds, largest worlds 194, 169 and 144, purity 0.858	The audit script behind these numbers was truncated at line 44 and did not parse	The numbers do not reproduce. The rewritten audit gives 1,133 worlds and purity 0.883 (Sec. III-A)
V13	Ablation plan, lines 29 and 36	Domain and world probes with three-fold cross-validation	The earlier results file and the audit docstring say five-fold, and the earlier probe code is lost	Inconsistent specification. Nested regularized probes replace them (Table S7)
V14	Design referred to by the ablation plan	Dataset-identity probe on benign traces	Not run in version 0.3 or in the revision	The two-sample test on B for R4 (0.992 [0.991, 0.993]) answers the same question for B (Tables II and S2)

Line numbers refer to the plan files provided with this supplement. The design of V14 is Part 1 of `ABLATION_RESULTS.md` (line 86), to which the ablation plan points.

TABLE S3
DEVIATIONS FROM THE PRE-SPECIFIED PLANS, PART B, THE GRU PLAN

ID	Plan and location	Planned	Executed	Effect and where reported
V15	GRU plan, line 28	R1 early stopping on `data/val.jsonl` (1,880 trajectories)	A stratified 10% hold-out of the R1 training data, as for every other setting. The GRU audit file and results file recorded the planned source	No effect on any number. Both records now carry correction notes (Sec. V-E)
V16	GRU plan, lines 29 and 70	Three training seeds per fit, including within AgentDojo	One training seed per cross-validation repetition within AgentDojo, seeds 0 to 4	Seed variance cannot be separated from fold-assignment variance within AgentDojo (Sec. IV-A)
V17	GRU plan, line 36	R1-wd with five split seeds and three training seeds	Split seed 0 only	The revision fits all five split seeds with three training seeds (GRU AUROC 0.949 to 0.962, Table S8)
V18	GRU plan, line 55	Independent-bootstrap AUROC change of each shifted setting against R1	Replaced by the non-overlap of intervals	Valid and conservative. GRU R3 − R1dm is now distinguishable after Holm correction in five of five folds (Table IV)
V19	GRU plan, line 39	R4 on P0, C1 and model-balanced as in the R4 control code, which uses native counts	Model-balanced GRU only at 10% prevalence	Native-count model-balanced transfer is 0.536 [0.501, 0.569] under the stratified interval, so Outcome 1 is indeterminate at the boundary (Sec. VI-E)
V20	GRU plan, line 61	Outcome 1 requires near-chance transfer on P0 and model-balanced	Assessed on the 10% subsample only	At the boundary, as in V19 (Sec. VI-E)
V21	GRU plan, line 64	Outcome 4 requires reporting the inconsistency and the seed variance	The results file dismissed Outcome 4, although thresholds varied strongly by seed	Under criteria defined after the fact, Outcome 4 is triggered on the operating point (Sec. VI-E, Table S12)
V22	GRU plan, lines 24 to 31 and 73	Training procedure fixed, with no values beyond those listed	Gradient norm clipped at 1.0 (`gru_detector.py`, line 83), which the plan does not list	An addition in every version 0.3 and revision GRU fit, found during the check. Its effect was not measured (Sec. IV-A)
V23	GRU plan, line 19	Optional GRU on the tool sequence only	The diagnostic also added a linear head on B trained with the GRU procedure, after the GRU results were known	Post hoc and reported as exploratory (Table S12)
V24	GRU plan, line 77	Every deviation recorded in the results file and in the plan	The results file stated that the plan was implemented exactly	Withdrawn by the correction notes appended to both files
V25	GRU plan, lines 12 and 22	Vocabulary from the tool names of benign training trajectories	The code matches the plan	No deviation. The version 0.3 manuscript misstated the vocabulary source (Sec. V-E)

The GRU results file and the split audit are `GRU_RESULTS.md` and `gru_audit.json`. The audit file is left unchanged as script output, and `revision/gru_audit_CORRECTION.md` records its error.

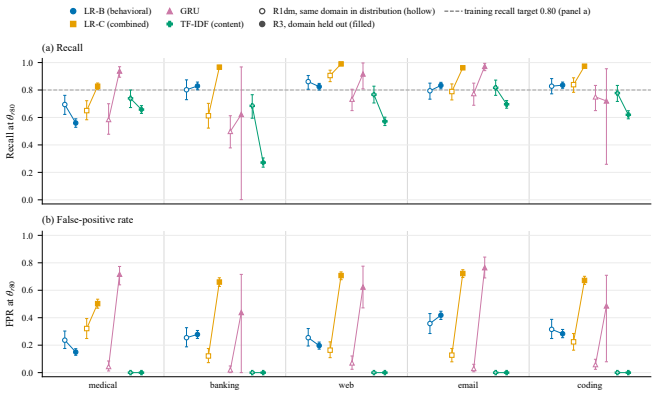


Fig. S1. Recall (a) and FPR (b) at the training threshold θ_{r80} on each held-out domain for LR-B, LR-C, the GRU and TF-IDF. Hollow markers give the R1 model on the same domain (R1dm) and filled markers the model trained without it (R3). Bars are 95% intervals, and the dashed line in (a) is the training recall target of 0.8.

changes in recall and FPR with their intervals. Part B gives the recall and FPR themselves for LR-B, LR-C, the GRU and TF-IDF. Threshold failure is common but its direction depends on the detector. LR-C, LR-C-noobs and the Markov model over-alert, with recall and FPR rising together, whereas LR-A, the Isolation Forest, GRU-drop0.2 and TF-IDF under-alert. LR-B fails only in medical, in the recall direction. The pooled GRU fails in three folds after Holm correction, but each seed fails in every fold, because the seeds fail in different directions in banking and coding. Fig. S1 shows the same comparison for LR-B, LR-C, the GRU and TF-IDF.

Table S10 evaluates a label-free remedy. For each held-out domain, the threshold is set on 200 unlabeled held-out trajectories so that it reproduces the alert rate the detector had in R1. Reading (a), which was pre-specified, draws the sample at the native prevalence of the held-out pool and targets the native R1 alert rate. Reading (b), added after reading (a) was seen, draws traffic at 2% prevalence and targets the R1 alert rate at 2%. Both readings lower LR-C's FPR well below its value at θ_{r80}. Under reading (b), however, the recall of the GRU collapses, and LR-B's FPR is higher than at θ_{r80} in four of five folds. The remedy restores an alert budget, not ranking quality, and it assumes that the prevalence of the calibration sample is known.

S8. Mechanism of the Domain-Shift Failures

Table S11 gives the numbers behind Fig. 4. Inside R1, with no change of domain, a fixed fraction of the 103 tool names of the benign training vocabulary is masked at test time. The masked tools then reach the same untrained inputs as an unseen domain. Level L masks round((1 − L) × 103) randomly drawn tool names, with five draws at each intermediate level. The achieved share of tool occurrences is given in parentheses. As coverage falls, LR-C loses its advantage over LR-B at a coverage of about one half and ends below it. The GRU ends 0.156 below LR-B. The dropout variants keep their FPR nearly flat.

Table S12 reports the GRU per training seed. In every held-out domain the GRU ranks below LR-C for each seed, and the difference is distinguishable from zero under three interval estimands. The mixture interval, which the GRU plan specified, draws one seed per replicate and describes a randomly drawn single-seed deployment. The two-level interval

TABLE S3
DEVIATIONS FROM THE PRE-SPECIFIED PLANS, PART C, EXECUTION OF THE REVISION PLAN

Plan item	Status	Deviation or interpretation	Effect on conclusions
1.1 Positional pairing and unit test	Done	None, and the alignment test passes	None
1.2 No truncation	Done	Observations are uncut in both benchmarks, whereas the review's corrected estimates still used the 2,000-character cut	Explains why the corrected within-AgentDojo LR-C value (0.797) is below the review's (0.846)
1.3 Unsuccessful attacks, exclusions and DoS flag	Done	None	None
1.4 Re-run of the version 0.3 scripts, unchanged	Done	The prediction caches reproduce the re-run points to within 5e-5, which is four-decimal rounding	None
2 Shared prediction cache	Done, with restrictions	Tied scores are made exact. No GRU dropout arm in R1-wd, R2 or UT, and no GRU in LOSO. GRU fits under IT and UT-F were added later with one seed per repetition	Exact ties move the R4 Markov AUROC from 0.508 to 0.590 and R3 LR-A from 0.499 to 0.500, and no conclusion depends on them
2 GRU seeds and validation data	As in the version 0.3 code	The same as V15, V16 and V22	None beyond those items
2 Added detectors	Addition	TF-IDF on tool names, TF-IDF with anonymization, LR-obsonly (post hoc), LR-B with annotation-based features, GRU fits without <UNK>-bearing traces and without DoS positives	Sensitivity analyses only
3.1 Prevalence-weighted AUPRC as primary	Done	Subsampling kept as secondary. Rounding scores to 12 decimals merges saturated AgentDojo scores, so unrounded AgentDojo values are quoted	Changes of up to 0.0056. The LR-C AUPRC criterion of the R4 rule flips (0.2985 weighted against 0.322 subsampled)
3.2 Domain-matched R1 comparator	Done	The R1dm and R3 pools share 345 trajectories per domain (276 in banking)	Conservative intervals, and a joint bootstrap changes no decision
3.3 Threshold failure	Done	Headline counts use Holm correction, whereas the plan's definition is unadjusted, and both are reported	The counts differ only for the pooled GRU (3 against 4) and LR-B-noobs (1 against 2)
3.4 Reversal rule	Done	AUROC primary and wAUPRC@10% secondary. The rule is also applied to the dropout pairs, and the GRU has three interval estimands	The reversal folds are identical under all three estimands
3.5 Cluster bootstrap	Done, extended	Also used for R4 transfer and LOSO, with the cross-validation repetition shared across detectors	Borderline results depend on the resampling unit, for example R4 TF-IDF − LR-C
3.6 Holm correction	Done	Bootstrap p-values with different floors across streams, and family (d) read in three ways	A recomputation with a per-detector family changed no decision
3.7 Label-free threshold	Done, with a post hoc reading	The pre-specified reading (a) and a post hoc reading (b) at 2% prevalence are both reported, and the R1 alert rate had to be interpreted	Reading (b), not (a), matches the figures of the review
3.8 FPR at a fixed TPR and FPR budgets	Done, with a restriction	FPR at a test TPR of 0.8 is descriptive, and the budgets are computed in R1 only	The budget question is answered for R1 only
4 Observation length	Done	Padding appends a neutral sentence, the GRU is padded in R1 only, and only lengthening is tested	The within-AgentDojo padding effect reported by the review is not reproduced
4 Keyword features	Done	One non-blind annotator	Annotation-based features change LR-B by at most 0.024
4 Negative-class and pattern breakdown	Restriction	Computed for R1, R2, the five R3 folds, R4 and UT, not for R1-wd, IT, UT-F or LOSO	No headline number depends on the missing settings
4 Composition and templates in R4	Done	Interpretations of the Case rule, an exploratory attempt target and a per-template design	The LR-C outcome lies at the boundary (0.003 AUROC)
4 Mechanism	Done	Graded coverage masks tool names rather than domains, the R1 GRU was refitted, and the GRU training negatives include failed attacks	None
4 Leave-one-suite-out	Done	Linear detectors only	None
4 Shift size	Done, extended	Added for R1-wd seeds 1 to 4, LOSO and every cross-validation pair	None
4 Diagnostics	Done	The audit script was rewritten, probes are nested and regularized, and an exploratory world-key search was added	None
5 Decision rules	Applied	The Rule 4 estimand was corrected to the R1-referenced difference	Wording only
Outside the plan (review M2)	Done in part	A frozen-encoder reimplementation (SE, Section S13), no released guard model	The request is met only in part, with TF-IDF and SE

Item numbers follow the sections of the revision plan. The stream logs behind each row are in `revision/DEVIATIONS.md`, Part E.

resamples seeds and then trajectories, and the seed-averaged interval is conditional on the three seeds. In email the per-seed differences range from -0.096 to -0.025. The operating point varies far more across seeds than the ranking does. In banking, seed 1 flags almost nothing (FPR 0.000, recall 0.006), whereas seeds 0 and 2 over-alert. This is why Outcome 4 of the GRU plan is triggered on the operating point.

The last two columns reproduce the post hoc diagnostic of the earlier version, which was not re-run. A GRU that reads only the tool sequence scores 0.567 to 0.606 in R3, close to what trajectory length alone achieves. A linear head on representation B trained with the GRU procedure scores 0.786 to 0.898. The earlier version called the second variant GRU-B although it has no recurrence.

Table S13 isolates the observation-length cue. Without the maximum and mean observation length, LR-B loses 0.112 to 0.221 AUROC in every held-out domain, while the two features alone (LR-obsonly) come within 0.053 of the full LR-B. In R4, dropping them raises transfer to 0.630 for LR-B-noobs and 0.657 for LR-C-noobs. The noobs variants and LR-obsonly were defined after related results were known, however, so these results are post hoc. The padding test appends a neutral English sentence, repeated and cut to the target length, to the observation of the injection step of every positive, with models and thresholds unchanged. In R1 it lowers recall at θ_{r80} for every detector, down to 0.033 for LR-C and 0.000 for the GRU at 1,200 characters. Within AgentDojo it slightly raises the recall of LR-B and LR-C, because there the cue points the other way. We did not test whether padded injections still succeed, and we tested only lengthening.

Table S14 repeats the coverage analysis on recorded traces through AgentDojo leave-one-suite-out. Two suites share no tool with the others (banking and Slack), and two share a

TABLE S3
DEVIATIONS FROM THE PRE-SPECIFIED PLANS, PART D, CORRECTIONS MADE IN THE CONSOLIDATION OF THE REVISION ANALYSES

ID	Correction	Effect
X1	The breakdown by negative class and hijack pattern covers R1, R2, the five R3 folds, R4 and UT only, although the caches for the other settings exist	No headline depends on the missing settings, and the gap is stated in Sec. VIII
X2	TPR at FPR budgets of 1% and 5% is reported for R1 only, for all detectors including TF-IDF	The budget question is answered in R1 only
X3	No GRU in LOSO and no GRU dropout arm in R1-wd, R2 or UT	Stated in Sec. VIII
X4	The pre-specified Rule 4 estimand is the R1-referenced degradation difference, +0.057 [+0.042, +0.073]. The within-AgentDojo-referenced value (+0.064) is supplementary	The outcome is the same
X5	One world key on the training and validation data (user, date and sorted contact names) reproduces 1,029 worlds, purity 0.858 and 54 worlds with at least 30 trajectories, but not the largest world sizes. The README's ext-audit share is approximately reproduced (0.281)	A lead, not proof. The paper reports 1,133 worlds and purity 0.883
X6	Test FPR exceeds a 1% budget only for the GRU (0.070) and TF-IDF (0.033), not for LR-C (0.017). The Isolation Forest is close to chance rather than at chance, and in R4 only LR-A is slightly above chance on wAUPRC@10%	Wording
X7	R4 TF-IDF − LR-C, +0.047 [−0.002, +0.096] with p = 0.057, is borderline. The best single-token rule (0.780) is the maximum over ten rules selected on training data, and the rule ranked first gives 0.753	Wording
X8	Bootstrap p-value floors differ between streams	No Holm decision changes
X9	Streams read Holm family (d) in three ways	No decision changes. LR-C under IT is borderline in every reading (Holm p = 0.051)
X10	One stream summary rounds four-decimal values a second time, so a few printed values differ by 0.001	Values at a decision boundary are quoted to four decimals (0.5655, 0.2985)
X11	Unrounded AgentDojo rank values are quoted, for example LR-C transfer AUROC 0.473	None
X12	Paired-difference centers in the R4 analysis are bootstrap means	Third-decimal differences between tables
X13	The world-cluster sensitivity of the reversal analysis was run in the domain-shift stream only	Same reversal folds
X14	Gradient clipping at norm 1.0 is part of every GRU fit and is not in the GRU plan (the same as V22)	Disclosed as a deviation
X15	The fall in LR-C alert rates under LOSO holds only in the two zero-coverage suites, and the graded-coverage GRU intervals are seed-averaged	Wording
X16	No released guard model was run. The only frozen sentence encoder is the SE reimplementation of Section S13	Review request M2 is met only in part
X17	The within-AgentDojo effect of the 2,000-character cut (0.846 against 0.797) has no paired interval. Shortened observations also occur in hard negatives (median 73 against 98 characters). LR-C-noobs ranks below LR-B-noobs in five of five folds on wAUPRC@10% and in four of five on AUROC	Both metrics are reported
X18	The stratified lower bound of the model-balanced GRU transfer (0.5007) lies within Monte Carlo error of 0.5	GRU Outcome 1 is reported as indeterminate at the boundary

These corrections come from the independent verification of each analysis stream and take precedence over the stream logs.

TABLE S4
KEYWORD STATE-CHANGE FLAG AGAINST A TOOL ANNOTATION, AND LR-B WITH ANNOTATION-BASED FEATURES OR WITHOUT THE KEYWORD FEATURES

Keyword flag vs annotation	Tool-level precision	Tool-level recall	Call-level precision	Call-level recall
AgentDrift (train and test)	22/23 (0.957)	22/50 (0.440)	0.970 [0.966, 0.973]	0.455 [0.448, 0.461]
AgentDojo (P0 and unsuccessful)	15/18 (0.833)	15/22 (0.682)	0.867 [0.784, 0.944]	0.831 [0.773, 0.887]
AgentDojo P0	15/17 (0.882)	15/22 (0.682)	0.974 [0.951, 0.991]	0.830 [0.783, 0.881]
LR-B AUROC	**LR-B**	**LR-B-annot**	**LR-B-minus**	**LR-B-annot − LR-B**
R1	0.830 [0.811, 0.849]	0.826 [0.806, 0.846]	0.827 [0.808, 0.846]	−0.004 [−0.010, +0.001]
R3 medical	0.808 [0.788, 0.826]	0.807 [0.788, 0.826]	0.803 [0.783, 0.822]	−0.001 [−0.004, +0.003]
R3 banking	0.854 [0.836, 0.873]	0.850 [0.831, 0.869]	0.839 [0.820, 0.858]	−0.004 [−0.009, +0.001]
R3 web	0.896 [0.882, 0.910]	0.899 [0.885, 0.912]	0.899 [0.886, 0.911]	+0.003 [−0.002, +0.008]
R3 email	0.792 [0.772, 0.812]	0.768 [0.747, 0.790]	0.807 [0.787, 0.827]	−0.024* [−0.031, −0.016]
R3 coding	0.852 [0.835, 0.868]	0.859 [0.842, 0.874]	0.849 [0.832, 0.865]	+0.007* [+0.002, +0.012]
R4 transfer	0.482 [0.434, 0.522]	0.479 [0.431, 0.520]	0.478 [0.431, 0.518]	−0.002 [−0.006, +0.001]

Tool-level values count tools of the vocabulary, and call-level intervals resample trajectories (clusters in AgentDojo). LR-B-annot replaces the two keyword features with the same counts based on the annotation, and LR-B-minus drops them. * marks a difference distinguishable from zero after Holm correction over the five R3 folds.

TABLE S5
ALTERNATIVE OPERATING POINTS IN R1, AND THE THRESHOLD 0.5 UNDER DOMAIN SHIFT AND TRANSFER

	FPR at test TPR 0.8	1% budget		5% budget		$\theta = 0.5$, R1		$\theta = 0.5$, R3 (range)		$\theta = 0.5$, R4	
Detector	(descriptive)	TPR	FPR	TPR	FPR	Recall	FPR	Recall	FPR	Recall	FPR
LR-A (tool identity)	0.411 [0.364, 0.472]	0.173 [0.147, 0.199]	0.013 [0.006, 0.022]	0.400 [0.367, 0.433]	0.065 [0.048, 0.082]	0.658 [0.628, 0.687]	0.230 [0.202, 0.259]	0.000 to 0.000	0.000 to 0.000	0.067	0.008
LR-B (behavioral)	0.288 [0.250, 0.335]	0.194 [0.166, 0.221]	0.011 [0.005, 0.019]	0.373 [0.341, 0.407]	0.067 [0.051, 0.085]	0.758 [0.730, 0.787]	0.247 [0.219, 0.274]	0.527 to 0.807	0.140 to 0.384	0.178	0.224
LR-C (combined)	0.210 [0.175, 0.244]	0.276 [0.244, 0.307]	0.017 [0.008, 0.027]	0.529 [0.495, 0.563]	0.065 [0.048, 0.082]	0.786 [0.759, 0.815]	0.198 [0.172, 0.224]	0.864 to 0.990	0.538 to 0.733	0.226	0.224
GRU	0.082 [0.061, 0.108]	0.770 [0.717, 0.809]	0.070 [0.050, 0.092]	0.889 [0.866, 0.910]	0.156 [0.128, 0.182]	0.856 [0.819, 0.893]	0.119 [0.084, 0.159]	0.686 to 0.994	0.531 to 0.832	0.301	0.212
TF-IDF (content)	0.000 [0.000, 0.000]	0.998 [0.994, 1.000]	0.033 [0.021, 0.045]	0.999 [0.996, 1.000]	0.081 [0.063, 0.102]	0.990 [0.983, 0.996]	0.008 [0.002, 0.016]	0.559 to 0.951	0.000 to 0.145	0.175	0.217
Markov	0.472 [0.417, 0.524]	0.054 [0.039, 0.070]	0.017 [0.008, 0.027]	0.243 [0.214, 0.272]	0.069 [0.052, 0.087]	n/a	n/a	n/a	n/a	n/a	n/a
IForest	0.634 [0.594, 0.676]	0.006 [0.001, 0.012]	0.013 [0.006, 0.022]	0.046 [0.032, 0.060]	0.052 [0.038, 0.068]	n/a	n/a	n/a	n/a	n/a	n/a
GRU at θ_{val80} (R1)	n/a	n/a	n/a	n/a	n/a	0.775 [0.745, 0.804]	0.073 [0.055, 0.092]	n/a	n/a	n/a	n/a

FPR at a test TPR of 0.8 uses test labels and is descriptive only. Budget thresholds flag at most 1% or 5% of the benign and hard-negative training trajectories, scored in sample. R3 cells give the range of points over the five folds, and R4 cells are points. Markov and Isolation Forest scores are not probabilities, so they have no threshold of 0.5. The last row gives the GRU at the validation threshold θ_{val80}, pooled over three seeds.

TABLE S6
OPERATING POINT AT THE TRAINING THRESHOLD BY NEGATIVE CLASS AND HIJACK PATTERN IN R1, R3 AND AGENTDOJO

	FPR		Failed	Recall				Precision at 2%, negative mix		
Detector	benign	hard neg.	flagged	pool E	full	partial	delayed	benign only	pool E	hard only
R1 (pool E with 831 attacked, 600 benign and 225 hard-negative trajectories)										
LR-A (tool identity)	0.295 [0.258, 0.333]	0.440 [0.373, 0.502]	0.351 [0.289, 0.413]	0.744 [0.716, 0.771]	0.893 [0.863, 0.921]	0.502 [0.437, 0.567]	0.642 [0.560, 0.716]	0.049	0.043	0.033
LR-B (behavioral)	0.193 [0.163, 0.225]	0.524 [0.458, 0.591]	0.524 [0.462, 0.587]	0.795 [0.768, 0.822]	0.758 [0.717, 0.796]	0.848 [0.801, 0.892]	0.836 [0.776, 0.896]	0.077	0.054	0.030
LR-C (combined)	0.118 [0.092, 0.147]	0.387 [0.320, 0.449]	0.218 [0.164, 0.271]	0.771 [0.744, 0.800]	0.830 [0.796, 0.863]	0.727 [0.671, 0.784]	0.642 [0.567, 0.724]	0.117	0.076	0.039
GRU	0.024 [0.010, 0.040]	0.101 [0.062, 0.142]	0.043 [0.013, 0.084]	0.682 [0.645, 0.716]	0.722 [0.670, 0.768]	0.606 [0.532, 0.675]	0.674 [0.575, 0.761]	0.373	0.238	0.122
TF-IDF (content)	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.763 [0.739, 0.787]	0.974 [0.959, 0.987]	0.541 [0.476, 0.606]	0.410 [0.328, 0.493]	1.000 ($\geq$ 0.711)	1.000 ($\geq$ 0.772)	1.000 ($\geq$ 0.481)
Markov	0.412 [0.372, 0.452]	0.671 [0.604, 0.729]	0.618 [0.556, 0.680]	0.810 [0.786, 0.836]	0.775 [0.738, 0.811]	0.827 [0.775, 0.874]	0.903 [0.851, 0.948]	0.039	0.033	0.024
IForest	0.628 [0.590, 0.667]	0.609 [0.542, 0.676]	0.791 [0.738, 0.844]	0.785 [0.756, 0.811]	0.811 [0.775, 0.843]	0.697 [0.636, 0.758]	0.843 [0.784, 0.903]	0.025	0.025	0.026
R3 held-out domains										
LR-B, medical	0.090 [0.069, 0.112]	0.310 [0.255, 0.365]	0.396 [0.337, 0.455]	0.560 [0.529, 0.589]	0.468 [0.427, 0.506]	0.830 [0.781, 0.879]	0.497 [0.419, 0.574]	0.113	0.071	0.036
LR-B, banking	0.150 [0.124, 0.176]	0.620 [0.561, 0.678]	0.737 [0.682, 0.788]	0.829 [0.800, 0.856]	0.659 [0.594, 0.719]	0.958 [0.931, 0.981]	0.852 [0.792, 0.906]	0.101	0.057	0.027
LR-B, web	0.088 [0.069, 0.110]	0.486 [0.424, 0.549]	0.424 [0.361, 0.482]	0.825 [0.801, 0.847]	0.840 [0.811, 0.870]	0.833 [0.787, 0.878]	0.747 [0.673, 0.814]	0.160	0.079	0.033
LR-B, email	0.357 [0.322, 0.394]	0.580 [0.522, 0.639]	0.424 [0.361, 0.486]	0.833 [0.811, 0.856]	0.820 [0.790, 0.851]	0.864 [0.822, 0.902]	0.834 [0.775, 0.887]	0.045	0.039	0.028
LR-B, coding	0.213 [0.182, 0.244]	0.471 [0.412, 0.533]	0.565 [0.502, 0.627]	0.835 [0.813, 0.859]	0.834 [0.804, 0.863]	0.853 [0.807, 0.896]	0.811 [0.750, 0.866]	0.074	0.057	0.035
LR-C, medical	0.451 [0.413, 0.488]	0.639 [0.580, 0.694]	0.576 [0.514, 0.635]	0.827 [0.805, 0.849]	0.759 [0.725, 0.791]	0.951 [0.923, 0.976]	0.903 [0.852, 0.948]	0.036	0.033	0.026
LR-C, banking	0.599 [0.562, 0.634]	0.824 [0.776, 0.867]	0.831 [0.784, 0.875]	0.966 [0.952, 0.979]	0.917 [0.876, 0.949]	0.996 [0.988, 1.000]	0.987 [0.966, 1.000]	0.032	0.029	0.023
LR-C, web	0.632 [0.599, 0.669]	0.906 [0.871, 0.941]	0.745 [0.690, 0.800]	0.990 [0.984, 0.996]	0.990 [0.982, 0.997]	0.992 [0.981, 1.000]	0.987 [0.967, 1.000]	0.031	0.028	0.022
LR-C, email	0.685 [0.650, 0.718]	0.820 [0.773, 0.867]	0.616 [0.557, 0.675]	0.962 [0.950, 0.973]	0.950 [0.932, 0.967]	0.977 [0.958, 0.992]	0.980 [0.954, 1.000]	0.028	0.026	0.023
LR-C, coding	0.610 [0.575, 0.647]	0.835 [0.792, 0.879]	0.753 [0.698, 0.804]	0.974 [0.964, 0.983]	0.977 [0.963, 0.988]	0.961 [0.938, 0.985]	0.982 [0.957, 1.000]	0.032	0.029	0.023
AgentDojo P0 (392 benign negatives, with unsuccessful attacks in the failed column)										
LR-C (combined), transfer	0.224 [0.186, 0.265]	n/a	0.056 [0.050, 0.062]	0.225 [0.208, 0.242]	n/a	n/a	n/a	0.020	0.020	n/a
GRU, transfer	0.168 [0.115, 0.230]	n/a	0.073 [0.049, 0.094]	0.256 [0.213, 0.296]	n/a	n/a	n/a	0.030	0.030	n/a
TF-IDF (content), transfer	0.000 [0.000, 0.000]	n/a	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n/a	n/a	n/a	no alert	no alert	n/a
LR-C (combined), within UT	0.216 [0.138, 0.281]	n/a	0.279 [0.253, 0.305]	0.720 [0.686, 0.744]	n/a	n/a	n/a	0.065	0.065	n/a
GRU, within UT	0.178 [0.115, 0.240]	n/a	0.227 [0.173, 0.276]	0.761 [0.719, 0.794]	n/a	n/a	n/a	0.082	0.082	n/a
TF-IDF (content), within UT	0.000 [0.000, 0.000]	n/a	0.308 [0.286, 0.331]	0.780 [0.758, 0.801]	n/a	n/a	n/a	1.000 ($\geq$ 0.623)	1.000 ($\geq$ 0.623)	n/a

The bootstrap resamples each combination of category and hijack pattern separately, so the class composition is that of the test set. Precision at 2% combines the pooled recall with the FPR of each negative mix through (4), and its values are points. For AgentDojo the negatives are benign runs only, and the failed column counts unsuccessful attacks. Where TF-IDF flags no negative, precision is 1.000 with the lower bound implied by the exact 95% upper bound on the FPR. "No alert" means that the detector flags nothing, so precision is undefined. The breakdown was not computed for R1-wd, IT, UT-F or LOSO.

TABLE S7
DOMAIN, WORLD AND MODEL-IDENTITY PROBES WITH THEIR FLOORS

Probe	A (tools)	B (behavioral)	B-minus	C (combined)	Majority	Domain only / suite
AgentDrift domain and world probes (accuracy)						
Domain, all pooled ($n = 10,656$)	1.000 [1.000, 1.000]	0.458 [0.449, 0.467]	0.381 [0.372, 0.390]	1.000 [1.000, 1.000]	0.207 [0.200, 0.215]	n/a
Domain, benign only ($n = 3,400$)	1.000 [1.000, 1.000]	0.505 [0.489, 0.522]	0.458 [0.441, 0.474]	n/a	0.200 [0.186, 0.213]	n/a
World, 54 worlds of 30 or more ($n = 2,724$)	0.096 [0.086, 0.107]	0.059 [0.050, 0.067]	0.049 [0.041, 0.056]	0.096 [0.085, 0.107]	0.044 [0.036, 0.052]	0.097 [0.086, 0.108]
World, non-email (supplementary) ($n = 4,646$)	0.039 [0.034, 0.045]	0.029 [0.025, 0.034]	0.025 [0.021, 0.030]	0.039 [0.034, 0.045]	0.026 [0.021, 0.031]	0.043 [0.038, 0.049]
AgentDojo model-identity probe on benign P0 (AUROC, with the suite floor in the last column)						
gpt-4o vs other models (92 vs 300)	n/a	0.488 [0.402, 0.578]	n/a	0.427 [0.342, 0.504]	0.500 [0.500, 0.500]	0.459 [0.383, 0.538]

AgentDrift probes standardize the features and fit an L2-regularized multinomial logistic regression. Its strength is chosen by an inner three-fold grid search inside outer five-fold cross-validation repeated with seeds 0, 1 and 2. Values are accuracies with paired trajectory-level intervals. The floors are the majority class and a probe on the domain alone. The AgentDojo probe separates the 92 benign `gpt-4o-2024-05-13` traces from the 300 benign traces of the other models with nested cross-validation (outer five folds repeated five times, inner grid over the regularization strength). It reports the pooled out-of-fold AUROC against the majority floor and a suite-only floor.

TABLE S8
WORLD-DISJOINT AND TASK-DISJOINT CONTROLS ON POOL E

Quantity	LR-A (tool identity)	LR-B (behavioral)	LR-C (combined)	GRU	TF-IDF (content)
World-disjoint control (R1-wd), pool E, five split seeds					
R1 AUROC	0.777 [0.756, 0.800]	0.830 [0.811, 0.850]	0.879 [0.863, 0.895]	0.936 [0.923, 0.947]	0.999 [0.999, 1.000]
R1-wd s0 − R1	$+0.043^{*}$ [+0.013, +0.072]	−0.010 [−0.038, +0.017]	+0.012 [−0.011, +0.034]	$+0.020^{*}$ [+0.003, +0.037]	+0.000 [−0.000, +0.001]
R1-wd s1 − R1	$+0.044^{*}$ [+0.014, +0.073]	−0.027 [−0.054, +0.001]	+0.011 [−0.011, +0.033]	+0.014 [−0.002, +0.029]	+0.000 [−0.000, +0.001]
R1-wd s2 − R1	$+0.045^{*}$ [+0.015, +0.074]	−0.008 [−0.036, +0.019]	$+0.022^{*}$ [+0.001, +0.044]	$+0.026^{*}$ [+0.011, +0.041]	+0.001 [+0.000, +0.001]
R1-wd s3 − R1	$+0.044^{*}$ [+0.014, +0.073]	−0.012 [−0.040, +0.014]	+0.010 [−0.013, +0.032]	$+0.023^{*}$ [+0.003, +0.041]	+0.000 [−0.000, +0.001]
R1-wd s4 − R1	$+0.039^{*}$ [+0.008, +0.068]	−0.012 [−0.039, +0.016]	+0.013 [−0.010, +0.034]	$+0.021^{*}$ [+0.002, +0.037]	+0.000 [−0.000, +0.001]
R1-wd AUROC, mean (range)	0.821 (0.817 to 0.823)	0.817 (0.803 to 0.822)	0.892 (0.888 to 0.901)	0.956 (0.949 to 0.962)	1.000 (1.000 to 1.000)
R1-wd s1 − R1, overlap-aware	$+0.044^{*}$ [+0.018, +0.071]	-0.027^{*} [−0.052, −0.002]	+0.011 [−0.009, +0.031]	+0.014 [−0.001, +0.028]	+0.000 [−0.000, +0.001]
Task-disjoint split (R2), 35 test tasks, task-cluster bootstrap					
R2 AUROC	n/a	0.849 [0.817, 0.880]	0.902 [0.879, 0.923]	0.915 [0.884, 0.941]	n/a
R2 − R1 AUROC	n/a	+0.018 [−0.017, +0.054]	+0.023 [−0.005, +0.049]	−0.020 [−0.054, +0.008]	n/a
R2 wAUPRC 10%	n/a	0.451 [0.377, 0.552]	0.615 [0.554, 0.678]	0.647 [0.538, 0.759]	n/a
R2 − R1 wAUPRC 10%	n/a	+0.049 [−0.044, +0.157]	$+0.092^{*}$ [+0.002, +0.174]	−0.051 [−0.188, +0.077]	n/a

R1-wd rows give R1-wd − R1 AUROC with independent bootstraps, and * marks a difference distinguishable from zero without correction. The overlap-aware row resamples the union of the two test pools, so that trajectories shared by R1 and R1-wd enter both estimates together. The GRU rows average three training seeds. R2 intervals resample the 35 test tasks. LR-A and TF-IDF have no task-cluster values in R2.

few (travel 0.084 and workspace 0.362). No suite meets the reversal rule. The in-distribution advantage of LR-C over LR-B is not distinguishable from zero under cluster resampling (+0.056 [−0.003, +0.111]), and the sign of the held-out difference differs between the two zero-coverage suites. The LR-C offset of (7) is negative in three suites. In the two zero-coverage suites it applies to every trajectory, so there LR-C raises fewer alerts than LR-B rather than more. With 22 to 47 clusters per suite, small reversals in single suites cannot be excluded.

S9. Cross-Benchmark Transfer Controls

Table S15 applies the composition controls of the R4 plan per representation. Removing the 65 benign traces in which an injection task was run as the user task (C1) lifts LR-C and LR-A above the reasonable bar within AgentDojo. Model balancing, in contrast, lowers the within values of the content-blind detectors slightly and moves transfer by only a few hundredths. Without the 101 DoS positives, within LR-C reaches 0.813, and transfer is unchanged. With the unsuccessful attacks as training negatives (UT-F), LR-A, LR-C and the GRU are reasonable when evaluated on P0F. Transfer on P0F (0.626 for LR-C and 0.645 for the GRU) is no longer near chance, however, so this is not transfer failure in the rule's sense. With positives from the important_instructions template only, the one template run for every model, transfer stays near 0.5, and across the 15 templates of `gpt-4o-2024-05-13` LR-C transfer ranges from 0.309 to 0.666. Per model, within-UT LR-C lies between 0.745 and 0.794 and transfer between 0.444 and 0.562.

The two attempt-target rows count every injection attempt as positive, which relabels the 4,987 unsuccessful attacks as positives at evaluation with the models unchanged. This target is exploratory and outside the plan. Within AgentDojo the content-blind representations then score only 0.584 to 0.613 under UT. Their transfer lies between 0.387 and 0.506,

TABLE S9
THRESHOLD FAILURE, PART A, STATUS OF EVERY DETECTOR AND SEED AND THE R3 − R1DM CHANGES OF RECALL AND FPR AT THE TRAINING THRESHOLD

Detector	medical	banking	web	email	coding	Folds failing (Holm / unadj.)
Status relative to domain-matched R1 after Holm over five folds						
LR-A (tool identity)	FPR↓ rec↓	FPR↓ rec↓	FPR↓ rec↓	FPR↓ rec↓	FPR↓ rec↓	5 / 5
LR-B (behavioral)	rec↓	holds	holds	holds	holds	1 / 1
LR-B-minus	FPR↓	holds	holds	holds	holds	1 / 1
LR-B-noobs	holds	holds	FPR↓ rec↓	holds	holds	1 / 2
LR-C (combined)	FPR↑ rec↑	FPR↑ rec↑	FPR↑ rec↑	FPR↑ rec↑	FPR↑ rec↑	5 / 5
LR-C-drop	FPR↓ rec↓	FPR↑ rec↑	rec↓	FPR↑	holds	4 / 4
LR-C-noobs	FPR↑ rec↑	FPR↑ rec↑	FPR↑ rec↑	FPR↑ rec↑	FPR↑ rec↑	5 / 5
GRU	FPR↑ rec↑	holds	FPR↑ rec↑	FPR↑ rec↑	holds	3 / 4
GRU, seed 0	FPR↑ rec↑	FPR↑ rec↑	FPR↑ rec↑	FPR↑ rec↑	FPR↑ rec↑	5 / 5
GRU, seed 1	FPR↑ rec↑	rec↓	FPR↑ rec↑	FPR↑ rec↑	FPR↑ rec↑	5 / 5
GRU, seed 2	FPR↑ rec↑	FPR↑ rec↑	FPR↑ rec↑	FPR↑ rec↑	rec↓	5 / 5
GRU at θ_{val80}	FPR↑ rec↑	holds	FPR↑ rec↑	FPR↑ rec↑	FPR↑	4 / 4
GRU-drop0.2	rec↓	rec↓	rec↓	rec↓	rec↓	5 / 5
GRU-drop0.1	rec↓	holds	rec↓	rec↓	rec↓	4 / 4
TF-IDF (content)	rec↓	rec↓	rec↓	rec↓	rec↓	5 / 5
TF-IDF no digits	holds	rec↓	rec↓	rec↓	rec↓	4 / 4
Markov	FPR↑ rec↑	FPR↑ rec↑	FPR↑ rec↑	FPR↑ rec↑	FPR↑ rec↑	5 / 5
IForest	FPR↓ rec↓	FPR↓ rec↓	FPR↓ rec↓	FPR↓ rec↓	FPR↓ rec↓	5 / 5
R3 minus domain-matched R1 at θ_{r80} (paired rates, independent bootstraps)						
LR-B (behavioral), Δ recall	-0.135^{*} $[-0.208, -0.059]$	$+0.027$ $[-0.048, +0.110]$	-0.037 $[-0.092, +0.023]$	$+0.039$ $[-0.025, +0.106]$	$+0.007$ $[-0.054, +0.069]$	
LR-B (behavioral), Δ FPR	-0.087 $[-0.159, -0.018]$	$+0.024$ $[-0.052, +0.094]$	-0.058 $[-0.129, +0.013]$	$+0.061$ $[-0.018, +0.138]$	-0.032 $[-0.109, +0.046]$	
LR-C (combined), Δ recall	$+0.177^{*}$ $[+0.102, +0.248]$	$+0.354^{*}$ $[+0.265, +0.447]$	$+0.085^{*}$ $[+0.045, +0.131]$	$+0.173^{*}$ $[+0.114, +0.235]$	$+0.135^{*}$ $[+0.083, +0.191]$	
LR-C (combined), Δ FPR	$+0.181^{*}$ $[+0.105, +0.262]$	$+0.539^{*}$ $[+0.478, +0.594]$	$+0.543^{*}$ $[+0.479, +0.603]$	$+0.595^{*}$ $[+0.539, +0.652]$	$+0.447^{*}$ $[+0.379, +0.513]$	
GRU, Δ recall	$+0.353^{*}$ $[+0.225, +0.466]$	$+0.123$ $[-0.575, +0.551]$	$+0.183^{*}$ $[+0.036, +0.326]$	$+0.199^{*}$ $[+0.117, +0.288]$	-0.030 $[-0.538, +0.283]$	
GRU, Δ FPR	$+0.672^{*}$ $[+0.582, +0.740]$	$+0.418$ $[-0.042, +0.702]$	$+0.552^{*}$ $[+0.381, +0.724]$	$+0.734^{*}$ $[+0.649, +0.818]$	$+0.427$ $[+0.002, +0.668]$	
TF-IDF (content), Δ recall	-0.081^{*} $[-0.147, -0.009]$	-0.413^{*} $[-0.503, -0.320]$	-0.195^{*} $[-0.261, -0.127]$	-0.121^{*} $[-0.184, -0.059]$	-0.158^{*} $[-0.224, -0.089]$	
TF-IDF (content), Δ FPR	$+0.000$ $[+0.000, +0.000]$	$+0.000$ $[+0.000, +0.000]$	$+0.000$ $[+0.000, +0.000]$	$+0.000$ $[+0.000, +0.000]$	$+0.000$ $[+0.000, +0.000]$	

The upper block gives the status of each detector relative to R1dm after Holm correction over the five folds. Arrows give the direction of a distinguishable change in FPR or recall, and "holds" means that neither changes. The last column counts failing folds with and without the correction. The lower block gives R3 − R1dm with independent-bootstrap intervals, and * marks a change distinguishable from zero after Holm correction. The pooled GRU rows draw one seed per replicate, which widens their intervals where the seeds disagree.

TABLE S9
THRESHOLD FAILURE, PART B, RECALL AND FPR AT THE TRAINING THRESHOLD θ_{r80} IN R1DM AND R3 FOR LR-B, LR-C, THE GRU AND TF-IDF

Detector	Rate	medical	banking	web	email	coding	Failing folds (Holm) and direction
LR-B (behavioral)	Recall	$0.694 \to 0.560^{*}$	$0.802 \to 0.829$	$0.861 \to 0.825$	$0.794 \to 0.833$	$0.828 \to 0.835$	1 of 5
	FPR	$0.236 \to 0.150$	$0.255 \to 0.278$	$0.255 \to 0.197$	$0.358 \to 0.418$	$0.315 \to 0.283$	recall down (medical only)
LR-C (combined)	Recall	$0.650 \to 0.828^{*}$	$0.613 \to 0.967^{*}$	$0.906 \to 0.990^{*}$	$0.789 \to 0.962^{*}$	$0.839 \to 0.974^{*}$	5 of 5
	FPR	$0.321 \to 0.503^{*}$	$0.121 \to 0.660^{*}$	$0.164 \to 0.707^{*}$	$0.127 \to 0.722^{*}$	$0.224 \to 0.672^{*}$	over-alerting (FPR and recall up)
GRU	Recall	$0.585 \to 0.938^{*}$	$0.498 \to 0.622$	$0.733 \to 0.917^{*}$	$0.774 \to 0.973^{*}$	$0.750 \to 0.720$	3 of 5
	FPR	$0.044 \to 0.717^{*}$	$0.020 \to 0.438$	$0.071 \to 0.623^{*}$	$0.030 \to 0.764^{*}$	$0.059 \to 0.486$	over-alerting (FPR and recall up)
	Per seed	3 of 3 seeds fail	3 of 3 seeds fail	3 of 3 seeds fail	3 of 3 seeds fail	3 of 3 seeds fail	5 of 5 for every seed
TF-IDF (content)	Recall	$0.739 \to 0.658^{*}$	$0.685 \to 0.272^{*}$	$0.767 \to 0.572^{*}$	$0.817 \to 0.696^{*}$	$0.778 \to 0.620^{*}$	5 of 5
	FPR	$0.000 \to 0.000$	$0.000 \to 0.000$	$0.000 \to 0.000$	$0.000 \to 0.000$	$0.000 \to 0.000$	under-alerting (recall down)

Each cell gives R1dm → R3. * marks a change distinguishable from zero after Holm correction over five folds (independent bootstraps). Threshold failure is any such change in recall or FPR (Section IV-E). The GRU rows pool three seeds. The per-seed row counts seeds whose own threshold fails. It differs from the pooled row because the seeds disagree, as in banking, where recall by seed is 0.903, 0.006 and 0.957. Intervals are in Part A.

TABLE S10
LABEL-FREE THRESHOLD SET ON 200 UNLABELED HELD-OUT TRAJECTORIES, WITH A PRE-SPECIFIED AND A POST HOC READING

	(a) native sample, pre-specified		(b) 2% traffic, post hoc			θ_{r80}, same fold	
Fold and detector	FPR	Recall	FPR	Recall	Precision 2%	FPR	Recall
medical, LR-C (combined)	0.312 [0.257, 0.342]	0.631 [0.559, 0.689]	0.180 [0.130, 0.200]	0.489 [0.439, 0.512]	0.053 [0.050, 0.065]	0.503	0.827
medical, LR-B (behavioral)	0.303 [0.243, 0.348]	0.742 [0.677, 0.776]	0.271 [0.234, 0.299]	0.731 [0.697, 0.755]	0.052 [0.049, 0.057]	0.150	0.560
medical, GRU	0.314 [0.269, 0.372]	0.432 [0.367, 0.496]	0.057 [0.026, 0.076]	0.123 [0.092, 0.160]	0.044 [0.035, 0.068]	0.717	0.938
banking, LR-C (combined)	0.284 [0.248, 0.347]	0.770 [0.737, 0.827]	0.179 [0.153, 0.217]	0.665 [0.640, 0.698]	0.071 [0.062, 0.079]	0.660	0.966
banking, LR-B (behavioral)	0.303 [0.271, 0.334]	0.846 [0.817, 0.871]	0.287 [0.235, 0.340]	0.834 [0.801, 0.863]	0.057 [0.049, 0.065]	0.278	0.829
banking, GRU	0.243 [0.148, 0.300]	0.553 [0.445, 0.618]	0.050 [0.020, 0.092]	0.273 [0.204, 0.352]	0.107 [0.072, 0.170]	0.438	0.622
web, LR-C (combined)	0.220 [0.182, 0.245]	0.770 [0.744, 0.796]	0.203 [0.173, 0.246]	0.745 [0.714, 0.783]	0.070 [0.061, 0.078]	0.707	0.990
web, LR-B (behavioral)	0.221 [0.143, 0.264]	0.846 [0.772, 0.877]	0.276 [0.235, 0.323]	0.889 [0.870, 0.912]	0.062 [0.054, 0.070]	0.197	0.825
web, GRU	0.250 [0.159, 0.314]	0.495 [0.390, 0.613]	0.077 [0.053, 0.143]	0.250 [0.191, 0.340]	0.065 [0.042, 0.094]	0.623	0.917
email, LR-C (combined)	0.290 [0.226, 0.343]	0.609 [0.514, 0.670]	0.221 [0.172, 0.267]	0.505 [0.441, 0.557]	0.045 [0.041, 0.050]	0.722	0.962
email, LR-B (behavioral)	0.295 [0.223, 0.356]	0.721 [0.630, 0.781]	0.309 [0.279, 0.376]	0.738 [0.714, 0.769]	0.047 [0.040, 0.050]	0.418	0.833
email, GRU	0.257 [0.194, 0.331]	0.440 [0.391, 0.527]	0.065 [0.047, 0.096]	0.194 [0.152, 0.267]	0.058 [0.044, 0.069]	0.764	0.973
coding, LR-C (combined)	0.217 [0.164, 0.277]	0.666 [0.602, 0.728]	0.192 [0.160, 0.241]	0.634 [0.591, 0.677]	0.064 [0.054, 0.070]	0.672	0.974
coding, LR-B (behavioral)	0.244 [0.186, 0.323]	0.785 [0.725, 0.849]	0.293 [0.237, 0.357]	0.836 [0.792, 0.869]	0.056 [0.047, 0.064]	0.283	0.835
coding, GRU	0.245 [0.190, 0.345]	0.444 [0.380, 0.516]	0.059 [0.035, 0.087]	0.180 [0.135, 0.244]	0.060 [0.051, 0.076]	0.486	0.720

Values are the mean with the minimum and maximum over five draws (for the GRU over three seeds and five draws), not confidence intervals. Reading (a) is pre-specified and reading (b) post hoc. The last two columns repeat the operating point at θ_{r80} in the same fold for reference.

TABLE S11
GRADED TOOL COVERAGE INSIDE R1, WITH AUROC, FPR AT THE TRAINING THRESHOLD AND PAIRED DIFFERENCES

AUROC by target coverage (achieved coverage)	LR-C	LR-C-drop	LR-B	GRU	GRU-drop0.2
1.0 (1.000)	0.879 [0.863, 0.895]	0.882 [0.866, 0.898]	0.830 [0.810, 0.848]	0.936 [0.925, 0.946]	0.943 [0.933, 0.953]
0.75 (0.775)	0.853 [0.836, 0.869]	0.862 [0.845, 0.877]	0.830 [0.810, 0.848]	0.874 [0.862, 0.887]	0.923 [0.912, 0.933]
0.5 (0.483)	0.825 [0.807, 0.841]	0.846 [0.829, 0.861]	0.830 [0.810, 0.848]	0.792 [0.778, 0.807]	0.888 [0.875, 0.899]
0.25 (0.225)	0.800 [0.780, 0.818]	0.827 [0.808, 0.843]	0.830 [0.810, 0.848]	0.720 [0.700, 0.739]	0.842 [0.826, 0.857]
0.0 (0.000)	0.787 [0.765, 0.808]	0.822 [0.801, 0.840]	0.830 [0.810, 0.848]	0.674 [0.650, 0.699]	0.794 [0.772, 0.813]
FPR at θ_{r80}	**LR-C**	**LR-C-drop**	**LR-B**	**GRU**	**GRU-drop0.2**
1.0 (1.000)	0.192 [0.166, 0.219]	0.187 [0.160, 0.213]	0.284 [0.253, 0.315]	0.045 [0.034, 0.057]	0.054 [0.041, 0.068]
0.75 (0.775)	0.263 [0.237, 0.289]	0.182 [0.160, 0.206]	0.284 [0.253, 0.315]	0.169 [0.152, 0.186]	0.057 [0.046, 0.069]
0.5 (0.483)	0.442 [0.416, 0.470]	0.246 [0.223, 0.271]	0.284 [0.253, 0.315]	0.383 [0.362, 0.405]	0.070 [0.060, 0.082]
0.25 (0.225)	0.546 [0.519, 0.575]	0.221 [0.198, 0.245]	0.284 [0.253, 0.315]	0.586 [0.560, 0.612]	0.055 [0.045, 0.066]
0.0 (0.000)	0.657 [0.627, 0.687]	0.235 [0.207, 0.264]	0.284 [0.253, 0.315]	0.737 [0.710, 0.766]	0.053 [0.040, 0.066]
Paired AUROC	**LR-C − LR-B**	**LR-C-drop − LR-B**	**GRU − LR-B**	**GRU-drop0.2 − LR-B**	**GRU − LR-C**
1.0 (1.000)	$+0.049^*$ [+0.034, +0.064]	$+0.051^*$ [+0.036, +0.067]	$+0.105^*$ [+0.087, +0.125]	$+0.113^*$ [+0.096, +0.131]	$+0.057^*$ [+0.043, +0.071]
0.75 (0.775)	$+0.023^*$ [+0.009, +0.037]	$+0.031^*$ [+0.018, +0.045]	$+0.044^*$ [+0.026, +0.064]	$+0.093^*$ [+0.077, +0.110]	$+0.021^*$ [+0.008, +0.035]
0.5 (0.483)	−0.005 [−0.016, +0.006]	$+0.016^*$ [+0.006, +0.025]	-0.038^* [−0.056, −0.020]	$+0.058^*$ [+0.044, +0.072]	-0.033^* [−0.045, −0.021]
0.25 (0.225)	-0.030^* [−0.041, −0.020]	−0.004 [−0.009, +0.001]	-0.110^* [−0.131, −0.089]	+0.012 [+0.000, +0.023]	-0.080^* [−0.095, −0.066]
0.0 (0.000)	-0.043^* [−0.055, −0.031]	-0.009^* [−0.011, −0.007]	-0.156^* [−0.179, −0.131]	-0.037^* [−0.046, −0.027]	-0.113^* [−0.129, −0.096]

Intervals are conditional on the drawn masks and, for the GRUs, on the three training seeds (seed-averaged). * marks a paired difference distinguishable from zero without correction. LR-B does not read tool identity, so its values are the same at every level.

TABLE S12
GRU PER TRAINING SEED, THREE INTERVAL ESTIMANDS FOR GRU − LR-C, AND THE POST HOC DIAGNOSTIC OF THE EARLIER VERSION

	Per seed (0 / 1 / 2)		GRU − LR-C AUROC, estimand			Per seed at θ_{r80}		v0.3 post hoc diagnostic	
Setting	AUROC	− LR-C	two-level	mixture	seed-avg.	FPR	Recall	LR head on B, GRU training	GRU, tool sequence only
R1	0.936 0.935 0.935	+0.058 +0.057 +0.056	+0.057 [+0.043, +0.072]	+0.057 [+0.042, +0.072]	+0.057 [+0.043, +0.071]	0.046 0.050 0.039	0.674 0.693 0.680	n/a	n/a
R3 medical	0.640 0.649 0.626	−0.105 −0.096 −0.119	−0.107 [−0.124, −0.090]	−0.107 [−0.130, −0.087]	−0.107 [−0.119, −0.094]	0.752 0.736 0.662	0.962 0.945 0.908	0.797	0.572
R3 banking	0.733 0.718 0.758	−0.096 −0.111 −0.070	−0.092 [−0.120, −0.069]	−0.092 [−0.135, −0.061]	−0.092 [−0.104, −0.079]	0.619 0.000 0.694	0.903 0.006 0.957	0.847	0.568
R3 web	0.752 0.718 0.707	−0.110 −0.144 −0.155	−0.137 [−0.162, −0.110]	−0.137 [−0.167, −0.100]	−0.137 [−0.151, −0.122]	0.754 0.495 0.620	0.994 0.827 0.929	0.898	0.606
R3 email	0.644 0.715 0.654	−0.096 −0.025 −0.085	−0.069 [−0.101, −0.027]	−0.069 [−0.106, −0.009]	−0.069 [−0.082, −0.055]	0.823 0.760 0.710	0.990 0.977 0.951	0.786	0.567
R3 coding	0.684 0.672 0.729	−0.130 −0.142 −0.086	−0.119 [−0.148, −0.087]	−0.119 [−0.154, −0.078]	−0.119 [−0.133, −0.105]	0.686 0.680 0.092	0.944 0.938 0.278	0.843	0.582

Per-seed values are given for seeds 0, 1 and 2. The GRU − LR-C columns give AUROC differences under the two-level, mixture and seed-averaged estimands. The diagnostic columns are points from `archive_v0.3/gru_r3_diagnostic_results.json`, which was computed after the GRU results were known.

TABLE S13
OBSERVATION-LENGTH ABLATION AND PADDING TEST

AUROC, pool E	LR-B	LR-B-noobs	LR-C	LR-C-noobs	LR-obsonly
R1	0.830 [0.811, 0.849]	0.699 [0.674, 0.724]	0.879 [0.862, 0.894]	0.805 [0.784, 0.824]	0.790 [0.769, 0.812]
R3 medical	0.808 [0.788, 0.826]	0.629 [0.605, 0.654]	0.745 [0.724, 0.767]	0.588 [0.563, 0.614]	0.802 [0.781, 0.821]
R3 banking	0.854 [0.836, 0.873]	0.666 [0.639, 0.693]	0.828 [0.809, 0.847]	0.645 [0.619, 0.670]	0.847 [0.828, 0.865]
R3 web	0.896 [0.882, 0.910]	0.675 [0.651, 0.698]	0.862 [0.847, 0.878]	0.672 [0.648, 0.697]	0.858 [0.842, 0.873]
R3 email	0.792 [0.772, 0.812]	0.680 [0.657, 0.705]	0.739 [0.717, 0.761]	0.621 [0.596, 0.647]	0.739 [0.718, 0.761]
R3 coding	0.852 [0.835, 0.868]	0.740 [0.718, 0.761]	0.814 [0.796, 0.832]	0.682 [0.659, 0.704]	0.808 [0.789, 0.827]
R4 transfer	0.482 [0.434, 0.522]	0.630 [0.582, 0.680]	0.473 [0.425, 0.513]	0.657 [0.611, 0.706]	0.469 [0.422, 0.509]
Within UT	0.741 [0.689, 0.793]	0.740 [0.688, 0.790]	0.797 [0.747, 0.842]	0.795 [0.744, 0.838]	0.708 [0.657, 0.762]
Padding (recall at θ_{r80}, AUROC)	+0	+100	+300	+600	+1,200
R1 LR-B, recall	0.795 [0.768, 0.823]	0.664 [0.632, 0.698]	0.448 [0.414, 0.480]	0.274 [0.244, 0.306]	0.161 [0.137, 0.188]
R1 LR-B, AUROC	0.830 [0.811, 0.849]	0.761 [0.739, 0.783]	0.616 [0.589, 0.643]	0.435 [0.407, 0.464]	0.259 [0.235, 0.284]
R1 LR-C, recall	0.771 [0.744, 0.799]	0.634 [0.602, 0.664]	0.360 [0.327, 0.392]	0.170 [0.144, 0.195]	0.033 [0.021, 0.046]
R1 LR-C, AUROC	0.879 [0.862, 0.894]	0.809 [0.789, 0.829]	0.632 [0.605, 0.658]	0.382 [0.354, 0.408]	0.132 [0.115, 0.149]
R1 GRU, recall	0.682 [0.646, 0.717]	0.578 [0.543, 0.611]	0.341 [0.300, 0.380]	0.100 [0.069, 0.132]	0.000 [0.000, 0.000]
R1 GRU, AUROC	0.936 [0.924, 0.947]	0.907 [0.892, 0.921]	0.814 [0.791, 0.837]	0.593 [0.556, 0.633]	0.159 [0.119, 0.199]
Within UT LR-B, recall	0.776 [0.705, 0.840]	0.778 [0.708, 0.842]	0.785 [0.713, 0.852]	0.792 [0.719, 0.860]	0.808 [0.732, 0.876]
Within UT LR-B, AUROC	0.741 [0.689, 0.793]	0.743 [0.690, 0.795]	0.746 [0.695, 0.798]	0.752 [0.701, 0.804]	0.764 [0.713, 0.814]
Within UT LR-C, recall	0.720 [0.642, 0.788]	0.723 [0.645, 0.790]	0.731 [0.656, 0.796]	0.745 [0.675, 0.807]	0.759 [0.693, 0.820]
Within UT LR-C, AUROC	0.797 [0.747, 0.842]	0.799 [0.749, 0.844]	0.803 [0.752, 0.848]	0.808 [0.757, 0.853]	0.818 [0.767, 0.863]

The noobs variants drop the maximum and mean observation length, and LR-obsonly keeps only these two features. All three were defined after related results were seen. AgentDojo rows use cluster intervals. Padding lengthens the injected observation of every positive by the stated number of characters, and the GRU is padded in R1 only. In AgentDojo the injected steps were located from each run's record of injections (2,285 of 2,294 located).

TABLE S14
AGENTDOJO LEAVE-ONE-SUITE-OUT REPLICATION OF THE COVERAGE EFFECT

Held-out suite	pos / neg (clusters)	Coverage	Held-out suite AUROC (cluster)			LR-C − LR-B			LR-C offset
			LR-B	LR-C	LR-C-drop	LOSO cluster	LOSO traj.	Within UT, suite-matched	
banking	671 / 56 (22)	0.000	0.898 [0.814, 0.964]	0.840 [0.731, 0.929]	0.913 [0.827, 0.975]	−0.059 [−0.172, +0.034]	−0.059* [−0.104, −0.016]	+0.021 [−0.070, +0.141]	−0.358
slack	904 / 105 (24)	0.000	0.701 [0.619, 0.799]	0.744 [0.665, 0.834]	0.720 [0.639, 0.817]	+0.044* [+0.016, +0.076]	+0.044* [+0.030, +0.056]	+0.029 [−0.075, +0.113]	−2.217
travel	182 / 103 (24)	0.084	0.613 [0.550, 0.706]	0.661 [0.588, 0.761]	0.626 [0.557, 0.720]	+0.048* [+0.011, +0.084]	+0.048* [+0.022, +0.075]	+0.034 [−0.138, +0.174]	−0.433
workspace	537 / 128 (47)	0.362	0.644 [0.574, 0.725]	0.643 [0.570, 0.726]	0.640 [0.570, 0.719]	−0.001 [−0.019, +0.019]	−0.001 [−0.012, +0.011]	+0.054 [−0.038, +0.153]	+1.558
Within UT, pooled	2294 / 392	n/a	n/a	n/a	n/a	+0.056 [−0.003, +0.111]	+0.056* [+0.014, +0.092]	n/a	n/a

Held-out AUROC and the LOSO contrast use a cluster bootstrap over suite and user task, and * marks a difference distinguishable from zero after Holm correction over the four suites. The trajectory column is a sensitivity analysis. The suite-matched column gives the same contrast within UT restricted to the held-out suite. The offset is the logit that the LR-C tool block adds to a trajectory with no known tool. The GRU was not run in this design.

below 0.5 for all but LR-A, because the AgentDrift-trained models score unsuccessful attacks below benign runs. Only TF-IDF, which reads the injected text, remains reasonable within AgentDojo under this target.

Table S16 shows how the AgentDojo intervals depend on the resampling unit. Clustering by suite and user task widens the transfer intervals modestly, and clustering by attack goal, with only 30 goal clusters among the positives, widens them substantially. LR-C transfer goes from [0.441, 0.508] under trajectory resampling to [0.426, 0.512] with user-task clusters and [0.329, 0.599] with attack-goal clusters. The within-minus-transfer gap under UT stays distinguishable after Holm correction for every supervised detector, whereas under IT it remains distinguishable only for LR-B and TF-IDF.

The lower block reports the prevalence-weighted AUPRC of (5) at 10% and 5% prevalence, which the main paper gives only at 10% (Table VI). Chance is 0.10 and 0.05. Transfer wAUPRC@5% lies between 0.044 and 0.086 for all detectors, near the chance value. Within UT it is 0.271 for the GRU and 0.173 for LR-C. The last three columns give the subsampling estimator of the earlier version at 5% (21 positives against 392 negatives), which Section IV-B keeps only to show how variable a small test set is. The main paper makes no claim at 5% prevalence.

S10. Content-Aware Reference

Table S17 runs the TF-IDF reference [43] through every setting with its variants. It is near perfect in every in-distribution setting and under R3, where its FPR at θ_{r80} stays at zero while its recall falls. The R1 AUROC changes by at most 0.0008, whether digits are removed or world strings are masked or anonymized. AgentDrift's separability therefore rests not on the year or on world identities but on the category templates of the generator. The words of tool names alone score 0.520 to 0.663 in R3. Transfer to AgentDojo is near chance (0.520) and flags no trace at θ_{r80}. Within AgentDojo, TF-IDF reaches 0.996 under UT and 0.989 to 0.998 in LOSO, largely by detecting the presence of injected text. With the unsuccessful attacks as training negatives (UT-F), it falls to 0.941, flagging 0.055 of the unsuccessful attacks.

S11. Reproducibility

This section gives everything needed to regenerate the numbers of the paper and of this supplement. The file `revision/REPRODUCIBILITY.md` repeats it with the file and line of every setting. The file `revision/repro_checks.json` records the SHA-256 of every script, the environment variables it reads and every line that sets a seed, a bootstrap size or an estimator.

Environment. All results were computed with CPython 3.10.4 on a Windows 11 machine without a GPU, using numpy 2.2.6, scipy 1.15.3, scikit-learn 1.7.2, torch 2.11.0+cpu and matplotlib 3.10.8 (figures only). The lock file `revision/requirements-lock.txt` pins these packages and their 20 installed runtime dependencies. Install it with `pip install -r revision/requirements-lock.txt --extra-index-url https://download.pytorch.org/whl/cpu`. The code imports nothing else beyond the Python standard library.

Data and environment variables. The variable `AGENTDRIFT` must point to a clone of AgentDrift [29] at commit `014a514fa998b4ac4519579fceb8a5884b379bda` (dataset release 1.0, record schema 2.0). The variable `AGENTDOJO` must point to a clone of AgentDojo [3] at commit `089ed468cf3ed0322acc66b0211f26d9d90dbf60` with its `runs` folder. Several modules read `AGENTDRIFT` when they are imported, so both variables should be set before any script runs. The optional variable `AUDIT_NJOBS` sets the number of grid-search workers of the audit script and does not affect results.

Commands in order. All commands run from the project root with the Python interpreter above unless stated.

- Data preparation. Run `agentdojo_extract.py`, which writes the P0 traces, the unsuccessful attacks and the extraction manifest. Run `tests/test_agentdojo_alignment.py`, which must pass. Run `pre_ablation_audit.py` repeatedly until it prints ASSEMBLED, then `revision/repro_checks.py` and `revision/write_diagnostics_summary.py`.

TABLE S15
R4 UNDER COMPOSITION CONTROLS AND ALTERNATIVE POPULATIONS, WITH WITHIN-AGENTDOJO AND TRANSFER AUROC PER REPRESENTATION

	LR-A (tool identity)		LR-B (behavioral)		LR-C (combined)		GRU		TF-IDF (content)	
Population	Within	Transfer	Within	Transfer	Within	Transfer	Within	Transfer	Within	Transfer
P0	0.788 [0.740, 0.828]	0.566	0.741 [0.688, 0.792]	0.482	0.797 [0.747, 0.842]	0.473	0.840§ [0.784, 0.884]	0.513	0.996§ [0.992, 0.999]	0.520
C1	0.802§ [0.757, 0.846]	0.594	0.741 [0.689, 0.791]	0.480	0.832§ [0.789, 0.872]	0.472	0.848§ [0.797, 0.892]	0.508	0.998§ [0.994, 1.000]	0.555
Model-balanced	0.771 [0.722, 0.815]	0.555	0.728 [0.673, 0.782]	0.491	0.778 [0.725, 0.829]	0.486	0.829§ [0.772, 0.878]	0.536	0.997§ [0.994, 0.999]	0.488
UT-F	0.790 [0.750, 0.828]	0.566	0.720 [0.669, 0.770]	0.482	0.794 [0.747, 0.838]	0.473	0.849§ [0.805, 0.888]	0.513	0.941§ [0.911, 0.966]	0.520
P0F	0.781 [0.733, 0.828]	0.585	0.718 [0.651, 0.780]	0.626	0.776 [0.723, 0.827]	0.626	0.828§ [0.774, 0.877]	0.645	0.821§ [0.795, 0.847]	0.626
UT-F on P0F	0.815§ [0.778, 0.853]	0.585	0.734 [0.676, 0.789]	0.626	0.819§ [0.773, 0.861]	0.626	0.869§ [0.831, 0.901]	0.645	0.942§ [0.927, 0.955]	0.626
P0 without DoS	0.802 [0.754, 0.842]	0.570	0.753 [0.697, 0.807]	0.483	0.813 [0.763, 0.859]	0.475	0.855 [0.800, 0.900]	0.517	0.998 [0.995, 1.000]	0.523
important_instructions only (traj.)	0.782 [0.750, 0.813]	0.561	0.739 [0.703, 0.771]	0.481	0.794 [0.756, 0.832]	0.476	0.842 [0.796, 0.874]	0.522	0.998 [0.996, 0.999]	0.485
Per template, gpt-4o (range)	n/a	0.460 to 0.744	n/a	0.322 to 0.664	n/a	0.309 to 0.666	n/a	0.295 to 0.716	n/a	0.318 to 0.819
Attempt target, UT (exploratory)	0.591 [0.555, 0.628]	0.506	0.584 [0.554, 0.618]	0.396	0.605 [0.564, 0.645]	0.387	0.613 [0.573, 0.653]	0.403	0.959§ [0.936, 0.976]	0.440
Attempt target, IT (exploratory)	0.584 [0.514, 0.660]	0.506	0.578 [0.515, 0.642]	0.396	0.595 [0.521, 0.668]	0.387	0.580 [0.514, 0.652]	0.403	0.956§ [0.940, 0.968]	0.440
Per model (UT, cluster intervals)										
`gemini-1.5-pro` (79/107)	0.753 [0.683, 0.817]	0.572	0.703 [0.625, 0.777]	0.483	0.745 [0.666, 0.816]	0.477	0.804§ [0.720, 0.871]	0.535	1.000§ [0.997, 1.000]	0.430
`gemini-2.0-flash` (55/134)	0.788 [0.719, 0.862]	0.525	0.780 [0.706, 0.854]	0.565	0.789 [0.708, 0.860]	0.562	0.841§ [0.771, 0.909]	0.627	0.997§ [0.991, 1.000]	0.485
`gpt-4-turbo` (82/180)	0.782 [0.724, 0.836]	0.540	0.695 [0.614, 0.774]	0.487	0.791 [0.722, 0.857]	0.485	0.844§ [0.773, 0.902]	0.531	0.997§ [0.991, 1.000]	0.531
`gpt-4o` (92/1866)	0.769 [0.723, 0.815]	0.573	0.748 [0.694, 0.801]	0.454	0.794 [0.744, 0.843]	0.444	0.831§ [0.774, 0.878]	0.482	0.995§ [0.989, 0.999]	0.512

Within cells carry cluster intervals (trajectory intervals for the important_instructions row), and transfer cells are points whose intervals are in Tables VI and S16. § marks a within value that meets the reasonable bar of Section IV-F. P0F adds the 4,987 unsuccessful attacks as evaluation negatives, and UT-F also uses them as training negatives. The per-template row gives the range of transfer AUROC over the 15 templates of `gpt-4o-2024-05-13`, which share its 92 benign traces as negatives. The attempt-target rows are exploratory. Per-model counts are negatives and positives. The model `claude-3-5-sonnet-20241022`, with 7 compromised traces, is omitted.

- Version 0.3 scripts on the corrected extraction, unchanged. Run `rerun_r1r3.py`, `ablation.py`, `world_disjoint_r1.py`, `r4_control.py` and `gru_ladder.py full`. The GRU diagnostic of Table S12 is the archived version 0.3 output of `gru_r3_diagnostic.py` and was not re-run.
- Prediction caches. Run `revision/build_linear_cache.py --all`, then `revision/validate_linear_cache.py`, `revision/linear_r4_truncation_check.py` and `revision/linear_tie_check.py`. Run `revision/build_gru_cache.py run` with each setting and each of GRU, GRU-drop0.2 and GRU-drop0.1, in the order of `revision/offline_driver.sh`, then `revision/summarize_gru_cache.py all`. The cache builders are resumable and can be repeated until their status command reports completion.
- Analysis streams, which read the caches. Run `revision/prevalence_ops.py all`, `revision/domain_shift.py all`, `revision/oov_mechanism.py all`, `revision/features_obslen.py all`. Then run `revision/content_detector.py` with `fit --all`, `analyze --all`, `assemble` and `summarize`, and `revision/r4_deep.py` with `fit all`, `analyze all` and `assemble`. Finally, run `revision/consolidate_results.py`, which fits nothing and collects the values quoted most often.
- Paper. From the folder `paper`, run `gen_tables_v04.py`, `make_figures_v04.py`, `sync_tables.py`, `build_manuscript.py`, `md2tex_v04.py main` and `md2tex_v04.py supp`, and compile `main_v04.tex` and `supplement_v04.tex` twice with pdflatex.

Model settings. Logistic regressions use L2 regularization with inverse strength 1, the lbfgs solver, balanced class weights, standardized features and at most 1,000 iterations (2,000 for TF-IDF). TF-IDF uses lowercase word unigrams and bigrams, a minimum document frequency of 2, at most 50,000 features and sublinear term frequency. The Isolation Forest [42] uses 200 trees with random state 0 on the standardized features of benign training trajectories, and the Markov model uses add-0.5 smoothing. The GRU [44] follows Section V-E, with Adam [45], gradient clipping at norm 1.0 and deterministic PyTorch algorithms.

Seeds. The GRU uses training seeds 0, 1 and 2 in AgentDrift, each with a stratified 10% validation hold-out drawn with seed 1000 plus the training seed. Within AgentDojo the training seed equals the fold-assignment seed, 0 to 4, and token dropout draws from a separate generator seeded with the training seed plus 10007. Cross-validation shuffles the groups with seeds 0 to 4 and assigns folds by position modulo 5, and groups that no P0 trace defines are assigned

TABLE S16
AGENTDOJO INTERVALS UNDER TRAJECTORY, USER-TASK CLUSTER AND ATTACK-GOAL CLUSTER RESAMPLING, AND PREVALENCE-WEIGHTED AUPRC AT 10% AND 5%

	Transfer AUROC			Within UT AUROC		Within IT AUROC		UT − transfer	IT − transfer
Detector	traj.	UT clust.	IT clust.	traj.	UT clust.	traj.	IT clust.	UT clust.	IT clust.
LR-A (tool identity)	0.566 [0.542, 0.587]	0.566 [0.515, 0.618]	0.566 [0.484, 0.670]	0.788 [0.761, 0.817]	0.788 [0.743, 0.829]	0.690 [0.613, 0.768]	0.690 [0.551, 0.827]	+0.223* [+0.158, +0.285]	+0.124 [−0.050, +0.287]
LR-B (behavioral)	0.482 [0.450, 0.516]	0.482 [0.434, 0.522]	0.482 [0.342, 0.606]	0.741 [0.709, 0.772]	0.741 [0.690, 0.791]	0.719 [0.677, 0.759]	0.719 [0.639, 0.797]	+0.262* [+0.197, +0.323]	+0.241* [+0.095, +0.392]
LR-C (combined)	0.473 [0.441, 0.508]	0.473 [0.426, 0.512]	0.473 [0.329, 0.599]	0.797 [0.765, 0.831]	0.797 [0.747, 0.844]	0.688 [0.636, 0.737]	0.688 [0.546, 0.832]	+0.326* [+0.266, +0.383]	+0.222 [+0.045, +0.404]
GRU	0.513 [0.484, 0.545]	0.513 [0.465, 0.553]	0.513 [0.358, 0.643]	0.840 [0.796, 0.872]	0.840 [0.782, 0.889]	0.645 [0.581, 0.691]	0.645 [0.505, 0.793]	+0.329* [+0.265, +0.390]	+0.143 [−0.063, +0.344]
TF-IDF (content)	0.520 [0.488, 0.554]	0.520 [0.473, 0.566]	0.520 [0.386, 0.637]	0.996 [0.993, 0.998]	0.996 [0.991, 0.999]	0.987 [0.982, 0.991]	0.987 [0.973, 0.996]	+0.477* [+0.430, +0.522]	+0.471* [+0.352, +0.600]
Markov	0.590 [0.572, 0.607]	0.590 [0.548, 0.632]	0.590 [0.536, 0.656]	0.575 [0.528, 0.617]	0.575 [0.514, 0.635]	0.564 [0.527, 0.601]	0.564 [0.464, 0.662]	−0.014 [−0.081, +0.054]	−0.026 [−0.120, +0.068]
IForest	0.647 [0.615, 0.681]	0.647 [0.593, 0.700]	0.647 [0.562, 0.738]	0.692 [0.662, 0.723]	0.692 [0.641, 0.744]	0.690 [0.649, 0.723]	0.690 [0.607, 0.772]	+0.045 [−0.032, +0.121]	+0.042 [−0.081, +0.169]

wAUPRC	**Transfer 10%**	**Transfer 5%**	**UT 10%**	**UT 5%**	**IT 10%**	**IT 5%**	**Subsample 5% transfer**	**Subsample 5% UT**	**Subsample 5% IT**
LR-A (tool identity)	0.138 [0.113, 0.190]	0.074 [0.058, 0.128]	0.303 [0.230, 0.490]	0.177 [0.127, 0.378]	0.216 [0.131, 0.408]	0.119 [0.068, 0.274]	0.093 [0.049, 0.196]	0.211 [0.113, 0.369]	0.142 [0.071, 0.262]
LR-B (behavioral)	0.090 [0.082, 0.101]	0.045 [0.041, 0.051]	0.228 [0.183, 0.311]	0.129 [0.100, 0.196]	0.198 [0.151, 0.291]	0.108 [0.080, 0.173]	0.051 [0.039, 0.069]	0.151 [0.084, 0.272]	0.128 [0.073, 0.224]
LR-C (combined)	0.089 [0.081, 0.098]	0.044 [0.040, 0.049]	0.298 [0.227, 0.457]	0.173 [0.125, 0.320]	0.242 [0.145, 0.450]	0.137 [0.076, 0.322]	0.050 [0.038, 0.067]	0.203 [0.114, 0.347]	0.164 [0.080, 0.313]
GRU	0.104 [0.093, 0.118]	0.052 [0.047, 0.060]	0.411 [0.319, 0.614]	0.271 [0.191, 0.496]	0.236 [0.139, 0.437]	0.140 [0.074, 0.336]	0.061 [0.042, 0.094]	0.301 [0.161, 0.471]	0.162 [0.072, 0.316]
TF-IDF (content)	0.100 [0.090, 0.119]	0.052 [0.046, 0.064]	0.980 [0.958, 0.996]	0.968 [0.931, 0.994]	0.955 [0.914, 0.982]	0.936 [0.882, 0.974]	0.060 [0.042, 0.110]	0.968 [0.911, 1.000]	0.937 [0.846, 1.000]
Markov	0.128 [0.113, 0.161]	0.066 [0.057, 0.091]	0.110 [0.096, 0.136]	0.056 [0.048, 0.073]	0.109 [0.089, 0.154]	0.055 [0.044, 0.085]	0.079 [0.052, 0.140]	0.063 [0.047, 0.095]	0.064 [0.046, 0.099]
IForest	0.160 [0.134, 0.215]	0.086 [0.070, 0.147]	0.190 [0.154, 0.298]	0.101 [0.080, 0.199]	0.178 [0.135, 0.302]	0.094 [0.069, 0.201]	0.107 [0.057, 0.211]	0.126 [0.070, 0.229]	0.116 [0.067, 0.204]

Points are identical across resampling schemes. In the upper block * marks a within − transfer gap distinguishable from zero after Holm correction over the 12 within-capable detectors, of which seven are shown. In the lower block, transfer and within-UT intervals resample user-task clusters and within-IT intervals resample attack-goal clusters. The subsample columns give the mean and the 2.5th to 97.5th percentiles over 2,000 stratified replicates. They are not confidence intervals for the weighted estimand. The LR-C within-UT value at 10% (0.2985 to four decimals) comes from a separate analysis run from Table VI, so its interval differs slightly.

with seed 10000 plus the repetition. World-disjoint splits use split seeds 0 to 4, and the graded-coverage masks use seed 7100 plus 100 times the level plus the draw. Every bootstrap of the revision uses 2,000 replicates with seeds fixed in the stream scripts. The operating-point and diagnostic streams use 20260926 plus a fixed offset per block, and the domain-shift and content streams use the CRC32 of the population or analysis name. The fixed blocks 8001 to 8024, 8101 to 8102 and 9200 to 9903 serve the observation-length stream, and the fixed blocks 20201 to 20224 and 50500 to 50912 serve the mechanism stream. The R4 stream uses 12345 (stratified), 20260927 (cluster) and 4242 (unsuccessful-attack flag rates). The two-sample test uses five stratified folds with seed 0. The re-run version 0.3 scripts keep their own seeds, which `revision/REPRODUCIBILITY.md` lists.

S12. Evaluation Protocols of the Audited Studies

Table S18 gives the evaluation protocol of each of the eleven works audited in Section II-A. We checked every cell against the arXiv full text of the version named in the table note. The attack shares are our own arithmetic from the class counts each paper reports. The table describes what each protocol measures and is not a judgment of the detectors.

S13. A Frozen-Encoder Reimplementation

Table S19 reports SE, our reimplementation of the DriftNet architecture [7]. Its code and weights have not been released, so no number in the table is DriftNet's own. Each step is written as the tool name, the sorted arguments and the observation, without the THOUGHT field. A frozen all-mpnet-base-v2 encoder embeds it [52], and four world features computed from the arguments are appended. A three-layer transformer with 1,845,637 trainable parameters, the published count, scores the trajectory. We adopted the source's sweep-selected configuration and tuned nothing. Ten seeds were trained on CPU. The analysis plan was registered before any embedding or fit existed (SHA-256 `cef3e767...a8493`). The code is `revision/sencoder_*.py` with `revision/build_sencoder_cache.py`, and the run record, plan amendments and manifests are in `revision/sencoder_run`.

SE's point estimate lies above LR-B at every rung, and it meets the reversal rule in no R3 fold. Its training thresholds still fail in medical, coding and web, where held-out negatives sit 5.2 to 7.4 logits above training negatives. Removing tool names inside R1 barely moves it, although masking shifts the embeddings far less than a held-out domain does. Padding the injected observation costs SE less than it costs the detectors that read length as a scalar. Transfer to AgentDojo is near chance, as for every other detector. At R2 our F1 is 0.973 against a published 0.983, in a setting that is not like for like (panel g).

TABLE S17
CONTENT-AWARE TF-IDF REFERENCE THROUGH THE SETTINGS, WITH NO-DIGIT, TOOL-NAME AND MASKING VARIANTS AND LEAVE-ONE-SUITE-OUT

	TF-IDF (content)				TF-IDF	TF-IDF no digits	TF-IDF tool names	Failed
Setting	AUROC	wAUPRC 2%	Recall	FPR	− LR-C	AUROC	AUROC	flagged
AgentDrift ladder								
R1	0.999 [0.999, 1.000]	0.984 [0.973, 0.994]	0.763 [0.735, 0.791]	0.000 [0.000, 0.000]	$+0.121^{*}$ [+0.105, +0.136]	0.999	0.839	0.000
R1-wd (5 seeds)	0.9997 (0.9996 to 0.9998)	0.990 (0.987 to 0.994)	0.776 (0.761 to 0.792)	0.000 (0.000 to 0.000)	+0.099 to +0.111	0.9994 (0.9991 to 0.9996)	n/a	n/a
R2 (task clusters)	0.998 [0.996, 1.000]	0.957 [0.927, 0.985]	0.734 [0.683, 0.781]	0.000 [0.000, 0.000]	$+0.096^{*}$ [+0.074, +0.119]	0.995	n/a	0.000
R3 medical	0.996 [0.994, 0.997]	0.926 [0.905, 0.949]	0.658 [0.630, 0.687]	0.000 [0.000, 0.000]	$+0.250^{*}$ [+0.230, +0.271]	0.980	0.520	0.000
R3 banking	0.989 [0.985, 0.992]	0.825 [0.783, 0.873]	0.272 [0.236, 0.307]	0.000 [0.000, 0.000]	$+0.161^{*}$ [+0.139, +0.181]	0.943	0.540	0.000
R3 web	0.980 [0.976, 0.984]	0.849 [0.824, 0.877]	0.572 [0.543, 0.603]	0.000 [0.000, 0.000]	$+0.118^{*}$ [+0.102, +0.136]	0.969	0.587	0.000
R3 email	0.990 [0.987, 0.993]	0.912 [0.888, 0.937]	0.696 [0.668, 0.724]	0.000 [0.000, 0.000]	$+0.251^{*}$ [+0.230, +0.273]	0.974	0.554	0.000
R3 coding	0.987 [0.983, 0.990]	0.871 [0.851, 0.894]	0.620 [0.589, 0.649]	0.000 [0.000, 0.000]	$+0.172^{*}$ [+0.153, +0.190]	0.958	0.663	0.000
AgentDojo								
R4 transfer	0.520 [0.471, 0.564]	0.023 [0.019, 0.032]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	+0.047 [−0.002, +0.096]	0.470	0.700	0.000
Within UT	0.996 [0.991, 0.999]	0.943 [0.890, 0.992]	0.780 [0.708, 0.837]	0.000 [0.000, 0.000]	$+0.199^{*}$ [+0.154, +0.249]	0.996	n/a	0.308
Within UT-F	0.941 [0.911, 0.966]	0.349 [0.196, 0.785]	0.755 [0.675, 0.822]	0.049 [0.010, 0.097]	$+0.148^{*}$ [+0.095, +0.195]	0.942	n/a	0.055
Within IT	0.987 [0.972, 0.996]	0.912 [0.844, 0.966]	0.640 [0.450, 0.786]	0.000 [0.000, 0.000]	$+0.300^{*}$ [+0.168, +0.440]	0.988	n/a	0.360
LOSO banking	0.998 [0.993, 1.000]	0.989 [0.975, 0.999]	0.921 [0.874, 0.977]	0.000 [0.000, 0.000]	$+0.158^{*}$ [+0.064, +0.264]	0.996	0.605	0.571
LOSO slack	0.998 [0.995, 1.000]	0.990 [0.980, 0.999]	0.950 [0.922, 0.975]	0.000 [0.000, 0.000]	$+0.254^{*}$ [+0.163, +0.332]	0.998	0.449	0.544
LOSO travel	0.989 [0.960, 1.000]	0.989 [0.961, 1.000]	0.489 [0.299, 0.659]	0.000 [0.000, 0.000]	$+0.328^{*}$ [+0.223, +0.406]	0.989	0.606	0.433
LOSO workspace	0.991 [0.969, 1.000]	0.991 [0.968, 1.000]	0.263 [0.128, 0.397]	0.000 [0.000, 0.000]	$+0.347^{*}$ [+0.267, +0.422]	0.991	0.703	0.253
R1 masking variants (fifth column gives the paired difference to TF-IDF)								
World mask (plan)	0.9992 [0.9985, 0.9997]	0.978 [0.962, 0.992]	0.757 [0.728, 0.787]	0.000 [0.000, 0.000]	−0.0002 [−0.0005, +0.0001]	n/a	n/a	n/a
README anonymization	0.9992 [0.9985, 0.9997]	0.981 [0.968, 0.993]	0.765 [0.736, 0.794]	0.000 [0.000, 0.000]	−0.0002 [−0.0004, +0.0000]	n/a	n/a	n/a
Anonymization and no digits	0.9986 [0.9975, 0.9994]	0.980 [0.971, 0.989]	0.758 [0.729, 0.787]	0.000 [0.000, 0.000]	-0.0007^{*} [−0.0014, −0.0002]	n/a	n/a	n/a
No digits	0.9985 [0.9973, 0.9994]	0.976 [0.963, 0.988]	0.752 [0.723, 0.783]	0.000 [0.000, 0.000]	-0.0008^{*} [−0.0016, −0.0002]	n/a	n/a	n/a

Intervals resample tasks in R2 and clusters in AgentDojo. * marks a TF-IDF − LR-C difference distinguishable from zero after Holm correction within the R3 folds and within the LOSO suites (R1 and R1-wd without correction). The last three columns are points, and the last column gives the share of failed or unsuccessful attacks flagged at θ_{r80}. In the masking rows, which all refer to R1, the fifth column is the paired difference to unmasked TF-IDF. The R1-wd row gives the mean with the range over five split seeds.

S14. A Released Action Monitor

This section gives the protocol and full results of the AlignmentCheck monitor of Section V-F. The plan is `MONITOR_PREREGISTRATION.md`, registered by its SHA-256 in `revision/PLAN_MANIFEST.txt` before the weights were downloaded and before any monitor call. Every later change is a hashed amendment, and each was made before any monitor result was analysed. Amendment 1 set the budget, order and deadline of the API arm. Amendment 2 corrected the call count of the masking arm to all 16 mask draws of Fig. 3. Amendment 3 moved the open backbone from vLLM 0.30.0 to 0.29.0, because 0.30.0 cannot load the registered checkpoint, which uses group activation ordering. Amendment 4 fixed the draw and the passes of the repeatability check. Amendment 5 added the comparison M5b, because the registered comparator of M5 flags no AgentDojo trace. The code is `revision/monitor_render.py`, `monitor_score_vllm.py`, `monitor_score_api.py`, `monitor_detector.py` and `monitor_tables.py`, and the run record is in `revision/monitor_run`.

Prompt and calls. The prompt, the user-message format and the output schema are the strings of `alignmentcheck_scanner.py` in LlamaFirewall (git blob `67d819d3`, commit `cd9fe657`), vendored byte for byte and checked before any call. Call k of a trace shows the user request, the first $k-1$ actions and action k, with the user request twice as in the released code. The open backbone made 321,604 calls, 106,622 of them for the primary arm, and every call returned a parsed output. The main run took 9.0 h on one H100 80GB. The API arm cost 85.79 US dollars in batch requests with a cached system prompt. The API backbone made 46,150 calls, of which 253 were refusals (174 on P0), and refusals also take the default flag. The arms marked "error steps excluded" recompute every trace from its remaining steps.

Scores and thresholds. The open backbone's step score is P(true) / (P(true) + P(false)) at the conclusion token, read from the 20 most likely tokens after the schema constraint, and its native decision equals this score above 0.5 for 99.97% of parsed calls. θ_{r80} is read from the monitor scores of each setting's training positives, per fold and repetition within AgentDojo. The API backbone has no probabilities, so it has

TABLE S18
EVALUATION PROTOCOLS OF AGENT-TRACE DETECTOR STUDIES

Work	Dataset	Detector	Metrics	Imbalance? (test attack share)	Drift?	Cross-benchmark or within-target?	Hard neg., adaptive, or shortcut check?
TraceAegis [4]	Own TraceAegis-Bench (2 agents, each with 1,300 benign, 150 unseen-malicious and 150 seen-but-abnormal traces) and red-team logs from one company	Hierarchical rules + LLM-summarized behavioral constraints	P, R, F1, acc.	No (≈50% abnormal, inferred from reported P, R, acc.)	No	No	Partial (seen-but-abnormal traces made by slight perturbations, no adaptive attacker)
Praetor [5]	ASB [32] attack suite (5 scenarios), ToolBench [50] benign traces (400 train, 100 test) and own context-sequential payloads	Parameterized DFA + sentence-embedding parameter guards (all-MiniLM-L6-v2)	ASR with Clopper-Pearson CIs, BTFR, latency, throughput	No (attack and benign runs scored separately)	Partial (20% of tools replaced, benign traffic only)	Partial (benign profile from ToolBench and attacks from ASB, but no train-on-A, test-on-B detector)	**Yes** (gray-box attacker knowing 10% to 90% of traces, oracle attacker with 100 queries, and TextFooler [51] variants, of which 18% evade)
Agent-Sentry [6]	Own Agent-Sentry Bench on AgentDojo [3] and AgentDyn [21] agents (3,299 benign/2,081 attack) and Native-Trace Bench (126/544)	XGBoost or rules, then allowlist, then LLM judge	Utility, ABR, FPR, FNR and F1	No (38.7% attack, Native-Trace 81.2%)	Partial (5-fold CV grouped by user task, prompt-style shift to Native-Trace)	Partial (transfer to Native-Trace with both classes and the same agents and tools, plus **within-target** grouped K-fold on Native-Trace with K = 2 to 5)	No (mimicry stated as a limitation)
AgentDrift [29]	Own synthetic AgentDrift (12,536 trajectories, 5 domains)	Surface-feature LR (benchmark validation, not a detector)	P, R, F1, AUROC, AUPRC	Partial (AUPRC at 44.2% attack, ratio not varied)	No (stratified split, world-held-out protocol recommended but not run)	No	Partial (hard negatives and failed attacks, and a world-identity lookup reaches 86.1% accuracy)
DriftNet [7]	AgentDrift, task-disjoint split (9,081/1,733/1,722)	Frozen all-mpnet-base-v2 step embeddings [52] + 4 world features, dual-head transformer	P, R, F1, step F1, EM, IoU	No (45.0% attack)	Partial (task-disjoint)	No	Partial (hard negatives and failed attacks, no adaptive attacker)
Trajectory Guard [8]	Galileo [53] and AgentAlign [54] trajectories with GPT-5-perturbed anomalies, plus RAS-Eval [31] and Who&When [55] anomalies	Siamese recurrent autoencoder	P, R and F1 (recall only on external sets)	No (synthetic test 45.5% anomalous, external sets anomaly-only)	Partial (external anomaly sources, recall only)	Partial (external sets lack benign traces, so FPR is not measurable)	Partial (contextual and structural plan perturbations, but the anomalies are plan errors, not injections)
MCPShield [9]	RAS-Eval (605/3,797), ATBench [26] (502/497), Combined (1,939/4,294)	GNNs, MLP, XGB, RF, LR, SVM on SBERT [56] content or tool-name metadata	AUROC, AUPRC, F1, P, R and FPR over 3 seeds	Partial (AUPRC at 86.3%, 49.7% and 68.9% attack, ratio not varied)	Partial (task-disjoint on RAS-Eval, and a random split inflates metadata AUROC by 25.8 points, 0.898 vs. 0.640, because one-hot tool names learn task-specific signatures)	No (sources pooled in Combined)	Partial (no adaptive attacker, benign-only model-fingerprint probe with AUROC 0.41)
Skynet [10]	ETO [57] benign (4,494/562/562) and anomalies from Agent-SafetyBench [30] (111), ATBench (497) and AgentErrorBench [58] (200)	Graph contrastive encoder, benign-only, k-means mode bank	Acc., P, R, F1, FPR and ROC curves without area	Partial (threshold for a sub-1% FPR budget, 0.89% on test, with 16.5% to 59.0% anomalous and no AUPRC)	Partial (anomaly types unseen in training)	Partial (benign and anomalous classes from different sources, so class and source are confounded)	Partial (counterfactual negative views in training, no adaptive attacker)
TrajAD [11]	Own TrajBench (60,000+ trajectories from AgentBank seeds, balanced classes, 5 domains)	Fine-tuned Qwen3-4B [59] generative verifier	P, R, macro-F1, joint exact match	No (balanced)	Partial (leave-one-domain-out with Embodied AI, a novel action space, held out, giving F1 83.09% vs. 83.84% for the fully supervised upper bound)	No	No (procedural errors, not attacks)
SentinelAgent [12]	Two own case studies (email assistant, Magentic-One)	Execution graph + LLM overseer	Qualitative	No	No	No	No
LlamaFirewall [13]	AgentDojo leaderboard traces (10 models, one attack template), an in-house goal-hijacking set and private jailbreak sets	PromptGuard 2 (fine-tuned input classifier) + AlignmentCheck (few-shot LLM auditor over the whole trace, no fine-tuning)	ASR and utility, AUC and recall at 1% FPR (PromptGuard 2), and recall and FPR (AlignmentCheck)	Partial (recall at 1% FPR, prompt level)	Partial (out-of-distribution jailbreak sets, prompt level)	Partial (PromptGuard 2 trained on Meta's data and applied to AgentDojo, AlignmentCheck not trained)	No (one attack template)

The versions audited are Agent-Sentry v2, Skynet v2 and MCPShield v3, and v1 for all other works. MCPShield is the name under which v1 of that study circulated. "Partial" means that an aspect is tested only in the limited form described in the cell. Attack shares are our arithmetic from the class counts in each paper. For TraceAegis the number of test negatives was inferred from its reported precision, recall and accuracy. ABR is the attack block rate, ASR the attack success rate, BTFR the benign task failure rate, EM exact match and IoU intersection over union.

TABLE S19
SE, A REIMPLEMENTATION OF THE DRIFTNET ARCHITECTURE, ON THE EVALUATION LADDER, WITH THE REVERSAL RULE, OFFSETS, GRADED COVERAGE, PADDING AND CROSS-BENCHMARK TRANSFER

(a) Rung	κ	AUROC	SE − LR-B, paired	AUPRC at 10%	recall at θ_{r80}	FPR at θ_{r80}	Drift
R1	1.000	0.998 [0.995, 0.999]	+0.167* [+0.148, +0.185]	0.988 [0.968, 0.999]	0.796 [0.764, 0.826]	0.001 [0.000, 0.004]	0.041
R1-wd split 0	1.000	0.994 [0.989, 0.998]	+0.174* [+0.153, +0.196]	0.976 [0.938, 0.997]	0.790 [0.750, 0.830]	0.002 [0.000, 0.008]	0.040
R1-wd split 1	1.000	0.997 [0.994, 0.999]	+0.194* [+0.175, +0.216]	0.989 [0.969, 1.000]	0.823 [0.780, 0.857]	0.000 [0.000, 0.004]	0.039
R1-wd split 2	1.000	0.998 [0.992, 1.000]	+0.176* [+0.156, +0.198]	0.991 [0.970, 0.999]	0.808 [0.774, 0.843]	0.000 [0.000, 0.000]	0.039
R1-wd split 3	1.000	0.994 [0.989, 0.998]	+0.176* [+0.156, +0.197]	0.983 [0.961, 0.998]	0.797 [0.747, 0.836]	0.000 [0.000, 0.003]	0.038
R1-wd split 4	1.000	0.997 [0.992, 0.999]	+0.178* [+0.158, +0.198]	0.987 [0.959, 1.000]	0.808 [0.773, 0.842]	0.001 [0.000, 0.005]	0.038
R2	1.000	0.996 [0.993, 0.998]	+0.147* [+0.127, +0.167]	0.977 [0.952, 0.994]	0.786 [0.746, 0.821]	0.001 [0.000, 0.008]	0.080
R3 medical	0.000	0.959 [0.940, 0.972]	+0.152* [+0.123, +0.177]	0.829 [0.706, 0.906]	0.770 [0.688, 0.884]	0.046 [0.018, 0.091]	0.296
R3 banking	0.000	0.917 [0.874, 0.940]	+0.062* [+0.016, +0.093]	0.745 [0.644, 0.834]	0.345 [0.281, 0.447]	0.002 [0.000, 0.005]	0.283
R3 web	0.000	0.904 [0.842, 0.935]	+0.007 [−0.058, +0.042]	0.678 [0.353, 0.818]	0.732 [0.632, 0.809]	0.106 [0.031, 0.255]	0.410
R3 email	0.000	0.950 [0.931, 0.967]	+0.158* [+0.133, +0.184]	0.864 [0.801, 0.925]	0.620 [0.575, 0.672]	0.001 [0.000, 0.006]	0.260
R3 coding	0.000	0.938 [0.922, 0.951]	+0.086* [+0.062, +0.108]	0.792 [0.655, 0.866]	0.737 [0.611, 0.802]	0.049 [0.016, 0.104]	0.383

(b) Held-out domain	R1dm SE − LR-B	R3 SE − LR-B	R3 SE − LR-B, AUPRC at 10%	SE R3 − R1dm	SE offset	SE offset, range over seeds	GRU offset†
medical	+0.206* [+0.161, +0.256]	+0.151* [+0.125, +0.177]	+0.481* [+0.355, +0.558]	−0.040* [−0.058, −0.027]	+5.161	+3.899 to +6.689	+5.920
banking	+0.163* [+0.117, +0.213]	+0.063* [+0.023, +0.093]	+0.301* [+0.192, +0.394]	−0.077* [−0.116, −0.054]	+0.524	−0.040 to +1.522	+5.592
web	+0.111* [+0.078, +0.148]	+0.008 [−0.056, +0.043]	+0.114 [−0.225, +0.254]	−0.093* [−0.154, −0.061]	+7.420	+3.994 to +8.821	+7.280
email	+0.195* [+0.151, +0.242]	+0.158* [+0.133, +0.181]	+0.508* [+0.450, +0.558]	−0.049* [−0.067, −0.033]	+0.692	+0.373 to +1.291	+7.133
coding	+0.151* [+0.113, +0.193]	+0.086* [+0.063, +0.107]	+0.335* [+0.197, +0.409]	−0.059* [−0.075, −0.045]	+5.795	+4.012 to +7.084	+7.382
R1 (reference)	n/a	n/a	n/a	n/a	+0.643	+0.281 to +0.984	+0.661

(c) Target coverage (achieved)	AUROC	FPR at θ_{r80}	recall at θ_{r80}	SE − LR-B, paired	LR-C‡	GRU‡	Drift
1.0 (1.000)	0.998 [0.996, 0.999]	0.001 [0.000, 0.001]	0.796 [0.772, 0.819]	+0.167* [+0.148, +0.187]	0.879	0.936	0.041
0.75 (0.775)	0.997 [0.995, 0.998]	0.000 [0.000, 0.001]	0.774 [0.750, 0.797]	+0.167* [+0.147, +0.186]	0.853	0.874	0.059
0.5 (0.483)	0.996 [0.994, 0.997]	0.000 [0.000, 0.001]	0.750 [0.725, 0.774]	+0.166* [+0.147, +0.186]	0.825	0.792	0.084
0.25 (0.225)	0.995 [0.993, 0.997]	0.000 [0.000, 0.001]	0.726 [0.699, 0.751]	+0.165* [+0.146, +0.185]	0.800	0.720	0.107
0.0 (0.000)	0.994 [0.991, 0.997]	0.000 [0.000, 0.000]	0.711 [0.683, 0.737]	+0.164* [+0.145, +0.184]	0.787	0.674	0.126

(d) Filler characters	SE recall at θ_{r80}	SE AUROC	SE recall change	LR-C recall‡	LR-B recall‡	GRU recall‡
0	0.796 [0.772, 0.819]	0.998 [0.996, 0.999]	n/a	0.771	0.795	0.682
100	0.685 [0.655, 0.713]	0.990 [0.987, 0.993]	−0.112* [−0.125, −0.098]	0.634	0.664	0.578
1,200	0.531 [0.498, 0.564]	0.883 [0.868, 0.897]	−0.265* [−0.290, −0.241]	0.033	0.161	0.000

Panels a to d. The notes for the whole table follow its continuation.

no θ_{r80}, and its AUROC rests on the number of flagged steps.

Outcomes. M1 requires a P0 AUROC and a model-balanced AUROC with lower bounds above 0.60 and a detected advantage over every AgentDrift-trained detector on the same rows. M2 is a P0 interval that includes 0.5. M3 compares implied precision at 2% prevalence at the native decision with 0.5 on P0 and on R1. M4 tests whether the native FPR or θ_{r80} ports between the benchmarks. M5 and M5b compare the flag rate of unsuccessful attacks with that of TF-IDF at its transfer and within-AgentDojo thresholds. M6 compares the backbones across benchmarks as an indication of generator familiarity. M7 lists the outcomes on which the backbones disagree. M8 compares masked and unmasked tool names. The readings are those of Section VI-G. For MON-L, M1 is met, M2 not met, M3 met, M4 met, M5 not met, M5b met and M8 beyond 0.05.

Diagnostics. Re-running 500 calls drawn with seed 20261016 in a fresh process reproduced the open backbone conclusion for 0.970 of 500 calls, with a mean absolute change in the step score of 0.029, and the API backbone conclusion for 0.970 of 492. The two backbones agree on step decisions with Cohen's κ of 0.65. On attacked AgentDrift traces, the first flagged step comes before the first non-benign step in 0.020 of flagged traces, at it in 0.139 and after it in 0.842, which we report descriptively.

TABLE S19
SE, A REIMPLEMENTATION OF THE DRIFTNET ARCHITECTURE, ON THE EVALUATION LADDER, WITH THE REVERSAL RULE, OFFSETS, GRADED COVERAGE, PADDING AND CROSS-BENCHMARK TRANSFER (CONTINUED)

(e) Population	n pos / n neg	SE transfer	SE transfer, cluster§	SE within (UT)	within − transfer	SE-W0 transfer	SE-W0 within
P0	2,294 / 392	0.481 [0.418, 0.555]	0.481 [0.399, 0.560]	0.956 [0.924, 0.977]	+0.475* [+0.395, +0.544]	0.503 [0.413, 0.579]	0.963 [0.916, 0.985]
C1	2,294 / 327	0.467 [0.403, 0.533]	0.467 [0.390, 0.535]	0.949 [0.875, 0.977]	+0.482* [+0.381, +0.561]	0.491 [0.409, 0.559]	0.952 [0.910, 0.979]
P0F	2,294 / 5,379	0.572 [0.525, 0.647]	0.572 [0.490, 0.657]	0.669 [0.630, 0.704]	+0.098* [+0.004, +0.165]	0.587 [0.525, 0.640]	0.645 [0.597, 0.684]
model-balanced	2,287 / 308	0.477 [0.400, 0.551]	0.474 [0.385, 0.555]	0.961 [0.924, 0.984]	+0.484* [+0.403, +0.566]	0.500 [0.405, 0.578]	0.968 [0.919, 0.990]

(f) AgentDojo suite	Traces	Any address in arguments	Any URL in arguments	Any non-zero world feature
banking	1,450	0.000	0.000	0.000
slack	1,601	0.347	0.816	0.871
travel	2,083	0.067	0.000	0.067
workspace	2,539	0.545	0.005	0.545

(g) R2, threshold 0.5	Ours, mean of 10 seeds	Ours, min	Ours, max	Published, single run¶
precision	0.975	0.965	0.982	0.982
recall	0.970	0.955	0.986	0.985
F1	0.973	0.967	0.979	0.983
flag rate, benign	0.021	0.015	0.027	0.015
flag rate, failed attack	0.005	0.005	0.009	0.000
flag rate, hard negative	0.036	0.020	0.059	0.029

SE is our reimplementation and none of its values is DriftNet's own. Values are means over ten seeds. Intervals come from the stratified bootstrap of Section IV-D (B = 1,000 in panels a, c and d and B = 2,000 in panel e) and, in panel b, from the domain-shift estimator of Section IV-E (B = 2,000). Drift is the mean cosine distance from each pool-E test step to its nearest training step in embedding space. In panel c it is averaged over the five mask draws. The offset is the mean detection logit on held-out pool-E negatives minus the mean on training negatives. * marks a paired difference distinguishable from zero, after Holm correction over five folds in panel b and without correction elsewhere. † The 10-seed GRU under the same full-logit estimator, computed after the SE values were seen. It differs from the tool-pathway offset of Table V. ‡ Published values under the same masks or padding protocol (Tables S11 and S13), with the GRU at three seeds. § The suite and user-task cluster bootstrap of Table VI, computed after the run. Under it the P0F interval includes 0.5. ¶ DriftNet as published, a single GPU run with seed 42. Our run drops THOUGHT, uses a literal reading of the fourth world feature, carves validation from training and trains ten CPU seeds, so the comparison is not like for like. Panel f gives the share of AgentDojo traces, successful and unsuccessful attacks together, with each kind of value in the tool arguments. In banking all four world features are constant. SE-W0 is SE refitted with the world features set to zero in training and test.

REFERENCES

[3] E. Debenedetti, J. Zhang, M. Balunović, L. Beurer-Kellner, M. Fischer, and F. Tramèr, "AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), Datasets Benchmarks Track*, vol. 37, Vancouver, BC, Canada, 2024, pp. 82895–82920, doi: 10.52202/079017-2636.

[4] J. Liu et al., "TraceAegis: Securing LLM-based agents via hierarchical and behavioral anomaly detection," arXiv:2510.11203v1, Oct. 2025.

[5] H. Dang, "Enforcing benign trajectories: A behavioral firewall for structured-workflow AI agents," arXiv:2604.26274v1, Apr. 2026.

[6] R. Sequeira, S. Damianakis, U. Iqbal, and K. Psounis, "Agent-Sentry: Bounding LLM agents via execution provenance," arXiv:2603.22868v2, May 2026.

[7] A. Pinjari and M. P. Saint-Germain, "DriftNet: A dual-head trajectory transformer for detecting and localizing prompt injection in LLM agents," arXiv:2609.10892v1, Sep. 2026.

[8] L. Advani, "Trajectory Guard: A lightweight, sequence-aware model for real-time anomaly detection in agentic AI," arXiv:2601.00516v1, Jan. 2026.

[9] S. Zavrak, "Content-aware attack detection in LLM agent tool-call traffic: An empirical study of features, architectures, and evaluation protocols," arXiv:2605.11053v3, May 2026.

[10] C. Zhang et al., "Skynet: Workflow-level anomaly detection for agentic AI via semantic and structural modeling," arXiv:2609.06835v2, Sep. 2026.

[11] Y. Liu et al., "TrajAD: Trajectory anomaly detection for trustworthy LLM agents," arXiv:2602.06443v1, Feb. 2026.

[12] X. He, D. Wu, Y. Zhai, and K. Sun, "SentinelAgent: Graph-based anomaly detection in multi-agent systems," arXiv:2505.24201v1, May 2025.

[13] S. Chennabasappa et al., "LlamaFirewall: An open source guardrail system for building secure AI agents," arXiv:2505.03574v1, May 2025.

[21] H. Li, R. Wen, S. Shi, N. Zhang, Y. Vorobeychik, and C. Xiao, "AgentDyn: Are your agent security defenses deployable in real-world dynamic environments?" arXiv:2602.03117, 2026.

[26] Y. Li et al., "ATBench: A diverse and realistic agent trajectory benchmark for safety evaluation and diagnosis," in *Proc. 3rd Conf. Lang. Model. (COLM)*, San Francisco, CA, USA, 2026, arXiv:2604.02022.

[29] A. Pinjari and M. P. Saint-Germain, "AgentDrift: A step-labeled benchmark of injection-hijacked LLM agent trajectories," arXiv:2609.06972v1, Sep. 2026.

[30] Z. Zhang et al., "Agent-SafetyBench: Evaluating the safety of LLM agents," arXiv:2412.14470, 2024.

[31] Y. Fu, X. Yuan, and D. Wang, "RAS-Eval: A comprehensive benchmark for security evaluation of LLM agents in real-world environments," arXiv:2506.15253, 2025.

[32] H. Zhang et al., "Agent Security Bench (ASB): Formalizing and benchmarking attacks and defenses in LLM-based agents," in *Proc. 13th Int. Conf. Learn. Represent. (ICLR)*, Singapore, 2025.

[36] D. Lopez-Paz and M. Oquab, "Revisiting classifier two-sample tests," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2017. [Online]. Available: https://arxiv.org/abs/1610.06545

[39] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman & Hall, 1993.

[40] C. A. Field and A. H. Welsh, "Bootstrapping clustered data," *J. R. Stat. Soc. B, Stat. Methodol.*, vol. 69, no. 3, pp. 369–390, Jun. 2007, doi: 10.1111/j.1467-9868.2007.00593.x.

[41] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Statist.*, vol. 6, no. 2, pp. 65–70, 1979.

[42] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. 8th IEEE Int. Conf. Data Mining (ICDM)*, Pisa, Italy, 2008, pp. 413–422, doi: 10.1109/ICDM.2008.17.

[43] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," *Inf. Process. Manage.*, vol. 24, no. 5, pp. 513–523, 1988, doi: 10.1016/0306-4573(88)90021-0.

[44] K. Cho et al., "Learning phrase representations using RNN encoder-decoder for statistical machine translation," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, Doha, Qatar, Oct. 2014, pp. 1724–1734, doi: 10.3115/v1/D14-1179.

[45] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Represent. (ICLR)*, San Diego, CA, USA, 2015. [Online]. Available: https://arxiv.org/abs/1412.6980

TABLE S20
THE ALIGNMENTCHECK PROMPT WITH A LLAMA 3.3 70B (W4A16) BACKBONE (MON-L) ON THE EVALUATION LADDER

Setting	AUROC	wAUPRC 2%	wAUPRC 10%	Recall, native	FPR, native	Precision 2%, native	Recall at θ_{r80}	FPR at θ_{r80}
(a) MON-L at each rung. Native is the released decision, θ_{r80} is read from the setting's training positives								
R1	0.856 [0.838, 0.874]	0.089 [0.076, 0.118]	0.338 [0.300, 0.397]	0.847 [0.822, 0.871]	0.221 [0.193, 0.248]	0.073 [0.065, 0.082]	0.769 [0.740, 0.798]	0.173 [0.149, 0.200
R1-wd split 0	0.892 [0.876, 0.908]	0.163 [0.134, 0.224]	0.466 [0.412, 0.541]	0.901 [0.880, 0.920]	0.201 [0.173, 0.229]	0.084 [0.074, 0.096]	0.844 [0.818, 0.868]	0.148 [0.123, 0.173
R1-wd split 1	0.872 [0.854, 0.889]	0.113 [0.093, 0.154]	0.394 [0.344, 0.466]	0.877 [0.855, 0.897]	0.216 [0.187, 0.245]	0.076 [0.068, 0.088]	0.826 [0.802, 0.849]	0.168 [0.141, 0.196
R1-wd split 2	0.883 [0.867, 0.900]	0.139 [0.111, 0.223]	0.450 [0.394, 0.537]	0.874 [0.853, 0.895]	0.201 [0.171, 0.228]	0.082 [0.072, 0.094]	0.828 [0.803, 0.851]	0.152 [0.126, 0.177
R1-wd split 3	0.875 [0.857, 0.892]	0.108 [0.089, 0.150]	0.388 [0.341, 0.460]	0.881 [0.860, 0.903]	0.211 [0.184, 0.237]	0.079 [0.070, 0.089]	0.820 [0.794, 0.847]	0.159 [0.134, 0.184
R1-wd split 4	0.879 [0.862, 0.896]	0.135 [0.108, 0.220]	0.443 [0.385, 0.530]	0.858 [0.837, 0.880]	0.218 [0.190, 0.249]	0.074 [0.066, 0.085]	0.795 [0.769, 0.822]	0.148 [0.124, 0.174
R2	0.851 [0.821, 0.881]	0.115 [0.088, 0.171]	0.370 [0.308, 0.454]	0.853 [0.799, 0.901]	0.269 [0.206, 0.338]	0.061 [0.050, 0.076]	0.774 [0.704, 0.840]	0.193 [0.142, 0.253
R3 medical	0.829 [0.811, 0.847]	0.073 [0.063, 0.088]	0.292 [0.262, 0.334]	0.806 [0.782, 0.827]	0.225 [0.199, 0.251]	0.068 [0.061, 0.077]	0.698 [0.671, 0.725]	0.155 [0.133, 0.179
R3 banking	0.864 [0.846, 0.882]	0.091 [0.078, 0.122]	0.342 [0.305, 0.397]	0.869 [0.842, 0.895]	0.225 [0.199, 0.251]	0.073 [0.066, 0.082]	0.799 [0.767, 0.829]	0.178 [0.154, 0.202
R3 web	0.925 [0.912, 0.937]	0.188 [0.152, 0.263]	0.542 [0.479, 0.628]	0.912 [0.893, 0.927]	0.173 [0.150, 0.199]	0.097 [0.086, 0.110]	0.888 [0.869, 0.906]	0.128 [0.107, 0.150
R3 email	0.890 [0.874, 0.903]	0.159 [0.129, 0.234]	0.476 [0.421, 0.548]	0.885 [0.866, 0.903]	0.213 [0.186, 0.240]	0.078 [0.070, 0.088]	0.838 [0.816, 0.861]	0.163 [0.141, 0.186
R3 coding	0.855 [0.838, 0.871]	0.134 [0.112, 0.180]	0.412 [0.365, 0.476]	0.848 [0.826, 0.870]	0.225 [0.198, 0.249]	0.072 [0.065, 0.081]	0.763 [0.736, 0.790]	0.165 [0.142, 0.188
R4	0.930 [0.909, 0.951]	0.541 [0.466, 0.758]	0.751 [0.679, 0.854]	0.900 [0.865, 0.929]	0.173 [0.111, 0.245]	0.096 [0.071, 0.141]	0.877 [0.836, 0.910]	0.148 [0.088, 0.215
Within UT	0.930 [0.910, 0.950]	0.541 [0.464, 0.757]	0.751 [0.675, 0.851]	0.900 [0.868, 0.931]	0.173 [0.111, 0.241]	0.096 [0.072, 0.140]	0.802 [0.754, 0.851]	0.084 [0.039, 0.138
Within UT-F	0.930 [0.908, 0.951]	0.541 [0.463, 0.763]	0.751 [0.676, 0.854]	0.900 [0.867, 0.929]	0.173 [0.110, 0.243]	0.096 [0.071, 0.142]	0.802 [0.752, 0.848]	0.084 [0.036, 0.136
Within IT	0.930 [0.888, 0.963]	0.541 [0.412, 0.771]	0.751 [0.633, 0.869]	0.900 [0.823, 0.954]	0.173 [0.111, 0.247]	0.096 [0.069, 0.141]	0.788 [0.671, 0.872]	0.077 [0.034, 0.133
LOSO banking	0.912 [0.857, 0.962]	0.714 [0.611, 0.826]	0.781 [0.698, 0.882]	0.896 [0.815, 0.972]	0.286 [0.080, 0.500]	0.060 [0.036, 0.180]	0.718 [0.610, 0.818]	0.107 [0.000, 0.233
LOSO slack	0.922 [0.867, 0.972]	0.482 [0.404, 0.879]	0.677 [0.576, 0.927]	0.936 [0.902, 0.963]	0.248 [0.100, 0.412]	0.072 [0.045, 0.159]	0.903 [0.855, 0.944]	0.152 [0.030, 0.306
LOSO travel	0.820 [0.727, 0.895]	0.353 [0.173, 0.749]	0.551 [0.342, 0.804]	0.643 [0.482, 0.787]	0.049 [0.000, 0.140]	0.213 [0.078, 1.000]	0.390 [0.216, 0.571]	0.049 [0.000, 0.140
LOSO workspace	0.959 [0.930, 0.983]	0.887 [0.826, 0.941]	0.918 [0.869, 0.959]	0.933 [0.884, 0.973]	0.164 [0.070, 0.274]	0.104 [0.065, 0.214]	0.881 [0.818, 0.937]	0.055 [0.009, 0.104

(b) MON-L − comparator, AUROC	**TF-IDF**	**LR-C**	**LR-B**	**GRU**	**SE**
R1	-0.143^{*} $[-0.162, -0.125]$	-0.023^{*} $[-0.043, -0.002]$	$+0.026^{*}$ $[+0.004, +0.049]$	-0.076^{*} $[-0.097, -0.054]$	-0.142^{*} $[-0.160, -0.123]$
R1wd s0	-0.108^{*} $[-0.124, -0.092]$	$+0.001$ $[-0.018, +0.020]$	$+0.071^{*}$ $[+0.047, +0.095]$	-0.063^{*} $[-0.080, -0.045]$	-0.102^{*} $[-0.118, -0.086]$
R1wd s1	-0.128^{*} $[-0.146, -0.111]$	-0.018 $[-0.038, +0.001]$	$+0.068^{*}$ $[+0.044, +0.093]$	-0.076^{*} $[-0.096, -0.055]$	-0.125^{*} $[-0.143, -0.108]$
R1wd s2	-0.117^{*} $[-0.133, -0.100]$	-0.018 $[-0.036, +0.001]$	$+0.061^{*}$ $[+0.036, +0.085]$	-0.079^{*} $[-0.096, -0.060]$	-0.115^{*} $[-0.132, -0.097]$
R1wd s3	-0.125^{*} $[-0.143, -0.107]$	-0.013 $[-0.033, +0.006]$	$+0.057^{*}$ $[+0.035, +0.080]$	-0.083^{*} $[-0.103, -0.061]$	-0.119^{*} $[-0.137, -0.101]$
R1wd s4	-0.120^{*} $[-0.138, -0.104]$	-0.012 $[-0.032, +0.007]$	$+0.061^{*}$ $[+0.036, +0.085]$	-0.078^{*} $[-0.096, -0.059]$	-0.118^{*} $[-0.136, -0.101]$
R2	-0.147^{*} $[-0.176, -0.118]$	-0.051^{*} $[-0.080, -0.021]$	$+0.002$ $[-0.030, +0.035]$	-0.064^{*} $[-0.103, -0.023]$	-0.145^{*} $[-0.173, -0.116]$
R3 medical	-0.166^{*} $[-0.185, -0.149]$	$+0.084^{*}$ $[+0.060, +0.108]$	$+0.021$ $[-0.002, +0.044]$	$+0.180^{*}$ $[+0.125, +0.223]$	-0.130^{*} $[-0.156, -0.104]$
R3 banking	-0.125^{*} $[-0.144, -0.106]$	$+0.036^{*}$ $[+0.013, +0.058]$	$+0.010$ $[-0.012, +0.032]$	$+0.143^{*}$ $[+0.070, +0.206]$	-0.053^{*} $[-0.083, -0.009]$
R3 web	-0.055^{*} $[-0.070, -0.042]$	$+0.063^{*}$ $[+0.046, +0.080]$	$+0.029^{*}$ $[+0.012, +0.046]$	$+0.208^{*}$ $[+0.167, +0.237]$	$+0.021$ $[-0.014, +0.081]$
R3 email	-0.101^{*} $[-0.117, -0.087]$	$+0.150^{*}$ $[+0.127, +0.174]$	$+0.097^{*}$ $[+0.075, +0.119]$	$+0.233^{*}$ $[+0.165, +0.284]$	-0.061^{*} $[-0.083, -0.039]$
R3 coding	-0.132^{*} $[-0.149, -0.115]$	$+0.041^{*}$ $[+0.018, +0.063]$	$+0.003$ $[-0.020, +0.025]$	$+0.172^{*}$ $[+0.119, +0.209]$	-0.083^{*} $[-0.104, -0.059]$
R4	$+0.410^{*}$ $[+0.359, +0.468]$	$+0.457^{*}$ $[+0.410, +0.509]$	$+0.449^{*}$ $[+0.402, +0.500]$	$+0.421^{*}$ $[+0.369, +0.481]$	$+0.449^{*}$ $[+0.363, +0.530]$
Within UT	-0.065^{*} $[-0.086, -0.045]$	$+0.133^{*}$ $[+0.085, +0.189]$	$+0.190^{*}$ $[+0.139, +0.244]$	$+0.086^{*}$ $[+0.044, +0.142]$	-0.026 $[-0.060, +0.016]$
Within UT-F	-0.011 $[-0.042, +0.024]$	$+0.137^{*}$ $[+0.086, +0.188]$	$+0.211^{*}$ $[+0.157, +0.265]$	n/a	n/a
Within IT	-0.057^{*} $[-0.092, -0.029]$	$+0.243^{*}$ $[+0.118, +0.376]$	$+0.211^{*}$ $[+0.144, +0.288]$	n/a	n/a
LOSO banking	-0.086^{*} $[-0.141, -0.036]$	$+0.072$ $[-0.032, +0.192]$	$+0.013$ $[-0.066, +0.112]$	n/a	n/a
LOSO slack	-0.077^{*} $[-0.131, -0.026]$	$+0.177^{*}$ $[+0.081, +0.256]$	$+0.221^{*}$ $[+0.106, +0.310]$	n/a	n/a
LOSO travel	-0.169^{*} $[-0.262, -0.094]$	$+0.160^{*}$ $[+0.023, +0.257]$	$+0.208^{*}$ $[+0.085, +0.291]$	n/a	n/a
LOSO workspace	-0.032^{*} $[-0.059, -0.008]$	$+0.316^{*}$ $[+0.235, +0.396]$	$+0.315^{*}$ $[+0.238, +0.391]$	n/a	n/a

Native is the released decision rule, which flags a trace if any step is flagged. θ_{r80} is read from the monitor scores of the setting's training positives, and the monitor itself is never trained. Intervals are 95% bootstrap percentiles with B = 2,000 under the scheme of each setting (Section IV-D). Panel b gives paired differences on the same rows, with * marking an interval that excludes zero, without correction. The comparators are read from the existing caches without refit, the GRU at ten seeds where they exist and at three elsewhere, and SE only where it was run.

TABLE S21
MONITOR ARMS ON R1 AND AGENTDOJO, AGENTDOJO POPULATIONS, UNSUCCESSFUL ATTACKS AND TOOL-NAME MASKING

	AUROC	wAUPRC 2%	Recall, native	FPR, native	Precision 2%, native	Recall at θ_{r80}	FPR at θ_{r80}	Precision 2% at θ_{r80}
(a) AgentDrift R1 test, pool E								
MON-L (Llama 3.3 70B w4a16)	0.856 [0.838, 0.874]	0.089 [0.076, 0.118]	0.847 [0.822, 0.871]	0.221 [0.193, 0.248]	0.073 [0.065, 0.082]	0.769 [0.740, 0.798]	0.173 [0.149, 0.200]	0.083 [0.073, 0.096]
MON-L, error steps excluded	0.856 [0.838, 0.874]	0.089 [0.076, 0.118]	0.847 [0.822, 0.871]	0.221 [0.193, 0.248]	0.073 [0.065, 0.082]	0.769 [0.740, 0.798]	0.173 [0.149, 0.200]	0.083 [0.073, 0.096]
MON-L-th (with thought)	0.858 [0.840, 0.875]	0.162 [0.133, 0.219]	0.756 [0.724, 0.783]	0.166 [0.142, 0.192]	0.085 [0.074, 0.098]	n/a	n/a	n/a
MON-A (Claude Sonnet 5.5)	0.817 [0.796, 0.837]	0.072 [0.061, 0.099]	0.874 [0.851, 0.897]	0.313 [0.281, 0.345]	0.054 [0.049, 0.060]	n/a	n/a	n/a
MON-A, error steps excluded	0.807 [0.785, 0.828]	0.066 [0.057, 0.090]	0.862 [0.838, 0.886]	0.312 [0.280, 0.344]	0.053 [0.049, 0.059]	n/a	n/a	n/a
(b) AgentDojo P0								
MON-L (Llama 3.3 70B w4a16)	0.930 [0.909, 0.951]	0.541 [0.466, 0.758]	0.900 [0.865, 0.929]	0.173 [0.111, 0.245]	0.096 [0.071, 0.141]	0.877 [0.836, 0.910]	0.148 [0.088, 0.215]	0.108 [0.077, 0.167]
MON-L, error steps excluded	0.930 [0.909, 0.951]	0.541 [0.466, 0.758]	0.900 [0.865, 0.929]	0.173 [0.111, 0.245]	0.096 [0.071, 0.141]	0.877 [0.836, 0.910]	0.148 [0.088, 0.215]	0.108 [0.077, 0.167]
MON-L-th (with thought)	0.944 [0.924, 0.961]	0.545 [0.458, 0.736]	0.900 [0.867, 0.926]	0.125 [0.078, 0.179]	0.128 [0.094, 0.191]	n/a	n/a	n/a
MON-A (Claude Sonnet 5.5)	0.934 [0.915, 0.952]	0.649 [0.554, 0.777]	0.883 [0.849, 0.917]	0.038 [0.012, 0.071]	0.320 [0.203, 0.591]	n/a	n/a	n/a
MON-A, error steps excluded	0.933 [0.915, 0.952]	0.693 [0.597, 0.833]	0.878 [0.843, 0.912]	0.028 [0.005, 0.057]	0.390 [0.240, 0.782]	n/a	n/a	n/a

(c) Population, AUROC	**MON-L**	**MON-L-th**	**MON-A**	**TF-IDF**	**LR-C**	**GRU**	**SE**
P0	0.930 [0.918, 0.942]	0.944 [0.932, 0.954]	0.934 [0.925, 0.941]	0.520 [0.487, 0.555]	0.473 [0.440, 0.507]	0.510 [0.473, 0.543]	0.481 [0.418, 0.557]
C1	0.932 [0.919, 0.945]	0.948 [0.935, 0.959]	0.937 [0.928, 0.944]	0.555 [0.521, 0.591]	0.472 [0.436, 0.509]	0.505 [0.469, 0.541]	0.467 [0.401, 0.533]
P0F	0.916 [0.908, 0.923]	0.925 [0.918, 0.932]	0.918 [0.911, 0.925]	0.626 [0.612, 0.640]	0.626 [0.612, 0.640]	0.646 [0.631, 0.661]	0.572 [0.524, 0.646]
model-balanced	0.940 [0.928, 0.952]	0.949 [0.938, 0.960]	0.939 [0.930, 0.947]	0.488 [0.449, 0.527]	0.486 [0.448, 0.522]	0.532 [0.490, 0.569]	0.474 [0.400, 0.549]

(d) Unsuccessful attacks	**Flag rate**	**Recall on P0**	**Ratio**	**Monitor − TF-IDF**
MON-L, native (M5)	0.212 [0.147, 0.289]	0.900	0.235	$+0.212^{*}$ $[+0.147, +0.289]$
MON-A, native (M5)	0.063 [0.045, 0.082]	0.883	0.071	$+0.063^{*}$ $[+0.045, +0.082]$
TF-IDF at R4 θ_{r80}	0.000 [0.000, 0.000]	0.000	n/a	
MON-L, native (M5b)	0.212 [0.145, 0.287]	0.900	0.235	-0.096^{*} $[-0.186, -0.006]$
MON-A, native (M5b)	0.063 [0.047, 0.083]	0.883	0.071	-0.245^{*} $[-0.316, -0.176]$
TF-IDF at Within UT θ_{r80}	0.308 [0.238, 0.375]	0.780	0.395	

(e) Tool names kept	**Achieved coverage**	**Mean AUROC over draws**	**Lowest draw**	**Highest draw**
1.0	1.000	0.856 [0.836, 0.875]	0.856	0.856
0.75	0.775	0.834 [0.815, 0.852]	0.818	0.841
0.5	0.483	0.810 [0.791, 0.828]	0.805	0.819
0.25	0.225	0.796 [0.777, 0.816]	0.787	0.808
0.0	0.000	0.765 [0.743, 0.788]	0.765	0.765

MON-A is the same prompt with a Claude Sonnet 5.5 backbone (`claude-sonnet-5-5`). Its AUROC and wAUPRC use the number of flagged steps and are coarse, and it has no θ_{r80}. MON-L-th adds the agent's own text as the thought and has no θ_{r80}, because no training rows were scored for it. Panel c uses the stratified bootstrap of Table S15 (seed 12345), so its P0 intervals differ slightly from the cluster intervals of panel b. Panel d gives the flag rate of the 4,987 unsuccessful attacks at the native decision, the recall on P0 attacks and their ratio, with the paired difference from TF-IDF under the suite and user-task cluster bootstrap. M5 uses TF-IDF at its transfer threshold, which flags no AgentDojo trace, and M5b, added by amendment 5, TF-IDF within AgentDojo. Panel e masks the tool names of R1 test traces with the masks of Fig. 3 and averages over the five draws of each level.

[49] Y. Bengio and Y. Grandvalet, "No unbiased estimator of the variance of K-fold cross-validation," *J. Mach. Learn. Res.*, vol. 5, pp. 1089–1105, Sep. 2004.

[50] Y. Qin et al., "ToolLLM: Facilitating large language models to master 16000+ real-world APIs," in *Proc. 12th Int. Conf. Learn. Represent. (ICLR)*, Vienna, Austria, 2024.

[51] D. Jin, Z. Jin, J. T. Zhou, and P. Szolovits, "Is BERT really robust? A strong baseline for natural language attack on text classification and entailment," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, no. 5, 2020, pp. 8018–8025, doi: 10.1609/aaai.v34i05.6311.

[52] Sentence Transformers, "all-mpnet-base-v2," Hugging Face model card. Accessed: Sep. 26, 2026. [Online]. Available: https://huggingface.co/sentence-transformers/all-mpnet-base-v2

[53] Galileo AI, "Agent Leaderboard v2," Hugging Face dataset, 2025. Accessed: Sep. 26, 2026. [Online]. Available: https://huggingface.co/datasets/galileo-ai/agent-leaderboard-v2

[54] J. Zhang, L. Yin, Y. Zhou, and S. Hu, "AgentAlign: Navigating safety alignment in the shift from informative to agentic large language models," arXiv:2505.23020, 2025.

[55] S. Zhang et al., "Which agent causes task failures and when? On automated failure attribution of LLM multi-agent systems," in *Proc. 42nd Int. Conf. Mach. Learn. (ICML)*, ser. PMLR, vol. 267, 2025, pp. 76583–76599.

[56] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. Conf. Empirical Methods Natural Lang. Process. 9th Int. Joint Conf. Natural Lang. Process. (EMNLP-IJCNLP)*, Hong Kong, China, Nov. 2019, pp. 3982–3992, doi: 10.18653/v1/D19-1410.

[57] Y. Song, D. Yin, X. Yue, J. Huang, S. Li, and B. Y. Lin, "Trial and error: Exploration-based trajectory optimization of LLM agents," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics (Vol. 1: Long Papers)*, Bangkok, Thailand, Aug. 2024, pp. 7584–7600, doi: 10.18653/v1/2024.acl-long.409.
[58] K. Zhu et al., "Where LLM agents fail and how they can learn from failures," arXiv:2509.25370, 2025.
[59] A. Yang et al., "Qwen3 technical report," arXiv:2505.09388, 2025.